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EUROPEAN COMMISSION
Directorate-General for Communications Networks, Content and Technology

CNECT.G – Data
G.1 – Data Policy and Innovation

GRANT AGREEMENT

Project 101135927 — NOUS

PREAMBLE

This **Agreement** ('the Agreement') is **between** the following parties:

on the one part,

the **European Union** ('EU'), represented by the European Commission ('European Commission' or 'granting authority'),

and

on the other part,

1. 'the coordinator':

FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR), PIC 904614338, established in PAZA DEL AYUNTAMIENTO 1, CARBAJOSA DE LA SAGRADA 37188, Spain,

and the following other beneficiaries, if they sign their 'accession form' (see Annex 3 and Article 40):

2. **AETHON ENGINEERING SINGLE MEMBER PC (AETHON)**, PIC 918590292, established in EMMANOUIL BENAKI 25, ATHENS 106 78, Greece,

3. **UNIVERSIDAD DE SALAMANCA (USAL)**, PIC 999846610, established in CALLE PATIO DE ESCUELAS 1, SALAMANCA 37008, Spain,

4. **NATIONAL CENTER FOR SCIENTIFIC RESEARCH "DEMOKRITOS" (NCSR "D")**, PIC 999978239, established in END OF PATRIARCHOU GRIGORIOU E AND 27 NEAPOLEOS STREET, AGIA PARASKEVI 15341, Greece,

5. **CONSORZIO INTERUNIVERSITARIO PER L'OTTIMIZZAZIONE E LA RICERCA OPERATIVA (ICOOR)**, PIC 996385068, established in VIALE RISORGIMENTO 2, BOLOGNA 40136, Italy,

6. **TELECOM ITALIA SPA (TIM)**, PIC 999908496, established in VIA GAETANO NEGRI 1, MILANO 20123, Italy,

7. **NETCOMPANY-INTRASOFT SA (NET-INTRA)**, PIC 999702371, established in RUE NICOLAS BOVE 2B, LUXEMBOURG 1253, Luxembourg,

8. **IOTAM INTERNET OF THINGS APPLICATIONS AND MULTI LAYER DEVELOPMENT LTD (ITML CY)**, PIC 912549326, established in 56 GRIVA DIGENI STR ANNA TOWER FLAT/OFFICE 3, LIMASSOL 3101, Cyprus,

9. **UNIVERSITA DI PISA (UNIPI)**, PIC 999862712, established in LUNGARNO PACINOTTI 43/44, PISA 56126, Italy,

10. **UNPARALLEL INNOVATION LDA (UNPARALLEL)**, PIC 953318426, established in RUA DAS LENDAS ALGARVIAS LOTE 123, PORTIMAO 8500-794, Portugal,

11. **KATHOLIEKE UNIVERSITEIT LEUVEN (KU Leuven)**, PIC 999991334, established in OUDE MARKT 13, LEUVEN 3000, Belgium,

12. **HEWLETT - PACKARD HELLAS ETAIREIA PERIORISMENIS EFTHYNIS (HPE Hellas)**, PIC 913711871, established in TZAVELLA 1-3, ATHINA 152 31, Greece,

13. **ECLIPSE FOUNDATION EUROPE GMBH (ECL)**, PIC 944758564, established in BERLINER ALLEE 47, DARMSTADT 64295, Germany,

14. **F6S NETWORK IRELAND LIMITED (F6S IE)**, PIC 900885658, established in 39 FITZWILLIAM PLACE, DUBLIN D02 ND61, Ireland,

15. **DIMOSIA EPICHEIRISI ILEKTRISMOU ANONYMI ETAIREIA (PPC)**, PIC 999938954, established in CHALKOKONDYLI STREET 30, ATHINA 104 32, Greece,

16. **ARCTUR RACUNALNISKI INZENIRING DOO (ARCTUR d.o.o.)**, PIC 991568048, established in INDUSTRIJSKA CESTA 1A, NOVA GORICA 5000, Slovenia,

17. **AEGIS IT RESEARCH GMBH (AEGIS)**, PIC 911867804, established in HUMBOLDTSTR. 25, BRAUNSCHWEIG 38106, Germany,

18. **CS GROUP-FRANCE (CS GROUP-FRANCE)**, PIC 999938275, established in AVENUE GALILEE 22, LE PLESSIS ROBINSON 92350, France,

19. **INESC TEC - INSTITUTO DE ENGENHARIADE SISTEMAS E COMPUTADORES, TECNOLOGIA E CIENCIA (INESC TEC)**, PIC 999513706, established in RUA DR ROBERTO FRIAS CAMPUS DA FEUP, PORTO 4200 465, Portugal,

Unless otherwise specified, references to ‘beneficiary’ or ‘beneficiaries’ include the coordinator and affiliated entities (if any).

If only one beneficiary signs the grant agreement (‘mono-beneficiary grant’), all provisions referring to the ‘coordinator’ or the ‘beneficiaries’ will be considered — mutatis mutandis — as referring to the beneficiary.

The parties referred to above have agreed to enter into the Agreement.

By signing the Agreement and the accession forms, the beneficiaries accept the grant and agree to implement the action under their own responsibility and in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

The Agreement is composed of:

Preamble

Terms and Conditions (including Data Sheet)

Annex 1	Description of the action[1]
Annex 2	Estimated budget for the action
Annex 2a	Additional information on unit costs and contributions (if applicable)
Annex 3	Accession forms (if applicable)[2]
Annex 3a	Declaration on joint and several liability of affiliated entities (if applicable)[3]
Annex 4	Model for the financial statements
Annex 5	Specific rules (if applicable)

[1] Template published on Portal Reference Documents.
[2] Template published on Portal Reference Documents.
[3] Template published on Portal Reference Documents.

TERMS AND CONDITIONS

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DATA SHEET

1. General data

Project summary:

Project summary
NOUS will develop the architecture of a European Cloud Service that allows computational and data storage resources to be used from edge devices as well as supercomputers, through the HPC network, and Quantum Computers. NOUS will be an Infrastructure-as-a-Service (IaaS)/Platform-as-a-Service (PaaS) cloud provider, harnessing edge computing and decentralisation paradigms to incorporate a wide array of devices and machines in its computational flow to provide leaps in Europe's capability to process vast amounts of data. The pipeline of the NOUS in the project will include three types of components: i) computational components that are responsible for executing computations, ii) edge components that are responsible for communicating with edge devices (such as IoT sensors/ actuators/ devices), iii) data storage components that are responsible for data storage and storage management. Components are researched individually, expecting to yield breakthroughs, and jointly, to create the architecture and cloud-level services such as syndication with other platforms and virtual labs. The project has defined 4 use-cases that will allow the testing of the developed technologies in real-world scenarios that industry leaders face. The NOUS architecture will be made open source to allow the capitalization by companies and organisations. Furthermore, a set of workshops and collaboration activities is envisioned with Data Spaces Support Centre, Gaia-X, FIWARE and EOSC powered by a strong consortium of 21 partners from 11 European countries.

Keywords:

- Data curation
- Linked open data
- Cloud architecture & services, HPC, Quantum computing, Edge computing, Distributed/Federated concepts, AI, ML, Blockchain, Data management, Data Spaces, sharing, life cycle & economy, Cybersecurity

Project number: 101135927

Project name: A catalyst for EuropeaN ClOUd Services in the era of data spaces, high-performance and edge computing

Project acronym: NOUS

Call: HORIZON-CL4-2023-DATA-01

Topic: HORIZON-CL4-2023-DATA-01-02

Type of action: HORIZON Research and Innovation Actions

Granting authority: European Commission-EU

Grant managed through EU Funding & Tenders Portal: Yes (eGrants)

Project starting date: fixed date: 1 January 2024

Project end date: 31 December 2026

Project duration: 36 months

Consortium agreement: Yes

2. Participants

List of participants:

N°	Role	Short name	Legal name	Ctry	PIC	Total eligible costs (BEN and AE)	Max grant amount
1	COO	AIR	FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION	ES	904614338	596 750.00	596 750.00
2	BEN	AETHON	AETHON ENGINEERING SINGLE MEMBER PC	EL	918590292	554 250.00	554 250.00

N°	Role	Short name	Legal name	Ctry	PIC	Total eligible costs (BEN and AE)	Max grant amount
3	BEN	USAL	UNIVERSIDAD DE SALAMANCA	ES	999846610	218 500.00	218 500.00
4	BEN	NCSR "D"	NATIONAL CENTER FOR SCIENTIFIC RESEARCH "DEMOKRITOS"	EL	999978239	651 875.00	651 875.00
5	BEN	ICOOR	CONSORZIO INTERUNIVERSITARIO PER L'OTTIMIZZAZIONE E LA RICERCA OPERATIVA	IT	996385068	42 500.00	42 500.00
5.1	AE	POLITO	POLITECNICO DI TORINO	IT	999977754	400 000.00	400 000.00
5.2	AE	UNIMORE	UNIVERSITA DEGLI STUDI DI MODENA E REGGIO EMILIA	IT	999840887	298 750.00	298 750.00
6	BEN	TIM	TELECOM ITALIA SPA	IT	999908496	351 875.00	351 875.00
7	BEN	NET-INTRA	NETCOMPANY-INTRASOFT SA	LU	999702371	586 250.00	586 250.00
8	BEN	ITML CY	IOTAM INTERNET OF THINGS APPLICATIONS AND MULTI LAYER DEVELOPMENT LTD	CY	912549326	504 375.00	504 375.00
9	BEN	UNIPI	UNIVERSITA DI PISA	IT	999862712	325 625.00	325 625.00
10	BEN	UNPARALLEL	UNPARALLEL INNOVATION LDA	PT	953318426	375 312.50	375 312.50
11	BEN	KU Leuven	KATHOLIEKE UNIVERSITEIT LEUVEN	BE	999991334	487 500.00	487 500.00
12	BEN	HPE Hellas	HEWLETT - PACKARD HELLAS ETAIREIA PERIORISMENIS EFTHYNIS	EL	913711871	545 625.00	545 625.00
13	BEN	ECL	ECLIPSE FOUNDATION EUROPE GMBH	DE	944758564	359 125.00	359 125.00
14	BEN	F6S IE	F6S NETWORK IRELAND LIMITED	IE	900885658	337 500.00	337 500.00
15	BEN	PPC	DIMOSIA EPICHEIRISI ILEKTRISMOU ANONYMI ETAIREIA	EL	999938954	333 437.50	333 437.50
16	BEN	ARCTUR d.o.o.	ARCTUR RACUNALNISKI INZENIRING DOO	SI	991568048	323 500.00	323 500.00
17	BEN	AEGIS	AEGIS IT RESEARCH GMBH	DE	911867804	398 000.00	398 000.00
18	BEN	CS GROUP-FRANCE	CS GROUP-FRANCE	FR	999938275	348 125.00	348 125.00
19	BEN	INESC TEC	INESC TEC - INSTITUTO DE ENGENHARIADE SISTEMAS E COMPUTADORES, TECNOLOGIA E CIENCIA	PT	999513706	390 000.00	390 000.00
Total						8 428 875.00	8 428 875.00

Coordinator:

– FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR)

3. Grant

Maximum grant amount, total estimated eligible costs and contributions and funding rate:

Total eligible costs (BEN and AE)	Funding rate (%)	Maximum grant amount (Annex 2)	Maximum grant amount (award decision)
8 428 875.00	100	8 428 875.00	8 428 875.00

Grant form: Budget-based

Grant mode: Action grant

Budget categories/activity types:

- A. Personnel costs
  - A.1 Employees, A.2 Natural persons under direct contract, A.3 Seconded persons
  - A.4 SME owners and natural person beneficiaries

- B. Subcontracting costs

- C. Purchase costs
  - C.1 Travel and subsistence
  - C.2 Equipment
  - C.3 Other goods, works and services

- D. Other cost categories
  - D.2 Internally invoiced goods and services

- E. Indirect costs

Cost eligibility options:

- In-kind contributions eligible costs

- Parental leave

- Project-based supplementary payments

- Average personnel costs (unit cost according to usual cost accounting practices)

- Limitation for subcontracting

- Travel and subsistence:
  - Travel: Actual costs
  - Accommodation: Actual costs
  - Subsistence: Actual costs

- Equipment: depreciation only

- Indirect cost flat-rate: 25% of the eligible direct costs (categories A-D, except volunteers costs, subcontracting costs, financial support to third parties and exempted specific cost categories, if any)

- VAT: Yes

- Other ineligible costs

Budget flexibility: Yes (no flexibility cap)

4. Reporting, payments and recoveries

4.1 Continuous reporting (art 21)

Deliverables: see Funding & Tenders Portal Continuous Reporting tool

4.2 Periodic reporting and payments

Reporting and payment schedule (art 21, 22):

Reporting					Payments	
Reporting periods			Type	Deadline	Type	Deadline (time to pay)
RP No	Month from	Month to				
					Initial prefinancing	30 days from entry into force/10 days before starting date – whichever is the latest
1	1	18	Periodic report	60 days after end of reporting period	Interim payment	90 days from receiving periodic report
2	19	36	Periodic report	60 days after end of reporting period	Final payment	90 days from receiving periodic report

Prefinancing payments and guarantees:

Prefinancing payment	
Type	Amount
Prefinancing 1 (initial)	6 743 100.00

Reporting and payment modalities (art 21, 22):

Mutual Insurance Mechanism (MIM): Yes

MIM contribution: 5% of the maximum grant amount (421 443.75), retained from the initial prefinancing

Restrictions on distribution of initial prefinancing: The prefinancing may be distributed only if the minimum number of beneficiaries set out in the call condititions (if any) have acceded to the Agreement and only to beneficiaries that have acceded.

Interim payment ceiling (if any): 90% of the maximum grant amount

Exception for revenues: Yes

No-profit rule: Yes

Late payment interest: ECB + 3.5%

Bank account for payments:

ES3121032200180030157510

Conversion into euros: Double conversion

Reporting language: Language of the Agreement

4.3 Certificates (art 24):

Certificates on the financial statements (CFS):

Conditions:

Schedule: only at final payment, if threshold is reached

Standard threshold (beneficiary-level):

- financial statement: requested EU contribution to costs ≥ EUR 430 000.00

Special threshold for beneficiaries with a systems and process audit(see Article 24): financial statement: requested EU contribution to costs ≥ EUR 725 000.00

4.4 Recoveries (art 22)

First-line liability for recoveries:

Beneficiary termination: Beneficiary concerned

Final payment: Each beneficiary for their own debt

After final payment: Beneficiary concerned

Joint and several liability for enforced recoveries (in case of non-payment):

Individual financial responsibility: Each beneficiary is liable only for its own debts (and those of its affiliated entities, if any)

Joint and several liability of affiliated entities — n/a

5. Consequences of non-compliance, applicable law & dispute settlement forum

Suspension and termination:

Additional suspension grounds (art 31)

Additional termination grounds (art 32)

Applicable law (art 43):

Standard applicable law regime: EU law + law of Belgium

Dispute settlement forum (art 43):

Standard dispute settlement forum:

EU beneficiaries: EU General Court + EU Court of Justice (on appeal)

Non-EU beneficiaries: Courts of Brussels, Belgium (unless an international agreement provides for the enforceability of EU court judgements)

6. Other

Specific rules (Annex 5): Yes

Standard time-limits after project end:

Confidentiality (for X years after final payment): 5

Record-keeping (for X years after final payment): 5 (or 3 for grants of not more than EUR 60 000)

Reviews (up to X years after final payment): 2

Audits (up to X years after final payment): 2

Extension of findings from other grants to this grant (no later than X years after final payment): 2

Impact evaluation (up to X years after final payment): 5 (or 3 for grants of not more than EUR 60 000)

CHAPTER 1 GENERAL

ARTICLE 1 — SUBJECT OF THE AGREEMENT

This Agreement sets out the rights and obligations and terms and conditions applicable to the grant awarded for the implementation of the action set out in Chapter 2.

ARTICLE 2 — DEFINITIONS

For the purpose of this Agreement, the following definitions apply:

Actions — The project which is being funded in the context of this Agreement.

Grant — The grant awarded in the context of this Agreement.

EU grants — Grants awarded by EU institutions, bodies, offices or agencies (including EU executive agencies, EU regulatory agencies, EDA, joint undertakings, etc.).

Participants — Entities participating in the action as beneficiaries, affiliated entities, associated partners, third parties giving in-kind contributions, subcontractors or recipients of financial support to third parties.

Beneficiaries (BEN) — The signatories of this Agreement (either directly or through an accession form).

Affiliated entities (AE) — Entities affiliated to a beneficiary within the meaning of Article 187 of EU Financial Regulation 2018/1046[4] which participate in the action with similar rights and obligations as the beneficiaries (obligation to implement action tasks and right to charge costs and claim contributions).

Associated partners (AP) — Entities which participate in the action, but without the right to charge costs or claim contributions.

Purchases — Contracts for goods, works or services needed to carry out the action (e.g. equipment, consumables and supplies) but which are not part of the action tasks (see Annex 1).

Subcontracting — Contracts for goods, works or services that are part of the action tasks (see Annex 1).

In-kind contributions — In-kind contributions within the meaning of Article 2(36) of EU Financial

[4] For the definition, see Article 187 Regulation (EU, Euratom) 2018/1046 of the European Parliament and of the Council of 18 July 2018 on the financial rules applicable to the general budget of the Union, amending Regulations (EU) No 1296/2013, (EU) No 1301/2013, (EU) No 1303/2013, (EU) No 1304/2013, (EU) No 1309/2013, (EU) No 1316/2013, (EU) No 223/2014, (EU) No 283/2014, and Decision No 541/2014/EU and repealing Regulation (EU, Euratom) No 966/2012 ('EU Financial Regulation') (OJ L 193, 30.7.2018, p. 1): "**affiliated entities** [are]:
(a) entities that form a sole beneficiary [(i.e. where an entity is formed of several entities that satisfy the criteria for being awarded a grant, including where the entity is specifically established for the purpose of implementing an action to be financed by a grant)];
(b) entities that satisfy the eligibility criteria and that do not fall within one of the situations referred to in Article 136(1) and 141(1) and that have a link with the beneficiary, in particular a legal or capital link, which is neither limited to the action nor established for the sole purpose of its implementation".

Regulation 2018/1046, i.e. non-financial resources made available free of charge by third parties.

Fraud — Fraud within the meaning of Article 3 of EU Directive 2017/1371[5] and Article 1 of the Convention on the protection of the European Communities' financial interests, drawn up by the Council Act of 26 July 1995[6], as well as any other wrongful or criminal deception intended to result in financial or personal gain.

Irregularities — Any type of breach (regulatory or contractual) which could impact the EU financial interests, including irregularities within the meaning of Article 1(2) of EU Regulation 2988/95[7].

Grave professional misconduct — Any type of unacceptable or improper behaviour in exercising one's profession, especially by employees, including grave professional misconduct within the meaning of Article 136(1)(c) of EU Financial Regulation 2018/1046.

Applicable EU, international and national law — Any legal acts or other (binding or non-binding) rules and guidance in the area concerned.

Portal — EU Funding & Tenders Portal; electronic portal and exchange system managed by the European Commission and used by itself and other EU institutions, bodies, offices or agencies for the management of their funding programmes (grants, procurements, prizes, etc.).

CHAPTER 2 ACTION

ARTICLE 3 — ACTION

The grant is awarded for the action **101135927 — NOUS** ('action'), as described in Annex 1.

ARTICLE 4 — DURATION AND STARTING DATE

The duration and the starting date of the action are set out in the Data Sheet (see Point 1).

CHAPTER 3 GRANT

ARTICLE 5 — GRANT

5.1 Form of grant

The grant is an action grant[8] which takes the form of a budget-based mixed actual cost grant (i.e. a

[5] Directive (EU) 2017/1371 of the European Parliament and of the Council of 5 July 2017 on the fight against fraud to the Union's financial interests by means of criminal law (OJ L 198, 28.7.2017, p. 29).

[6] OJ C 316, 27.11.1995, p. 48.

[7] Council Regulation (EC, Euratom) No 2988/95 of 18 December 1995 on the protection of the European Communities financial interests (OJ L 312, 23.12.1995, p. 1).

[8] For the definition, see Article 180(2)(a) EU Financial Regulation 2018/1046: '**action grant**' means an EU grant to finance "an action intended to help achieve a Union policy objective".

grant based on actual costs incurred, but which may also include other forms of funding, such as unit costs or contributions, flat-rate costs or contributions, lump sum costs or contributions or financing not linked to costs).

5.2 Maximum grant amount

The maximum grant amount is set out in the Data Sheet (see Point 3) and in the estimated budget (Annex 2).

5.3 Funding rate

The funding rate for costs is 100% of the action's eligible costs.

Contributions are not subject to any funding rate.

5.4 Estimated budget, budget categories and forms of funding

The estimated budget for the action is set out in Annex 2.

It contains the estimated eligible costs and contributions for the action, broken down by participant and budget category.

Annex 2 also shows the types of costs and contributions (forms of funding)[9] to be used for each budget category.

If unit costs or contributions are used, the details on the calculation will be explained in Annex 2a.

5.5 Budget flexibility

The budget breakdown may be adjusted — without an amendment (see Article 39) — by transfers (between participants and budget categories), as long as this does not imply any substantive or important change to the description of the action in Annex 1.

However:

- changes to the budget category for volunteers (if used) always require an amendment
- changes to budget categories with lump sums costs or contributions (if used; including financing not linked to costs) always require an amendment
- changes to budget categories with higher funding rates or budget ceilings (if used) always require an amendment
- addition of amounts for subcontracts not provided for in Annex 1 either require an amendment or simplified approval in accordance with Article 6.2
- other changes require an amendment or simplified approval, if specifically provided for in Article 6.2
- flexibility caps: not applicable.

[9] See Article 125 EU Financial Regulation 2018/1046.

ARTICLE 6 — ELIGIBLE AND INELIGIBLE COSTS AND CONTRIBUTIONS

In order to be eligible, costs and contributions must meet the **eligibility** conditions set out in this Article.

6.1 General eligibility conditions

The **general eligibility conditions** are the following:

(a) for actual costs:

(i) they must be actually incurred by the beneficiary

(ii) they must be incurred in the period set out in Article 4 (with the exception of costs relating to the submission of the final periodic report, which may be incurred afterwards; see Article 21)

(iii) they must be declared under one of the budget categories set out in Article 6.2 and Annex 2

(iv) they must be incurred in connection with the action as described in Annex 1 and necessary for its implementation

(v) they must be identifiable and verifiable, in particular recorded in the beneficiary's accounts in accordance with the accounting standards applicable in the country where the beneficiary is established and with the beneficiary's usual cost accounting practices

(vi) they must comply with the applicable national law on taxes, labour and social security and

(vii) they must be reasonable, justified and must comply with the principle of sound financial management, in particular regarding economy and efficiency

(b) for unit costs or contributions (if any):

(i) they must be declared under one of the budget categories set out in Article 6.2 and Annex 2

(ii) the units must:

- be actually used or produced by the beneficiary in the period set out in Article 4 (with the exception of units relating to the submission of the final periodic report, which may be used or produced afterwards; see Article 21)

- be necessary for the implementation of the action and

(iii) the number of units must be identifiable and verifiable, in particular supported by records and documentation (see Article 20)

(c) for flat-rate costs or contributions (if any):

(i) they must be declared under one of the budget categories set out in Article 6.2 and Annex 2

(ii) the costs or contributions to which the flat-rate is applied must:

- be eligible
- relate to the period set out in Article 4 (with the exception of costs or contributions relating to the submission of the final periodic report, which may be incurred afterwards; see Article 21)

(d) for lump sum costs or contributions (if any):

(i) they must be declared under one of the budget categories set out in Article 6.2 and Annex 2

(ii) the work must be properly implemented by the beneficiary in accordance with Annex 1

(iii) the deliverables/outputs must be achieved in the period set out in Article 4 (with the exception of deliverables/outputs relating to the submission of the final periodic report, which may be achieved afterwards; see Article 21)

(e) for unit, flat-rate or lump sum costs or contributions according to usual cost accounting practices (if any):

(i) they must fulfil the general eligibility conditions for the type of cost concerned

(ii) the cost accounting practices must be applied in a consistent manner, based on objective criteria, regardless of the source of funding

(f) for financing not linked to costs (if any): the results must be achieved or the conditions must be fulfilled as described in Annex 1.

In addition, for direct cost categories (e.g. personnel, travel & subsistence, subcontracting and other direct costs) only costs that are directly linked to the action implementation and can therefore be attributed to it directly are eligible. They must not include any indirect costs (i.e. costs that are only indirectly linked to the action, e.g. via cost drivers).

In-kind contributions provided by third parties free of charge may be declared as eligible direct costs by the beneficiaries which use them (under the same conditions as if they were their own, provided that they concern only direct costs and that the third parties and their in-kind contributions are set out in Annex 1 (or approved ex post in the periodic report, if their use does not entail changes to the Agreement which would call into question the decision awarding the grant or breach the principle of equal treatment of applicants; 'simplified approval procedure').

6.2 Specific eligibility conditions for each budget category

For each budget category, the **specific eligibility conditions** are as follows:

<u>Direct costs</u>

A. Personnel costs

A.1 Costs for employees (or equivalent) are eligible as personnel costs if they fulfil the general eligibility conditions and are related to personnel working for the beneficiary under an employment contract (or equivalent appointing act) and assigned to the action.

They must be limited to salaries (including net payments during parental leave), social security contributions, taxes and other costs linked to the remuneration, if they arise from national law or the employment contract (or equivalent appointing act) and be calculated on the basis of the costs actually incurred, in accordance with the following method:

{daily rate for the person

multiplied by

number of day-equivalents worked on the action (rounded up or down to the nearest half-day)}.

The daily rate must be calculated as:

{annual personnel costs for the person

divided by

215}.

The number of day-equivalents declared for a person must be identifiable and verifiable (see Article 20).

The actual time spent on parental leave by a person assigned to the action may be deducted from the 215 days indicated in the above formula.

The total number of day-equivalents declared in EU grants, for a person for a year, cannot be higher than 215, minus time spent on parental leave (if any).

For personnel which receives supplementary payments for work in projects (project-based remuneration), the personnel costs must be calculated at a rate which:

- corresponds to the actual remuneration costs paid by the beneficiary for the time worked by the person in the action over the reporting period
- does not exceed the remuneration costs paid by the beneficiary for work in similar projects funded by national schemes (‘national projects reference’)
- is defined based on objective criteria allowing to determine the amount to which the person is entitled

and

- reflects the usual practice of the beneficiary to pay consistently bonuses or supplementary payments for work in projects funded by national schemes.

The national projects reference is the remuneration defined in national law, collective labour agreement or written internal rules of the beneficiary applicable to work in projects funded by national schemes.

If there is no such national law, collective labour agreement or written internal rules or if the project-based remuneration is not based on objective criteria, the national project reference will be the average

remuneration of the person in the last full calendar year covered by the reporting period, excluding remuneration paid for work in EU actions.

If the beneficiary uses average personnel costs (unit cost according to usual cost accounting practices), the personnel costs must fulfil the general eligibility conditions for such unit costs and the daily rate must be calculated:

- using the actual personnel costs recorded in the beneficiary's accounts and excluding any costs which are ineligible or already included in other budget categories; the actual personnel costs may be adjusted on the basis of budgeted or estimated elements, if they are relevant for calculating the personnel costs, reasonable and correspond to objective and verifiable information

and

- according to usual cost accounting practices which are applied in a consistent manner, based on objective criteria, regardless of the source of funding.

A.2 and **A.3 Costs for natural persons working under a direct contract** other than an employment contract and costs for **seconded persons by a third party against payment** are also eligible as personnel costs, if they are assigned to the action, fulfil the general eligibility conditions and:

(a) work under conditions similar to those of an employee (in particular regarding the way the work is organised, the tasks that are performed and the premises where they are performed) and

(b) the result of the work belongs to the beneficiary (unless agreed otherwise).

They must be calculated on the basis of a rate which corresponds to the costs actually incurred for the direct contract or secondment and must not be significantly different from those for personnel performing similar tasks under an employment contract with the beneficiary.

A.4 The work of **SME owners** for the action (i.e. owners of beneficiaries that are small and medium-sized enterprises[10] not receiving a salary) or **natural person beneficiaries** (i.e. beneficiaries that are natural persons not receiving a salary) may be declared as personnel costs, if they fulfil the general eligibility conditions and are calculated as unit costs in accordance with the method set out in Annex 2a.

B. Subcontracting costs

Subcontracting costs for the action (including related duties, taxes and charges, such as non-deductible or non-refundable value added tax (VAT)) are eligible, if they are calculated on the basis of the costs actually incurred, fulfil the general eligibility conditions and are awarded using the

[10] For the definition, see Commission Recommendation 2003/361/EC: micro, small or medium-sized enterprise (SME) are enterprises

- engaged in an economic activity, irrespective of their legal form (including, in particular, self- employed persons and family businesses engaged in craft or other activities, and partnerships or associations regularly engaged in an economic activity) and

- employing fewer than 250 persons (expressed in 'annual working units' as defined in Article 5 of the Recommendation) and which have an annual turnover not exceeding EUR 50 million, and/or an annual balance sheet total not exceeding EUR 43 million.

beneficiary's usual purchasing practices — provided these ensure subcontracts with best value for money (or if appropriate the lowest price) and that there is no conflict of interests (see Article 12).

Beneficiaries that are 'contracting authorities/entities' within the meaning of the EU Directives on public procurement must also comply with the applicable national law on public procurement.

Subcontracting may cover only a limited part of the action.

The tasks to be subcontracted and the estimated cost for each subcontract must be set out in Annex 1 and the total estimated costs of subcontracting per beneficiary must be set out in Annex 2 (or may be approved ex post in the periodic report, if the use of subcontracting does not entail changes to the Agreement which would call into question the decision awarding the grant or breach the principle of equal treatment of applicants; 'simplified approval procedure').

C. Purchase costs

Purchase costs for the action (including related duties, taxes and charges, such as non-deductible or non-refundable value added tax (VAT)) are eligible if they fulfil the general eligibility conditions and are bought using the beneficiary's usual purchasing practices — provided these ensure purchases with best value for money (or if appropriate the lowest price) and that there is no conflict of interests (see Article 12).

Beneficiaries that are 'contracting authorities/entities' within the meaning of the EU Directives on public procurement must also comply with the applicable national law on public procurement.

C.1 Travel and subsistence

Purchases for **travel, accommodation** and **subsistence** must be calculated as follows:

- travel: on the basis of the costs actually incurred and in line with the beneficiary's usual practices on travel
- accommodation: on the basis of the costs actually incurred and in line with the beneficiary's usual practices on travel
- subsistence: on the basis of the costs actually incurred and in line with the beneficiary's usual practices on travel .

C.2 Equipment

Purchases of **equipment, infrastructure or other assets** used for the action must be declared as depreciation costs, calculated on the basis of the costs actually incurred and written off in accordance with international accounting standards and the beneficiary's usual accounting practices.

Only the portion of the costs that corresponds to the rate of actual use for the action during the action duration can be taken into account.

Costs for **renting or leasing** equipment, infrastructure or other assets are also eligible, if they do not exceed the depreciation costs of similar equipment, infrastructure or assets and do not include any financing fees.

C.3 Other goods, works and services

Purchases of **other goods, works and services** must be calculated on the basis of the costs actually incurred.

Such goods, works and services include, for instance, consumables and supplies, promotion, dissemination, protection of results, translations, publications, certificates and financial guarantees, if required under the Agreement.

D. Other cost categories

D.2 Internally invoiced goods and services

Costs for internally invoiced goods and services directly used for the action may be declared as unit cost according to usual cost accounting practices, if and as declared eligible in the call conditions, if they fulfil the general eligibility conditions for such unit costs and the amount per unit is calculated:

- using the actual costs for the good or service recorded in the beneficiary's accounts, attributed either by direct measurement or on the basis of cost drivers, and excluding any cost which are ineligible or already included in other budget categories; the actual costs may be adjusted on the basis of budgeted or estimated elements, if they are relevant for calculating the costs, reasonable and correspond to objective and verifiable information

and

- according to usual cost accounting practices which are applied in a consistent manner, based on objective criteria, regardless of the source of funding.

'Internally invoiced goods and services' means goods or services which are provided within the beneficiary's organisation directly for the action and which the beneficiary values on the basis of its usual cost accounting practices.

This cost will not be taken into account for the indirect cost flat-rate.

<u>Indirect costs</u>

E. Indirect costs

Indirect costs will be reimbursed at the flat-rate of 25% of the eligible direct costs (categories A-D, except volunteers costs, subcontracting costs, financial support to third parties and exempted specific cost categories, if any).

<u>Contributions</u>

Not applicable

6.3 Ineligible costs and contributions

The following costs or contributions are **ineligible**:

(a) costs or contributions that do not comply with the conditions set out above (Article 6.1 and 6.2), in particular:

(i) costs related to return on capital and dividends paid by a beneficiary

(ii) debt and debt service charges

(iii) provisions for future losses or debts

(iv) interest owed

(v) currency exchange losses

(vi) bank costs charged by the beneficiary's bank for transfers from the granting authority

(vii) excessive or reckless expenditure

(viii) deductible or refundable VAT (including VAT paid by public bodies acting as public authority)

(ix) costs incurred or contributions for activities implemented during grant agreement suspension (see Article 31)

(x) in-kind contributions by third parties: not applicable

(b) costs or contributions declared under other EU grants (or grants awarded by an EU Member State, non-EU country or other body implementing the EU budget), except for the following cases:

(i) Synergy actions: not applicable

(ii) if the action grant is combined with an operating grant[11] running during the same period and the beneficiary can demonstrate that the operating grant does not cover any (direct or indirect) costs of the action grant

(c) costs or contributions for staff of a national (or regional/local) administration, for activities that are part of the administration's normal activities (i.e. not undertaken only because of the grant)

(d) costs or contributions (especially travel and subsistence) for staff or representatives of EU institutions, bodies or agencies

(e) other :

(i) country restrictions for eligible costs: not applicable

(ii) costs or contributions declared specifically ineligible in the call conditions.

6.4 Consequences of non-compliance

If a beneficiary declares costs or contributions that are ineligible, they will be rejected (see Article 27).

This may also lead to other measures described in Chapter 5.

[11] For the definition, see Article 180(2)(b) of EU Financial Regulation 2018/1046: '**operating grant**' means an EU grant to finance "the functioning of a body which has an objective forming part of and supporting an EU policy".

CHAPTER 4 GRANT IMPLEMENTATION

SECTION 1 CONSORTIUM: BENEFICIARIES, AFFILIATED ENTITIES AND OTHER PARTICIPANTS

ARTICLE 7 — BENEFICIARIES

The beneficiaries, as signatories of the Agreement, are fully responsible towards the granting authority for implementing it and for complying with all its obligations.

They must implement the Agreement to their best abilities, in good faith and in accordance with all the obligations and terms and conditions it sets out.

They must have the appropriate resources to implement the action and implement the action under their own responsibility and in accordance with Article 11. If they rely on affiliated entities or other participants (see Articles 8 and 9), they retain sole responsibility towards the granting authority and the other beneficiaries.

They are jointly responsible for the *technical* implementation of the action. If one of the beneficiaries fails to implement their part of the action, the other beneficiaries must ensure that this part is implemented by someone else (without being entitled to an increase of the maximum grant amount and subject to an amendment; see Article 39). The *financial* responsibility of each beneficiary in case of recoveries is governed by Article 22.

The beneficiaries (and their action) must remain eligible under the EU programme funding the grant for the entire duration of the action. Costs and contributions will be eligible only as long as the beneficiary and the action are eligible.

The **internal roles and responsibilities** of the beneficiaries are divided as follows:

(a) Each beneficiary must:

(i) keep information stored in the Portal Participant Register up to date (see Article 19)

(ii) inform the granting authority (and the other beneficiaries) immediately of any events or circumstances likely to affect significantly or delay the implementation of the action (see Article 19)

(iii) submit to the coordinator in good time:

- the prefinancing guarantees (if required; see Article 23)
- the financial statements and certificates on the financial statements (CFS) (if required; see Articles 21 and 24.2 and Data Sheet, Point 4.3)
- the contribution to the deliverables and technical reports (see Article 21)
- any other documents or information required by the granting authority under the Agreement

(iv) submit via the Portal data and information related to the participation of their affiliated entities.

(b) The coordinator must:

(i) monitor that the action is implemented properly (see Article 11)

(ii) act as the intermediary for all communications between the consortium and the granting authority, unless the Agreement or granting authority specifies otherwise, and in particular:

- submit the prefinancing guarantees to the granting authority (if any)
- request and review any documents or information required and verify their quality and completeness before passing them on to the granting authority
- submit the deliverables and reports to the granting authority
- inform the granting authority about the payments made to the other beneficiaries (report on the distribution of payments; if required, see Articles 22 and 32)

(iii) distribute the payments received from the granting authority to the other beneficiaries without unjustified delay (see Article 22).

The coordinator may not delegate or subcontract the above-mentioned tasks to any other beneficiary or third party (including affiliated entities).

However, coordinators which are public bodies may delegate the tasks set out in Point (b)(ii) last indent and (iii) above to entities with 'authorisation to administer' which they have created or which are controlled by or affiliated to them. In this case, the coordinator retains sole responsibility for the payments and for compliance with the obligations under the Agreement.

Moreover, coordinators which are 'sole beneficiaries'[12] (or similar, such as European research infrastructure consortia (ERICs)) may delegate the tasks set out in Point (b)(i) to (iii) above to one of their members. The coordinator retains sole responsibility for compliance with the obligations under the Agreement.

The beneficiaries must have **internal arrangements** regarding their operation and co-ordination, to ensure that the action is implemented properly.

If required by the granting authority (see Data Sheet, Point 1), these arrangements must be set out in a written **consortium agreement** between the beneficiaries, covering for instance:

- the internal organisation of the consortium
- the management of access to the Portal
- different distribution keys for the payments and financial responsibilities in case of recoveries (if any)
- additional rules on rights and obligations related to background and results (see Article 16)

[12] For the definition, see Article 187(2) EU Financial Regulation 2018/1046: "Where several entities satisfy the criteria for being awarded a grant and together form one entity, that entity may be treated as the **sole beneficiary**, including where it is specifically established for the purpose of implementing the action financed by the grant."

- settlement of internal disputes
- liability, indemnification and confidentiality arrangements between the beneficiaries.

The internal arrangements must not contain any provision contrary to this Agreement.

ARTICLE 8 — AFFILIATED ENTITIES

The following entities which are linked to a beneficiary will participate in the action as 'affiliated entities':

- **POLITECNICO DI TORINO (POLITO)**, PIC 999977754, linked to CONSORZIO INTERUNIVERSITARIO PER L'OTTIMIZZAZIONE E LA RICERCA OPERATIVA (ICOOR)
- **UNIVERSITA DEGLI STUDI DI MODENA E REGGIO EMILIA (UNIMORE)**, PIC 999840887, linked to CONSORZIO INTERUNIVERSITARIO PER L'OTTIMIZZAZIONE E LA RICERCA OPERATIVA (ICOOR)

Affiliated entities can charge costs and contributions to the action under the same conditions as the beneficiaries and must implement the action tasks attributed to them in Annex 1 in accordance with Article 11.

Their costs and contributions will be included in Annex 2 and will be taken into account for the calculation of the grant.

The beneficiaries must ensure that all their obligations under this Agreement also apply to their affiliated entities.

The beneficiaries must ensure that the bodies mentioned in Article 25 (e.g. granting authority, OLAF, Court of Auditors (ECA), etc.) can exercise their rights also towards the affiliated entities.

Breaches by affiliated entities will be handled in the same manner as breaches by beneficiaries. Recovery of undue amounts will be handled through the beneficiaries.

If the granting authority requires joint and several liability of affiliated entities (see Data Sheet, Point 4.4), they must sign the declaration set out in Annex 3a and may be held liable in case of enforced recoveries against their beneficiaries (see Article 22.2 and 22.4).

ARTICLE 9 — OTHER PARTICIPANTS INVOLVED IN THE ACTION

9.1 Associated partners

Not applicable

9.2 Third parties giving in-kind contributions to the action

Other third parties may give in-kind contributions to the action (i.e. personnel, equipment, other goods, works and services, etc. which are free-of-charge) if necessary for the implementation.

Third parties giving in-kind contributions do not implement any action tasks. They may not charge costs or contributions to the action, but the costs for the in-kind contributions are eligible and may be

charged by the beneficiaries which use them, under the conditions set out in Article 6. The costs will be included in Annex 2 as part of the beneficiaries' costs.

The third parties and their in-kind contributions should be set out in Annex 1.

The beneficiaries must ensure that the bodies mentioned in Article 25 (e.g. granting authority, OLAF, Court of Auditors (ECA), etc.) can exercise their rights also towards the third parties giving in-kind contributions.

9.3 Subcontractors

Subcontractors may participate in the action, if necessary for the implementation.

Subcontractors must implement their action tasks in accordance with Article 11. The costs for the subcontracted tasks (invoiced price from the subcontractor) are eligible and may be charged by the beneficiaries, under the conditions set out in Article 6. The costs will be included in Annex 2 as part of the beneficiaries' costs.

The beneficiaries must ensure that their contractual obligations under Articles 11 (proper implementation), 12 (conflict of interest), 13 (confidentiality and security), 14 (ethics), 17.2 (visibility), 18 (specific rules for carrying out action), 19 (information) and 20 (record-keeping) also apply to the subcontractors.

The beneficiaries must ensure that the bodies mentioned in Article 25 (e.g. granting authority, OLAF, Court of Auditors (ECA), etc.) can exercise their rights also towards the subcontractors.

9.4 Recipients of financial support to third parties

If the action includes providing financial support to third parties (e.g. grants, prizes or similar forms of support), the beneficiaries must ensure that their contractual obligations under Articles 12 (conflict of interest), 13 (confidentiality and security), 14 (ethics), 17.2 (visibility), 18 (specific rules for carrying out action), 19 (information) and 20 (record-keeping)also apply to the third parties receiving the support (recipients).

The beneficiaries must also ensure that the bodies mentioned in Article 25 (e.g. granting authority, OLAF, Court of Auditors (ECA), etc.) can exercise their rights also towards the recipients.

ARTICLE 10 — PARTICIPANTS WITH SPECIAL STATUS

10.1 Non-EU participants

Participants which are established in a non-EU country (if any) undertake to comply with their obligations under the Agreement and:

- to respect general principles (including fundamental rights, values and ethical principles, environmental and labour standards, rules on classified information, intellectual property rights, visibility of funding and protection of personal data)

- for the submission of certificates under Article 24: to use qualified external auditors which

are independent and comply with comparable standards as those set out in EU Directive 2006/43/EC[13]

- for the controls under Article 25: to allow for checks, reviews, audits and investigations (including on-the-spot checks, visits and inspections) by the bodies mentioned in that Article (e.g. granting authority, OLAF, Court of Auditors (ECA), etc.).

Special rules on dispute settlement apply (see Data Sheet, Point 5).

10.2 Participants which are international organisations

Participants which are international organisations (IOs; if any) undertake to comply with their obligations under the Agreement and:

- to respect general principles (including fundamental rights, values and ethical principles, environmental and labour standards, rules on classified information, intellectual property rights, visibility of funding and protection of personal data)

- for the submission of certificates under Article 24: to use either independent public officers or external auditors which comply with comparable standards as those set out in EU Directive 2006/43/EC

- for the controls under Article 25: to allow for the checks, reviews, audits and investigations by the bodies mentioned in that Article, taking into account the specific agreements concluded by them and the EU (if any).

For such participants, nothing in the Agreement will be interpreted as a waiver of their privileges or immunities, as accorded by their constituent documents or international law.

Special rules on applicable law and dispute settlement apply (see Article 43 and Data Sheet, Point 5).

10.3 Pillar-assessed participants

Pillar-assessed participants (if any) may rely on their own systems, rules and procedures, in so far as they have been positively assessed and do not call into question the decision awarding the grant or breach the principle of equal treatment of applicants or beneficiaries.

‘Pillar-assessment’ means a review by the European Commission on the systems, rules and procedures which participants use for managing EU grants (in particular internal control system, accounting system, external audits, financing of third parties, rules on recovery and exclusion, information on recipients and protection of personal data; see Article 154 EU Financial Regulation 2018/1046).

Participants with a positive pillar assessment may rely on their own systems, rules and procedures, in particular for:

- record-keeping (Article 20): may be done in accordance with internal standards, rules and procedures

[13] Directive 2006/43/EC of the European Parliament and of the Council of 17 May 2006 on statutory audits of annual accounts and consolidated accounts or similar national regulations (OJ L 157, 9.6.2006, p. 87).

- currency conversion for financial statements (Article 21): may be done in accordance with usual accounting practices

- guarantees (Article 23): for public law bodies, prefinancing guarantees are not needed

- certificates (Article 24):

    - certificates on the financial statements (CFS): may be provided by their regular internal or external auditors and in accordance with their internal financial regulations and procedures

    - certificates on usual accounting practices (CoMUC): are not needed if those practices are covered by an ex-ante assessment

and use the following specific rules, for:

- recoveries (Article 22): in case of financial support to third parties, there will be no recovery if the participant has done everything possible to retrieve the undue amounts from the third party receiving the support (including legal proceedings) and non-recovery is not due to an error or negligence on its part

- checks, reviews, audits and investigations by the EU (Article 25): will be conducted taking into account the rules and procedures specifically agreed between them and the framework agreement (if any)

- impact evaluation (Article 26): will be conducted in accordance with the participant's internal rules and procedures and the framework agreement (if any)

- grant agreement suspension (Article 31): certain costs incurred during grant suspension are eligible (notably, minimum costs necessary for a possible resumption of the action and costs relating to contracts which were entered into before the pre-information letter was received and which could not reasonably be suspended, reallocated or terminated on legal grounds)

- grant agreement termination (Article 32): the final grant amount and final payment will be calculated taking into account also costs relating to contracts due for execution only after termination takes effect, if the contract was entered into before the pre-information letter was received and could not reasonably be terminated on legal grounds

- liability for damages (Article 33.2): the granting authority must be compensated for damage it sustains as a result of the implementation of the action or because the action was not implemented in full compliance with the Agreement only if the damage is due to an infringement of the participant's internal rules and procedures or due to a violation of third parties' rights by the participant or one of its employees or individual for whom the employees are responsible.

Participants whose pillar assessment covers procurement and granting procedures may also do purchases, subcontracting and financial support to third parties (Article 6.2) in accordance with their internal rules and procedures for purchases, subcontracting and financial support.

Participants whose pillar assessment covers data protection rules may rely on their internal standards, rules and procedures for data protection (Article 15).

The participants may however not rely on provisions which would breach the principle of equal treatment of applicants or beneficiaries or call into question the decision awarding the grant, such as in particular:

- eligibility (Article 6)
- consortium roles and set-up (Articles 7-9)
- security and ethics (Articles 13, 14)
- IPR (including background and results, access rights and rights of use), communication, dissemination and visibility (Articles 16 and 17)
- information obligation (Article 19)
- payment, reporting and amendments (Articles 21, 22 and 39)
- rejections, reductions, suspensions and terminations (Articles 27, 28, 29-32)

If the pillar assessment was subject to remedial measures, reliance on the internal systems, rules and procedures is subject to compliance with those remedial measures.

Participants whose assessment has not yet been updated to cover (the new rules on) data protection may rely on their internal systems, rules and procedures, provided that they ensure that personal data is:

- processed lawfully, fairly and in a transparent manner in relation to the data subject
- collected for specified, explicit and legitimate purposes and not further processed in a manner that is incompatible with those purposes
- adequate, relevant and limited to what is necessary in relation to the purposes for which they are processed
- accurate and, where necessary, kept up to date
- kept in a form which permits identification of data subjects for no longer than is necessary for the purposes for which the data is processed and
- processed in a manner that ensures appropriate security of the personal data.

Participants must inform the coordinator without delay of any changes to the systems, rules and procedures that were part of the pillar assessment. The coordinator must immediately inform the granting authority.

Pillar-assessed participants that have also concluded a framework agreement with the EU, may moreover — under the same conditions as those above (i.e. not call into question the decision awarding the grant or breach the principle of equal treatment of applicants or beneficiaries) — rely on the provisions set out in that framework agreement.

SECTION 2 RULES FOR CARRYING OUT THE ACTION

ARTICLE 11 — PROPER IMPLEMENTATION OF THE ACTION

11.1 Obligation to properly implement the action

The beneficiaries must implement the action as described in Annex 1 and in compliance with the provisions of the Agreement, the call conditions and all legal obligations under applicable EU, international and national law.

11.2 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 12 — CONFLICT OF INTERESTS

12.1 Conflict of interests

The beneficiaries must take all measures to prevent any situation where the impartial and objective implementation of the Agreement could be compromised for reasons involving family, emotional life, political or national affinity, economic interest or any other direct or indirect interest (‘conflict of interests’).

They must formally notify the granting authority without delay of any situation constituting or likely to lead to a conflict of interests and immediately take all the necessary steps to rectify this situation.

The granting authority may verify that the measures taken are appropriate and may require additional measures to be taken by a specified deadline.

12.2 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28) and the grant or the beneficiary may be terminated (see Article 32).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 13 — CONFIDENTIALITY AND SECURITY

13.1 Sensitive information

The parties must keep confidential any data, documents or other material (in any form) that is identified as sensitive in writing (‘sensitive information’) — during the implementation of the action and for at least until the time-limit set out in the Data Sheet (see Point 6).

If a beneficiary requests, the granting authority may agree to keep such information confidential for a longer period.

Unless otherwise agreed between the parties, they may use sensitive information only to implement the Agreement.

The beneficiaries may disclose sensitive information to their personnel or other participants involved in the action only if they:

(a) need to know it in order to implement the Agreement and

(b) are bound by an obligation of confidentiality.

The granting authority may disclose sensitive information to its staff and to other EU institutions and bodies.

It may moreover disclose sensitive information to third parties, if:

(a) this is necessary to implement the Agreement or safeguard the EU financial interests and

(b) the recipients of the information are bound by an obligation of confidentiality.

The confidentiality obligations no longer apply if:

(a) the disclosing party agrees to release the other party

(b) the information becomes publicly available, without breaching any confidentiality obligation

(c) the disclosure of the sensitive information is required by EU, international or national law.

Specific confidentiality rules (if any) are set out in Annex 5.

13.2 Classified information

The parties must handle classified information in accordance with the applicable EU, international or national law on classified information (in particular, Decision 2015/444[14] and its implementing rules).

Deliverables which contain classified information must be submitted according to special procedures agreed with the granting authority.

Action tasks involving classified information may be subcontracted only after explicit approval (in writing) from the granting authority.

Classified information may not be disclosed to any third party (including participants involved in the action implementation) without prior explicit written approval from the granting authority.

Specific security rules (if any) are set out in Annex 5.

13.3 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 14 — ETHICS AND VALUES

14.1 Ethics

[14] Commission Decision 2015/444/EC, Euratom of 13 March 2015 on the security rules for protecting EU classified information (OJ L 72, 17.3.2015, p. 53).

The action must be carried out in line with the highest ethical standards and the applicable EU, international and national law on ethical principles.

Specific ethics rules (if any) are set out in Annex 5.

14.2 Values

The beneficiaries must commit to and ensure the respect of basic EU values (such as respect for human dignity, freedom, democracy, equality, the rule of law and human rights, including the rights of minorities).

Specific rules on values (if any) are set out in Annex 5.

14.3 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 15 — DATA PROTECTION

15.1 Data processing by the granting authority

Any personal data under the Agreement will be processed under the responsibility of the data controller of the granting authority in accordance with and for the purposes set out in the Portal Privacy Statement.

For grants where the granting authority is the European Commission, an EU regulatory or executive agency, joint undertaking or other EU body, the processing will be subject to Regulation 2018/1725[15].

15.2 Data processing by the beneficiaries

The beneficiaries must process personal data under the Agreement in compliance with the applicable EU, international and national law on data protection (in particular, Regulation 2016/679[16]).

They must ensure that personal data is:

- processed lawfully, fairly and in a transparent manner in relation to the data subjects
- collected for specified, explicit and legitimate purposes and not further processed in a manner that is incompatible with those purposes

[15] Regulation (EU) 2018/1725 of the European Parliament and of the Council of 23 October 2018 on the protection of natural persons with regard to the processing of personal data by the Union institutions, bodies, offices and agencies and on the free movement of such data, and repealing Regulation (EC) No 45/2001 and Decision No 1247/2002/EC (OJ L 295, 21.11.2018, p. 39).

[16] Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (‘GDPR’) (OJ L 119, 4.5.2016, p. 1).

- adequate, relevant and limited to what is necessary in relation to the purposes for which they are processed
- accurate and, where necessary, kept up to date
- kept in a form which permits identification of data subjects for no longer than is necessary for the purposes for which the data is processed and
- processed in a manner that ensures appropriate security of the data.

The beneficiaries may grant their personnel access to personal data only if it is strictly necessary for implementing, managing and monitoring the Agreement. The beneficiaries must ensure that the personnel is under a confidentiality obligation.

The beneficiaries must inform the persons whose data are transferred to the granting authority and provide them with the Portal Privacy Statement.

15.3 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 16 — INTELLECTUAL PROPERTY RIGHTS (IPR) — BACKGROUND AND RESULTS —ACCESS RIGHTS AND RIGHTS OF USE

16.1 Background and access rights to background

The beneficiaries must give each other and the other participants access to the background identified as needed for implementing the action, subject to any specific rules in Annex 5.

‘Background’ means any data, know-how or information — whatever its form or nature (tangible or intangible), including any rights such as intellectual property rights — that is:

(a) held by the beneficiaries before they acceded to the Agreement and

(b) needed to implement the action or exploit the results.

If background is subject to rights of a third party, the beneficiary concerned must ensure that it is able to comply with its obligations under the Agreement.

16.2 Ownership of results

The granting authority does not obtain ownership of the results produced under the action.

‘Results’ means any tangible or intangible effect of the action, such as data, know-how or information, whatever its form or nature, whether or not it can be protected, as well as any rights attached to it, including intellectual property rights.

16.3 Rights of use of the granting authority on materials, documents and information received for policy, information, communication, dissemination and publicity purposes

The granting authority has the right to use non-sensitive information relating to the action and materials and documents received from the beneficiaries (notably summaries for publication, deliverables, as well as any other material, such as pictures or audio-visual material, in paper or electronic form) for policy, information, communication, dissemination and publicity purposes — during the action or afterwards.

The right to use the beneficiaries' materials, documents and information is granted in the form of a royalty-free, non-exclusive and irrevocable licence, which includes the following rights:

(a) **use for its own purposes** (in particular, making them available to persons working for the granting authority or any other EU service (including institutions, bodies, offices, agencies, etc.) or EU Member State institution or body; copying or reproducing them in whole or in part, in unlimited numbers; and communication through press information services)

(b) **distribution to the public** (in particular, publication as hard copies and in electronic or digital format, publication on the internet, as a downloadable or non-downloadable file, broadcasting by any channel, public display or presentation, communicating through press information services, or inclusion in widely accessible databases or indexes)

(c) **editing or redrafting** (including shortening, summarising, inserting other elements (e.g. meta-data, legends, other graphic, visual, audio or text elements), extracting parts (e.g. audio or video files), dividing into parts, use in a compilation)

(d) **translation**

(e) **storage** in paper, electronic or other form

(f) **archiving**, in line with applicable document-management rules

(g) the right to authorise **third parties** to act on its behalf or sub-license to third parties the modes of use set out in Points (b), (c), (d) and (f), if needed for the information, communication and publicity activity of the granting authority

(h) **processing**, analysing, aggregating the materials, documents and information received and **producing derivative works**.

The rights of use are granted for the whole duration of the industrial or intellectual property rights concerned.

If materials or documents are subject to moral rights or third party rights (including intellectual property rights or rights of natural persons on their image and voice), the beneficiaries must ensure that they comply with their obligations under this Agreement (in particular, by obtaining the necessary licences and authorisations from the rights holders concerned).

Where applicable, the granting authority will insert the following information:

> "© – [year] – [name of the copyright owner]. All rights reserved. Licensed to the [name of granting authority] under conditions."

16.4 Specific rules on IPR, results and background

Specific rules regarding intellectual property rights, results and background (if any) are set out in Annex 5.

16.5 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such a breach may also lead to other measures described in Chapter 5.

ARTICLE 17 — COMMUNICATION, DISSEMINATION AND VISIBILITY

17.1 Communication — Dissemination — Promoting the action

Unless otherwise agreed with the granting authority, the beneficiaries must promote the action and its results by providing targeted information to multiple audiences (including the media and the public), in accordance with Annex 1 and in a strategic, coherent and effective manner.

Before engaging in a communication or dissemination activity expected to have a major media impact, the beneficiaries must inform the granting authority.

17.2 Visibility — European flag and funding statement

Unless otherwise agreed with the granting authority, communication activities of the beneficiaries related to the action (including media relations, conferences, seminars, information material, such as brochures, leaflets, posters, presentations, etc., in electronic form, via traditional or social media, etc.), dissemination activities and any infrastructure, equipment, vehicles, supplies or major result funded by the grant must acknowledge EU support and display the European flag (emblem) and funding statement (translated into local languages, where appropriate):

Funded by the European Union

Co-funded by the European Union

Funded by the European Union

Co-funded by the European Union

The emblem must remain distinct and separate and cannot be modified by adding other visual marks, brands or text.

Apart from the emblem, no other visual identity or logo may be used to highlight the EU support.

When displayed in association with other logos (e.g. of beneficiaries or sponsors), the emblem must be displayed at least as prominently and visibly as the other logos.

For the purposes of their obligations under this Article, the beneficiaries may use the emblem without first obtaining approval from the granting authority. This does not, however, give them the right to exclusive use. Moreover, they may not appropriate the emblem or any similar trademark or logo, either by registration or by any other means.

17.3 Quality of information — Disclaimer

Any communication or dissemination activity related to the action must use factually accurate information.

Moreover, it must indicate the following disclaimer (translated into local languages where appropriate):

> "Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or [name of the granting authority]. Neither the European Union nor the granting authority can be held responsible for them."

17.4 Specific communication, dissemination and visibility rules

Specific communication, dissemination and visibility rules (if any) are set out in Annex 5.

17.5 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 18 — SPECIFIC RULES FOR CARRYING OUT THE ACTION

18.1 Specific rules for carrying out the action

Specific rules for implementing the action (if any) are set out in Annex 5.

18.2 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such a breach may also lead to other measures described in Chapter 5.

SECTION 3 GRANT ADMINISTRATION

ARTICLE 19 — GENERAL INFORMATION OBLIGATIONS

19.1 Information requests

The beneficiaries must provide — during the action or afterwards and in accordance with Article 7 — any information requested in order to verify eligibility of the costs or contributions declared, proper implementation of the action and compliance with the other obligations under the Agreement.

The information provided must be accurate, precise and complete and in the format requested, including electronic format.

19.2 Participant Register data updates

The beneficiaries must keep — at all times, during the action or afterwards — their information stored in the Portal Participant Register up to date, in particular, their name, address, legal representatives, legal form and organisation type.

19.3 Information about events and circumstances which impact the action

The beneficiaries must immediately inform the granting authority (and the other beneficiaries) of any of the following:

(a) **events** which are likely to affect or delay the implementation of the action or affect the EU's financial interests, in particular:

(i) changes in their legal, financial, technical, organisational or ownership situation (including changes linked to one of the exclusion grounds listed in the declaration of honour signed before grant signature)

(ii) linked action information: not applicable

(b) **circumstances** affecting:

(i) the decision to award the grant or

(ii) compliance with requirements under the Agreement.

19.4 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 20 — RECORD-KEEPING

20.1 Keeping records and supporting documents

The beneficiaries must — at least until the time-limit set out in the Data Sheet (see Point 6) — keep records and other supporting documents to prove the proper implementation of the action in line with the accepted standards in the respective field (if any).

In addition, the beneficiaries must — for the same period — keep the following to justify the amounts declared:

(a) for actual costs: adequate records and supporting documents to prove the costs declared (such

as contracts, subcontracts, invoices and accounting records); in addition, the beneficiaries' usual accounting and internal control procedures must enable direct reconciliation between the amounts declared, the amounts recorded in their accounts and the amounts stated in the supporting documents

(b) for flat-rate costs and contributions (if any): adequate records and supporting documents to prove the eligibility of the costs or contributions to which the flat-rate is applied

(c) for the following simplified costs and contributions: the beneficiaries do not need to keep specific records on the actual costs incurred, but must keep:

(i) for unit costs and contributions (if any): adequate records and supporting documents to prove the number of units declared

(ii) for lump sum costs and contributions (if any): adequate records and supporting documents to prove proper implementation of the work as described in Annex 1

(iii) for financing not linked to costs (if any): adequate records and supporting documents to prove the achievement of the results or the fulfilment of the conditions as described in Annex 1

(d) for unit, flat-rate and lump sum costs and contributions according to usual cost accounting practices (if any): the beneficiaries must keep any adequate records and supporting documents to prove that their cost accounting practices have been applied in a consistent manner, based on objective criteria, regardless of the source of funding, and that they comply with the eligibility conditions set out in Articles 6.1 and 6.2.

Moreover, the following is needed for specific budget categories:

(e) for personnel costs: time worked for the beneficiary under the action must be supported by declarations signed monthly by the person and their supervisor, unless another reliable time-record system is in place; the granting authority may accept alternative evidence supporting the time worked for the action declared, if it considers that it offers an adequate level of assurance

(f) additional record-keeping rules: not applicable

The records and supporting documents must be made available upon request (see Article 19) or in the context of checks, reviews, audits or investigations (see Article 25).

If there are on-going checks, reviews, audits, investigations, litigation or other pursuits of claims under the Agreement (including the extension of findings; see Article 25), the beneficiaries must keep these records and other supporting documentation until the end of these procedures.

The beneficiaries must keep the original documents. Digital and digitalised documents are considered originals if they are authorised by the applicable national law. The granting authority may accept non-original documents if they offer a comparable level of assurance.

20.2 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, costs or contributions insufficiently

substantiated will be ineligible (see Article 6) and will be rejected (see Article 27), and the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 21 — REPORTING

21.1 Continuous reporting

The beneficiaries must continuously report on the progress of the action (e.g. **deliverables, milestones, outputs/outcomes, critical risks, indicators,** etc; if any), in the Portal Continuous Reporting tool and in accordance with the timing and conditions it sets out (as agreed with the granting authority).

Standardised deliverables (e.g. progress reports not linked to payments, reports on cumulative expenditure, special reports, etc; if any) must be submitted using the templates published on the Portal.

21.2 Periodic reporting: Technical reports and financial statements

In addition, the beneficiaries must provide reports to request payments, in accordance with the schedule and modalities set out in the Data Sheet (see Point 4.2):

- for additional prefinancings (if any): an **additional prefinancing report**
- for interim payments (if any) and the final payment: a **periodic report**.

The prefinancing and periodic reports include a technical and financial part.

The technical part includes an overview of the action implementation. It must be prepared using the template available in the Portal Periodic Reporting tool.

The financial part of the additional prefinancing report includes a statement on the use of the previous prefinancing payment.

The financial part of the periodic report includes:

- the financial statements (individual and consolidated; for all beneficiaries/affiliated entities)
- the explanation on the use of resources (or detailed cost reporting table, if required)
- the certificates on the financial statements (CFS) (if required; see Article 24.2 and Data Sheet, Point 4.3).

The **financial statements** must detail the eligible costs and contributions for each budget category and, for the final payment, also the revenues for the action (see Articles 6 and 22).

All eligible costs and contributions incurred should be declared, even if they exceed the amounts indicated in the estimated budget (see Annex 2). Amounts that are not declared in the individual financial statements will not be taken into account by the granting authority.

By signing the financial statements (directly in the Portal Periodic Reporting tool), the beneficiaries confirm that:

- the information provided is complete, reliable and true
- the costs and contributions declared are eligible (see Article 6)
- the costs and contributions can be substantiated by adequate records and supporting documents (see Article 20) that will be produced upon request (see Article 19) or in the context of checks, reviews, audits and investigations (see Article 25)
- for the final periodic report: all the revenues have been declared (if required; see Article 22).

Beneficiaries will have to submit also the financial statements of their affiliated entities (if any). In case of recoveries (see Article 22), beneficiaries will be held responsible also for the financial statements of their affiliated entities.

21.3 Currency for financial statements and conversion into euros

The financial statements must be drafted in euro.

Beneficiaries with general accounts established in a currency other than the euro must convert the costs recorded in their accounts into euro, at the average of the daily exchange rates published in the C series of the *Official Journal of the European Union* (ECB website), calculated over the corresponding reporting period.

If no daily euro exchange rate is published in the *Official Journal* for the currency in question, they must be converted at the average of the monthly accounting exchange rates published on the European Commission website (InforEuro), calculated over the corresponding reporting period.

Beneficiaries with general accounts in euro must convert costs incurred in another currency into euro according to their usual accounting practices.

21.4 Reporting language

The reporting must be in the language of the Agreement, unless otherwise agreed with the granting authority (see Data Sheet, Point 4.2).

21.5 Consequences of non-compliance

If a report submitted does not comply with this Article, the granting authority may suspend the payment deadline (see Article 29) and apply other measures described in Chapter 5.

If the coordinator breaches its reporting obligations, the granting authority may terminate the grant or the coordinator's participation (see Article 32) or apply other measures described in Chapter 5.

ARTICLE 22 — PAYMENTS AND RECOVERIES — CALCULATION OF AMOUNTS DUE

22.1 Payments and payment arrangements

Payments will be made in accordance with the schedule and modalities set out in the Data Sheet (see Point 4.2).

They will be made in euro to the bank account indicated by the coordinator (see Data Sheet, Point 4.2)

and must be distributed without unjustified delay (restrictions may apply to distribution of the initial prefinancing payment; see Data Sheet, Point 4.2).

Payments to this bank account will discharge the granting authority from its payment obligation.

The cost of payment transfers will be borne as follows:

- the granting authority bears the cost of transfers charged by its bank
- the beneficiary bears the cost of transfers charged by its bank
- the party causing a repetition of a transfer bears all costs of the repeated transfer.

Payments by the granting authority will be considered to have been carried out on the date when they are debited to its account.

22.2 Recoveries

Recoveries will be made, if — at beneficiary termination, final payment or afterwards — it turns out that the granting authority has paid too much and needs to recover the amounts undue.

Each beneficiary's financial responsibility in case of recovery is in principle limited to their own debt and undue amounts of their affiliated entities.

In case of enforced recoveries (see Article 22.4), affiliated entities will be held liable for repaying debts of their beneficiaries, if required by the granting authority (see Data Sheet, Point 4.4).

22.3 Amounts due

22.3.1 Prefinancing payments

The aim of the prefinancing is to provide the beneficiaries with a float.

It remains the property of the EU until the final payment.

For **initial prefinancings** (if any), the amount due, schedule and modalities are set out in the Data Sheet (see Point 4.2).

For **additional prefinancings** (if any), the amount due, schedule and modalities are also set out in the Data Sheet (see Point 4.2). However, if the statement on the use of the previous prefinancing payment shows that less than 70% was used, the amount set out in the Data Sheet will be reduced by the difference between the 70% threshold and the amount used.

The contribution to the Mutual Insurance Mechanism will be retained from the prefinancing payments (at the rate and in accordance with the modalities set out in the Data Sheet, see Point 4.2) and transferred to the Mechanism.

Prefinancing payments (or parts of them) may be offset (without the beneficiaries' consent) against amounts owed by a beneficiary to the granting authority — up to the amount due to that beneficiary.

For grants where the granting authority is the European Commission or an EU executive agency, offsetting may also be done against amounts owed to other Commission services or executive agencies.

Payments will not be made if the payment deadline or payments are suspended (see Articles 29 and 30).

22.3.2 Amount due at beneficiary termination — Recovery

In case of beneficiary termination, the granting authority will determine the provisional amount due for the beneficiary concerned. Payments (if any) will be made with the next interim or final payment.

The **amount due** will be calculated in the following step:

Step 1 — Calculation of the total accepted EU contribution

Step 1 — Calculation of the total accepted EU contribution

The granting authority will first calculate the 'accepted EU contribution' for the beneficiary for all reporting periods, by calculating the 'maximum EU contribution to costs' (applying the funding rate to the accepted costs of the beneficiary), taking into account requests for a lower contribution to costs and CFS threshold cappings (if any; see Article 24.5) and adding the contributions (accepted unit, flat-rate or lump sum contributions and financing not linked to costs, if any).

After that, the granting authority will take into account grant reductions (if any). The resulting amount is the 'total accepted EU contribution' for the beneficiary.

The **balance** is then calculated by deducting the payments received (if any; see report on the distribution of payments in Article 32), from the total accepted EU contribution:

{total accepted EU contribution for the beneficiary

minus

{prefinancing and interim payments received (if any)}}.

If the balance is **positive**, the amount will be included in the next interim or final payment to the consortium.

If the balance is **negative**, it will be **recovered** in accordance with the following procedure:

The granting authority will send a **pre-information letter** to the beneficiary concerned:

- formally notifying the intention to recover, the amount due, the amount to be recovered and the reasons why and
- requesting observations within 30 days of receiving notification.

If no observations are submitted (or the granting authority decides to pursue recovery despite the observations it has received), it will confirm the amount to be recovered and ask this amount to be paid to the coordinator (**confirmation letter**).

If payment is not made to the coordinator by the date specified in the confirmation letter, the granting authority may call on the Mutual Insurance Mechanism to intervene, if continuation of the action is guaranteed and the conditions set out in the rules governing the Mechanism are met.

In this case, it will send a **beneficiary recovery letter**, together with a **debit note** with the terms and date for payment.

The debit note for the beneficiary will include the amount calculated for the affiliated entities which also had to end their participation (if any).

If payment is not made by the date specified in the debit note, the granting authority will **enforce recovery** in accordance with Article 22.4.

The amounts will later on also be taken into account for the next interim or final payment.

22.3.3 Interim payments

Interim payments reimburse the eligible costs and contributions claimed for the implementation of the action during the reporting periods (if any).

Interim payments (if any) will be made in accordance with the schedule and modalities set out the Data Sheet (see Point 4.2).

Payment is subject to the approval of the periodic report. Its approval does not imply recognition of compliance, authenticity, completeness or correctness of its content.

The **interim payment** will be calculated by the granting authority in the following steps:

Step 1 — Calculation of the total accepted EU contribution

Step 2 — Limit to the interim payment ceiling

<u>Step 1 — Calculation of the total accepted EU contribution</u>

The granting authority will calculate the 'accepted EU contribution' for the action for the reporting period, by first calculating the 'maximum EU contribution to costs' (applying the funding rate to the accepted costs of each beneficiary), taking into account requests for a lower contribution to costs, and CFS threshold cappings (if any; see Article 24.5) and adding the contributions (accepted unit, flat-rate or lump sum contributions and financing not linked to costs, if any).

After that, the granting authority will take into account grant reductions from beneficiary termination (if any). The resulting amount is the 'total accepted EU contribution'.

<u>Step 2 — Limit to the interim payment ceiling</u>

The resulting amount is then capped to ensure that the total amount of prefinancing and interim payments (if any) does not exceed the interim payment ceiling set out in the Data Sheet (see Point 4.2).

Interim payments (or parts of them) may be offset (without the beneficiaries' consent) against amounts owed by a beneficiary to the granting authority — up to the amount due to that beneficiary.

For grants where the granting authority is the European Commission or an EU executive agency, offsetting may also be done against amounts owed to other Commission services or executive agencies.

Payments will not be made if the payment deadline or payments are suspended (see Articles 29 and 30).

22.3.4 Final payment — Final grant amount — Revenues and Profit — Recovery

The final payment (payment of the balance) reimburses the remaining part of the eligible costs and contributions claimed for the implementation of the action (if any).

The final payment will be made in accordance with the schedule and modalities set out in the Data Sheet (see Point 4.2).

Payment is subject to the approval of the final periodic report. Its approval does not imply recognition of compliance, authenticity, completeness or correctness of its content.

The **final grant amount for the action** will be calculated in the following steps:

Step 1 — Calculation of the total accepted EU contribution

Step 2 — Limit to the maximum grant amount

Step 3 — Reduction due to the no-profit rule

Step 1 — Calculation of the total accepted EU contribution

The granting authority will first calculate the 'accepted EU contribution' for the action for all reporting periods, by calculating the 'maximum EU contribution to costs' (applying the funding rate to the total accepted costs of each beneficiary), taking into account requests for a lower contribution to costs, CFS threshold cappings (if any; see Article 24.5) and adding the contributions (accepted unit, flat-rate or lump sum contributions and financing not linked to costs, if any).

After that, the granting authority will take into account grant reductions (if any). The resulting amount is the 'total accepted EU contribution'.

Step 2 — Limit to the maximum grant amount

If the resulting amount is higher than the maximum grant amount set out in Article 5.2, it will be limited to the latter.

Step 3 — Reduction due to the no-profit rule

If the no-profit rule is provided for in the Data Sheet (see Point 4.2), the grant must not produce a profit (i.e. surplus of the amount obtained following Step 2 plus the action's revenues, over the eligible costs and contributions approved by the granting authority).

'Revenue' is all income generated by the action, during its duration (see Article 4), for beneficiaries that are profit legal entities (— with the exception of income generated by the exploitation of results, which are not considered as revenues).

If there is a profit, it will be deducted in proportion to the final rate of reimbursement of the eligible costs approved by the granting authority (as compared to the amount calculated following Steps 1 and 2 minus the contributions).

The **balance** (final payment) is then calculated by deducting the total amount of prefinancing and interim payments already made (if any), from the final grant amount:

{final grant amount

minus

{prefinancing and interim payments made (if any)}}.

If the balance is **positive**, it will be **paid** to the coordinator.

The amount retained for the Mutual Insurance Mechanism (see above) will be released and **paid** to the coordinator (in accordance with the rules governing the Mechanism).

The final payment (or part of it) may be offset (without the beneficiaries' consent) against amounts owed by a beneficiary to the granting authority — up to the amount due to that beneficiary.

For grants where the granting authority is the European Commission or an EU executive agency, offsetting may also be done against amounts owed to other Commission services or executive agencies.

Payments will not be made if the payment deadline or payments are suspended (see Articles 29 and 30).

If — despite the release of the Mutual Insurance Mechanism contribution — the balance is **negative**, it will be **recovered** in accordance with the following procedure:

The granting authority will send a **pre-information letter** to the coordinator:

- formally notifying the intention to recover, the final grant amount, the amount to be recovered and the reasons why
- requesting a report on the distribution of payments to the beneficiaries within 30 days of receiving notification and
- requesting observations within 30 days of receiving notification.

If no observations are submitted (or the granting authority decides to pursue recovery despite the observations it has received) and the coordinator has submitted the report on the distribution of payments, it will calculate the **share of the debt per beneficiary**, by:

(a) identifying the beneficiaries for which the amount calculated as follows is negative:

{{{total accepted EU contribution for the beneficiary

divided by

total accepted EU contribution for the action}

multiplied by

final grant amount for the action},

minus

{prefinancing and interim payments received by the beneficiary (if any)}}

and

(b) dividing the debt:

{{amount calculated according to point (a) for the beneficiary concerned

divided by

the sum of the amounts calculated according to point (a) for all the beneficiaries identified according to point (a)}

multiplied by

the amount to be recovered}.

and confirm the amount to be recovered from each beneficiary concerned (**confirmation letter**), together with **debit notes** with the terms and date for payment.

The debit notes for beneficiaries will include the amounts calculated for their affiliated entities (if any).

If the coordinator has not submitted the report on the distribution of payments, the granting authority will **recover** the full amount from the coordinator (**confirmation letter** and **debit note** with the terms and date for payment).

If payment is not made by the date specified in the debit note, the granting authority will **enforce recovery** in accordance with Article 22.4.

22.3.5 Audit implementation after final payment — Revised final grant amount — Recovery

If — after the final payment (in particular, after checks, reviews, audits or investigations; see Article 25) — the granting authority rejects costs or contributions (see Article 27) or reduces the grant (see Article 28), it will calculate the **revised final grant amount** for the beneficiary concerned.

The **beneficiary revised final grant amount** will be calculated in the following step:

Step 1 — Calculation of the revised total accepted EU contribution

<u>Step 1 — Calculation of the revised total accepted EU contribution</u>

The granting authority will first calculate the ‘revised accepted EU contribution’ for the beneficiary, by calculating the ‘revised accepted costs’ and ‘revised accepted contributions’.

After that, it will take into account grant reductions (if any). The resulting ‘revised total accepted EU contribution’ is the beneficiary revised final grant amount.

If the revised final grant amount is lower than the beneficiary’s final grant amount (i.e. its share in the final grant amount for the action), it will be **recovered** in accordance with the following procedure:

The **beneficiary final grant amount** (i.e. share in the final grant amount for the action) is calculated as follows:

{{total accepted EU contribution for the beneficiary

divided by

total accepted EU contribution for the action}

multiplied by

final grant amount for the action}.

The granting authority will send a **pre-information letter** to the beneficiary concerned:

- formally notifying the intention to recover, the amount to be recovered and the reasons why and
- requesting observations within 30 days of receiving notification.

If no observations are submitted (or the granting authority decides to pursue recovery despite the observations it has received), it will confirm the amount to be recovered (**confirmation letter**), together with a **debit note** with the terms and the date for payment.

Recoveries against affiliated entities (if any) will be handled through their beneficiaries.

If payment is not made by the date specified in the debit note, the granting authority will **enforce recovery** in accordance with Article 22.4.

22.4 Enforced recovery

If payment is not made by the date specified in the debit note, the amount due will be recovered:

(a) by offsetting the amount — without the coordinator or beneficiary's consent — against any amounts owed to the coordinator or beneficiary by the granting authority.

In exceptional circumstances, to safeguard the EU financial interests, the amount may be offset before the payment date specified in the debit note.

For grants where the granting authority is the European Commission or an EU executive agency, debts may also be offset against amounts owed by other Commission services or executive agencies.

(b) financial guarantee(s): not applicable

(c) joint and several liability of beneficiaries: not applicable

(d) by holding affiliated entities jointly and severally liable (if any, see Data Sheet, Point 4.4)

(e) by taking legal action (see Article 43) or, provided that the granting authority is the European Commission or an EU executive agency, by adopting an enforceable decision under Article 299 of the Treaty on the Functioning of the EU (TFEU) and Article 100(2) of EU Financial Regulation 2018/1046.

If the Mutual Insurance Mechanism was called on by the granting authority to intervene, recovery will be continued in the name of the Mutual Insurance Mechanism. If two debit notes were sent, the second one (in the name of the Mutual Insurance Mechanism) will be considered to replace the first one (in the name of the granting authority). Where the MIM intervened, offsetting, enforceable decisions or any other of the above-mentioned forms of enforced recovery may be used mutatis mutandis.

The amount to be recovered will be increased by **late-payment interest** at the rate set out in Article 22.5, from the day following the payment date in the debit note, up to and including the date the full payment is received.

Partial payments will be first credited against expenses, charges and late-payment interest and then against the principal.

Bank charges incurred in the recovery process will be borne by the beneficiary, unless Directive 2015/2366[17] applies.

For grants where the granting authority is an EU executive agency, enforced recovery by offsetting or enforceable decision will be done by the services of the European Commission (see also Article 43).

22.5 Consequences of non-compliance

22.5.1 If the granting authority does not pay within the payment deadlines (see above), the beneficiaries are entitled to **late-payment interest** at the rate applied by the European Central Bank (ECB) for its main refinancing operations in euros ('reference rate'), plus the rate specified in the Data Sheet (Point 4.2). The reference rate is the rate in force on the first day of the month in which the payment deadline expires, as published in the C series of the *Official Journal of the European Union*.

If the late-payment interest is lower than or equal to EUR 200, it will be paid to the coordinator only on request submitted within two months of receiving the late payment.

Late-payment interest is not due if all beneficiaries are EU Member States (including regional and local government authorities or other public bodies acting on behalf of a Member State for the purpose of this Agreement).

If payments or the payment deadline are suspended (see Articles 29 and 30), payment will not be considered as late.

Late-payment interest covers the period running from the day following the due date for payment (see above), up to and including the date of payment.

Late-payment interest is not considered for the purposes of calculating the final grant amount.

22.5.2 If the coordinator breaches any of its obligations under this Article, the grant may be reduced (see Article 28) and the grant or the coordinator may be terminated (see Article 32).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 23 — GUARANTEES

Not applicable

ARTICLE 24 — CERTIFICATES

24.1 Operational verification report (OVR)

Not applicable

24.2 Certificate on the financial statements (CFS)

If required by the granting authority (see Data Sheet, Point 4.3), the beneficiaries must provide

[17] Directive (EU) 2015/2366 of the European Parliament and of the Council of 25 November 2015 on payment services in the internal market, amending Directives 2002/65/EC, 2009/110/EC and 2013/36/EU and Regulation (EU) No 1093/2010, and repealing Directive 2007/64/EC (OJ L 337, 23.12.2015, p. 35).

certificates on their financial statements (CFS), in accordance with the schedule, threshold and conditions set out in the Data Sheet.

The coordinator must submit them as part of the periodic report (see Article 21).

The certificates must be drawn up using the template published on the Portal, cover the costs declared on the basis of actual costs and costs according to usual cost accounting practices (if any), and fulfil the following conditions:

(a) be provided by a qualified approved external auditor which is independent and complies with Directive 2006/43/EC[18] (or for public bodies: by a competent independent public officer)

(b) the verification must be carried out according to the highest professional standards to ensure that the financial statements comply with the provisions under the Agreement and that the costs declared are eligible.

The certificates will not affect the granting authority's right to carry out its own checks, reviews or audits, nor preclude the European Court of Auditors (ECA), the European Public Prosecutor's Office (EPPO) or the European Anti-Fraud Office (OLAF) from using their prerogatives for audits and investigations under the Agreement (see Article 25).

If the costs (or a part of them) were already audited by the granting authority, these costs do not need to be covered by the certificate and will not be counted for calculating the threshold (if any).

24.3 Certificate on the compliance of usual cost accounting practices (CoMUC)

Not applicable

24.4 Systems and process audit (SPA)

Beneficiaries which:

- use unit, flat rate or lump sum costs or contributions according to documented (i.e. formally approved and in writing) usual costs accounting practices (if any) or

- have formalised documentation on the systems and processes for calculating their costs and contributions (i.e. formally approved and in writing), have participated in at least 150 actions under Horizon 2020 or the Euratom Research and Training Programme (2014-2018 or 2019-2020) and participate in at least 3 ongoing actions under Horizon Europe or the Euratom Research and Training Programme (2021-2025 or 2026-2027)

may apply to the granting authority for a systems and process audit (SPA).

This audit will be carried out as follows:

Step 1 – Application by the beneficiary.

Step 2 – If the application is accepted, the granting authority will carry out the systems and process

[18] Directive 2006/43/EC of the European Parliament and of the Council of 17 May 2006 on statutory audits of annual accounts and consolidated accounts or similar national regulations (OJ L 157, 9.6.2006, p. 87).

audit, complemented by an audit of transactions (on a sample of the beneficiary's Horizon Europe or the Euratom Research and Training Programme financial statements).

Step 3 – The audit result will take the form of a risk assessment classification for the beneficiary: low, medium or high.

Low-risk beneficiaries will benefit from less (or less in-depth) ex-post audits (see Article 25) and a higher threshold for submitting certificates on the financial statements (CFS; see Articles 21 and 24.2 and Data Sheet, Point 4.3).

24.5 Consequences of non-compliance

If a beneficiary does not submit a certificate on the financial statements (CFS) or the certificate is rejected, the accepted EU contribution to costs will be capped to reflect the CFS threshold.

If a beneficiary breaches any of its other obligations under this Article, the granting authority may apply the measures described in Chapter 5.

ARTICLE 25 — CHECKS, REVIEWS, AUDITS AND INVESTIGATIONS — EXTENSION OF FINDINGS

25.1 Granting authority checks, reviews and audits

25.1.1 Internal checks

The granting authority may — during the action or afterwards — check the proper implementation of the action and compliance with the obligations under the Agreement, including assessing costs and contributions, deliverables and reports.

25.1.2 Project reviews

The granting authority may carry out reviews on the proper implementation of the action and compliance with the obligations under the Agreement (general project reviews or specific issues reviews).

Such project reviews may be started during the implementation of the action and until the time-limit set out in the Data Sheet (see Point 6). They will be formally notified to the coordinator or beneficiary concerned and will be considered to start on the date of the notification.

If needed, the granting authority may be assisted by independent, outside experts. If it uses outside experts, the coordinator or beneficiary concerned will be informed and have the right to object on grounds of commercial confidentiality or conflict of interest.

The coordinator or beneficiary concerned must cooperate diligently and provide — within the deadline requested — any information and data in addition to deliverables and reports already submitted (including information on the use of resources). The granting authority may request beneficiaries to provide such information to it directly. Sensitive information and documents will be treated in accordance with Article 13.

The coordinator or beneficiary concerned may be requested to participate in meetings, including with the outside experts.

For **on-the-spot visits**, the beneficiary concerned must allow access to sites and premises (including to the outside experts) and must ensure that information requested is readily available.

Information provided must be accurate, precise and complete and in the format requested, including electronic format.

On the basis of the review findings, a **project review report** will be drawn up.

The granting authority will formally notify the project review report to the coordinator or beneficiary concerned, which has 30 days from receiving notification to make observations.

Project reviews (including project review reports) will be in the language of the Agreement.

25.1.3 Audits

The granting authority may carry out audits on the proper implementation of the action and compliance with the obligations under the Agreement.

Such audits may be started during the implementation of the action and until the time-limit set out in the Data Sheet (see Point 6). They will be formally notified to the beneficiary concerned and will be considered to start on the date of the notification.

The granting authority may use its own audit service, delegate audits to a centralised service or use external audit firms. If it uses an external firm, the beneficiary concerned will be informed and have the right to object on grounds of commercial confidentiality or conflict of interest.

The beneficiary concerned must cooperate diligently and provide — within the deadline requested — any information (including complete accounts, individual salary statements or other personal data) to verify compliance with the Agreement. Sensitive information and documents will be treated in accordance with Article 13.

For **on-the-spot** visits, the beneficiary concerned must allow access to sites and premises (including for the external audit firm) and must ensure that information requested is readily available.

Information provided must be accurate, precise and complete and in the format requested, including electronic format.

On the basis of the audit findings, a **draft audit report** will be drawn up.

The auditors will formally notify the draft audit report to the beneficiary concerned, which has 30 days from receiving notification to make observations (contradictory audit procedure).

The **final audit report** will take into account observations by the beneficiary concerned and will be formally notified to them.

Audits (including audit reports) will be in the language of the Agreement.

25.2 European Commission checks, reviews and audits in grants of other granting authorities

Where the granting authority is not the European Commission, the latter has the same rights of checks, reviews and audits as the granting authority.

25.3 Access to records for assessing simplified forms of funding

The beneficiaries must give the European Commission access to their statutory records for the periodic assessment of simplified forms of funding which are used in EU programmes.

25.4 OLAF, EPPO and ECA audits and investigations

The following bodies may also carry out checks, reviews, audits and investigations — during the action or afterwards:

- the European Anti-Fraud Office (OLAF) under Regulations No 883/2013[19] and No 2185/96[20]
- the European Public Prosecutor's Office (EPPO) under Regulation 2017/1939
- the European Court of Auditors (ECA) under Article 287 of the Treaty on the Functioning of the EU (TFEU) and Article 257 of EU Financial Regulation 2018/1046.

If requested by these bodies, the beneficiary concerned must provide full, accurate and complete information in the format requested (including complete accounts, individual salary statements or other personal data, including in electronic format) and allow access to sites and premises for on-the-spot visits or inspections — as provided for under these Regulations.

To this end, the beneficiary concerned must keep all relevant information relating to the action, at least until the time-limit set out in the Data Sheet (Point 6) and, in any case, until any ongoing checks, reviews, audits, investigations, litigation or other pursuits of claims have been concluded.

25.5 Consequences of checks, reviews, audits and investigations — Extension of results of reviews, audits or investigations

25.5.1 Consequences of checks, reviews, audits and investigations in this grant

Findings in checks, reviews, audits or investigations carried out in the context of this grant may lead to rejections (see Article 27), grant reduction (see Article 28) or other measures described in Chapter 5.

Rejections or grant reductions after the final payment will lead to a revised final grant amount (see Article 22).

Findings in checks, reviews, audits or investigations during the action implementation may lead to a request for amendment (see Article 39), to change the description of the action set out in Annex 1.

Checks, reviews, audits or investigations that find systemic or recurrent errors, irregularities, fraud or breach of obligations in any EU grant may also lead to consequences in other EU grants awarded under similar conditions ('extension to other grants').

[19] Regulation (EU, Euratom) No 883/2013 of the European Parliament and of the Council of 11 September 2013 concerning investigations conducted by the European Anti-Fraud Office (OLAF) and repealing Regulation (EC) No 1073/1999 of the European Parliament and of the Council and Council Regulation (Euratom) No 1074/1999 (OJ L 248, 18/09/2013, p. 1).

[20] Council Regulation (Euratom, EC) No 2185/96 of 11 November 1996 concerning on-the-spot checks and inspections carried out by the Commission in order to protect the European Communities' financial interests against fraud and other irregularities (OJ L 292, 15/11/1996, p. 2).

Moreover, findings arising from an OLAF or EPPO investigation may lead to criminal prosecution under national law.

25.5.2 Extension from other grants

Results of checks, reviews, audits or investigations in other grants may be extended to this grant, if:

(a) the beneficiary concerned is found, in other EU grants awarded under similar conditions, to have committed systemic or recurrent errors, irregularities, fraud or breach of obligations that have a material impact on this grant and

(b) those findings are formally notified to the beneficiary concerned — together with the list of grants affected by the findings — within the time-limit for audits set out in the Data Sheet (see Point 6).

The granting authority will formally notify the beneficiary concerned of the intention to extend the findings and the list of grants affected.

If the extension concerns **rejections of costs or contributions**: the notification will include:

(a) an invitation to submit observations on the list of grants affected by the findings

(b) the request to submit revised financial statements for all grants affected

(c) the correction rate for extrapolation, established on the basis of the systemic or recurrent errors, to calculate the amounts to be rejected, if the beneficiary concerned:

  (i) considers that the submission of revised financial statements is not possible or practicable or

  (ii) does not submit revised financial statements.

If the extension concerns **grant reductions**: the notification will include:

(a) an invitation to submit observations on the list of grants affected by the findings and

(b) the **correction rate for extrapolation**, established on the basis of the systemic or recurrent errors and the principle of proportionality.

The beneficiary concerned has **60 days** from receiving notification to submit observations, revised financial statements or to propose a duly substantiated **alternative correction method/rate**.

On the basis of this, the granting authority will analyse the impact and decide on the implementation (i.e. start rejection or grant reduction procedures, either on the basis of the revised financial statements or the announced/alternative method/rate or a mix of those; see Articles 27 and 28).

25.6 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, costs or contributions insufficiently substantiated will be ineligible (see Article 6) and will be rejected (see Article 27), and the grant may be reduced (see Article 28).

Such breaches may also lead to other measures described in Chapter 5.

ARTICLE 26 — IMPACT EVALUATIONS

26.1 Impact evaluation

The granting authority may carry out impact evaluations of the action, measured against the objectives and indicators of the EU programme funding the grant.

Such evaluations may be started during implementation of the action and until the time-limit set out in the Data Sheet (see Point 6). They will be formally notified to the coordinator or beneficiaries and will be considered to start on the date of the notification.

If needed, the granting authority may be assisted by independent outside experts.

The coordinator or beneficiaries must provide any information relevant to evaluate the impact of the action, including information in electronic format.

26.2 Consequences of non-compliance

If a beneficiary breaches any of its obligations under this Article, the granting authority may apply the measures described in Chapter 5.

CHAPTER 5 CONSEQUENCES OF NON-COMPLIANCE

SECTION 1 REJECTIONS AND GRANT REDUCTION

ARTICLE 27 — REJECTION OF COSTS AND CONTRIBUTIONS

27.1 Conditions

The granting authority will — at beneficiary termination, interim payment, final payment or afterwards — reject any costs or contributions which are ineligible (see Article 6), in particular following checks, reviews, audits or investigations (see Article 25).

The rejection may also be based on the extension of findings from other grants to this grant (see Article 25).

Ineligible costs or contributions will be rejected.

27.2 Procedure

If the rejection does not lead to a recovery, the granting authority will formally notify the coordinator or beneficiary concerned of the rejection, the amounts and the reasons why. The coordinator or beneficiary concerned may — within 30 days of receiving notification — submit observations if it disagrees with the rejection (payment review procedure).

If the rejection leads to a recovery, the granting authority will follow the contradictory procedure with pre-information letter set out in Article 22.

27.3 Effects

If the granting authority rejects costs or contributions, it will deduct them from the costs or contributions declared and then calculate the amount due (and, if needed, make a recovery; see Article 22).

ARTICLE 28 — GRANT REDUCTION

28.1 Conditions

The granting authority may — at beneficiary termination, final payment or afterwards — reduce the grant for a beneficiary, if:

(a) the beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed:

(i) substantial errors, irregularities or fraud or

(ii) serious breach of obligations under this Agreement or during its award (including improper implementation of the action, non-compliance with the call conditions, submission of false information, failure to provide required information, breach of ethics or security rules (if applicable), etc.), or

(b) the beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed — in other EU grants awarded to it under similar conditions — systemic or recurrent errors, irregularities, fraud or serious breach of obligations that have a material impact on this grant (see Article 25).

The amount of the reduction will be calculated for each beneficiary concerned and proportionate to the seriousness and the duration of the errors, irregularities or fraud or breach of obligations, by applying an individual reduction rate to their accepted EU contribution.

28.2 Procedure

If the grant reduction does not lead to a recovery, the granting authority will formally notify the coordinator or beneficiary concerned of the reduction, the amount to be reduced and the reasons why. The coordinator or beneficiary concerned may — within 30 days of receiving notification — submit observations if it disagrees with the reduction (payment review procedure).

If the grant reduction leads to a recovery, the granting authority will follow the contradictory procedure with pre-information letter set out in Article 22.

28.3 Effects

If the granting authority reduces the grant, it will deduct the reduction and then calculate the amount due (and, if needed, make a recovery; see Article 22).

SECTION 2 SUSPENSION AND TERMINATION

ARTICLE 29 — PAYMENT DEADLINE SUSPENSION

29.1 Conditions

The granting authority may — at any moment — suspend the payment deadline if a payment cannot be processed because:

(a) the required report (see Article 21) has not been submitted or is not complete or additional information is needed

(b) there are doubts about the amount to be paid (e.g. ongoing audit extension procedure, queries about eligibility, need for a grant reduction, etc.) and additional checks, reviews, audits or investigations are necessary, or

(c) there are other issues affecting the EU financial interests.

29.2 Procedure

The granting authority will formally notify the coordinator of the suspension and the reasons why.

The suspension will **take effect** the day the notification is sent.

If the conditions for suspending the payment deadline are no longer met, the suspension will be **lifted** — and the remaining time to pay (see Data Sheet, Point 4.2) will resume.

If the suspension exceeds two months, the coordinator may request the granting authority to confirm if the suspension will continue.

If the payment deadline has been suspended due to the non-compliance of the report and the revised report is not submitted (or was submitted but is also rejected), the granting authority may also terminate the grant or the participation of the coordinator (see Article 32).

ARTICLE 30 — PAYMENT SUSPENSION

30.1 Conditions

The granting authority may — at any moment — suspend payments, in whole or in part for one or more beneficiaries, if:

(a) a beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed or is suspected of having committed:

  (i) substantial errors, irregularities or fraud or

  (ii) serious breach of obligations under this Agreement or during its award (including improper implementation of the action, non-compliance with the call conditions, submission of false information, failure to provide required information, breach of ethics or security rules (if applicable), etc.), or

(b) a beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed — in other EU grants awarded to it under similar conditions — systemic or recurrent errors, irregularities, fraud or serious breach of obligations that have a material impact on this grant.

If payments are suspended for one or more beneficiaries, the granting authority will make partial

payment(s) for the part(s) not suspended. If suspension concerns the final payment, the payment (or recovery) of the remaining amount after suspension is lifted will be considered to be the payment that closes the action.

30.2 Procedure

Before suspending payments, the granting authority will send a **pre-information letter** to the beneficiary concerned:

- formally notifying the intention to suspend payments and the reasons why and
- requesting observations within 30 days of receiving notification.

If the granting authority does not receive observations or decides to pursue the procedure despite the observations it has received, it will confirm the suspension (**confirmation letter**). Otherwise, it will formally notify that the procedure is discontinued.

At the end of the suspension procedure, the granting authority will also inform the coordinator.

The suspension will **take effect** the day after the confirmation notification is sent.

If the conditions for resuming payments are met, the suspension will be **lifted**. The granting authority will formally notify the beneficiary concerned (and the coordinator) and set the suspension end date.

During the suspension, no prefinancing will be paid to the beneficiaries concerned. For interim payments, the periodic reports for all reporting periods except the last one (see Article 21) must not contain any financial statements from the beneficiary concerned (or its affiliated entities). The coordinator must include them in the next periodic report after the suspension is lifted or — if suspension is not lifted before the end of the action — in the last periodic report.

ARTICLE 31 — GRANT AGREEMENT SUSPENSION

31.1 Consortium-requested GA suspension

31.1.1 Conditions and procedure

The beneficiaries may request the suspension of the grant or any part of it, if exceptional circumstances — in particular *force majeure* (see Article 35) — make implementation impossible or excessively difficult.

The coordinator must submit a request for **amendment** (see Article 39), with:

- the reasons why
- the date the suspension takes effect; this date may be before the date of the submission of the amendment request and
- the expected date of resumption.

The suspension will **take effect** on the day specified in the amendment.

Once circumstances allow for implementation to resume, the coordinator must immediately request

another **amendment** of the Agreement to set the suspension end date, the resumption date (one day after suspension end date), extend the duration and make other changes necessary to adapt the action to the new situation (see Article 39) — unless the grant has been terminated (see Article 32). The suspension will be **lifted** with effect from the suspension end date set out in the amendment. This date may be before the date of the submission of the amendment request.

During the suspension, no prefinancing will be paid. Costs incurred or contributions for activities implemented during grant suspension are not eligible (see Article 6.3).

31.2 EU-initiated GA suspension

31.2.1 Conditions

The granting authority may suspend the grant or any part of it, if:

(a) a beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed or is suspected of having committed:

  (i) substantial errors, irregularities or fraud or

  (ii) serious breach of obligations under this Agreement or during its award (including improper implementation of the action, non-compliance with the call conditions, submission of false information, failure to provide required information, breach of ethics or security rules (if applicable), etc.), or

(b) a beneficiary (or a person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed — in other EU grants awarded to it under similar conditions — systemic or recurrent errors, irregularities, fraud or serious breach of obligations that have a material impact on this grant

(c) other:

  (i) linked action issues: not applicable

  (ii) the action has lost its scientific or technological relevance, for EIC Accelerator actions: the action has lost its economic relevance, for challenge-based EIC Pathfinder actions and Horizon Europe Missions: the action has lost its relevance as part of the Portfolio for which it has been initially selected

31.2.2 Procedure

Before suspending the grant, the granting authority will send a **pre-information letter** to the coordinator:

- formally notifying the intention to suspend the grant and the reasons why and

- requesting observations within 30 days of receiving notification.

If the granting authority does not receive observations or decides to pursue the procedure despite the observations it has received, it will confirm the suspension (**confirmation letter**). Otherwise, it will formally notify that the procedure is discontinued.

The suspension will **take effect** the day after the confirmation notification is sent (or on a later date specified in the notification).

Once the conditions for resuming implementation of the action are met, the granting authority will formally notify the coordinator a **lifting of suspension letter**, in which it will set the suspension end date and invite the coordinator to request an amendment of the Agreement to set the resumption date (one day after suspension end date), extend the duration and make other changes necessary to adapt the action to the new situation (see Article 39) — unless the grant has been terminated (see Article 32). The suspension will be **lifted** with effect from the suspension end date set out in the lifting of suspension letter. This date may be before the date on which the letter is sent.

During the suspension, no prefinancing will be paid. Costs incurred or contributions for activities implemented during suspension are not eligible (see Article 6.3).

The beneficiaries may not claim damages due to suspension by the granting authority (see Article 33).

Grant suspension does not affect the granting authority's right to terminate the grant or a beneficiary (see Article 32) or reduce the grant (see Article 28).

ARTICLE 32 — GRANT AGREEMENT OR BENEFICIARY TERMINATION

32.1 Consortium-requested GA termination

32.1.1 Conditions and procedure

The beneficiaries may request the termination of the grant.

The coordinator must submit a request for **amendment** (see Article 39), with:

- the reasons why
- the date the consortium ends work on the action ('end of work date') and
- the date the termination takes effect ('termination date'); this date must be after the date of the submission of the amendment request.

The termination will **take effect** on the termination date specified in the amendment.

If no reasons are given or if the granting authority considers the reasons do not justify termination, it may consider the grant terminated improperly.

32.1.2 Effects

The coordinator must — within 60 days from when termination takes effect — submit a **periodic report** (for the open reporting period until termination).

The granting authority will calculate the final grant amount and final payment on the basis of the report submitted and taking into account the costs incurred and contributions for activities implemented before the end of work date (see Article 22). Costs relating to contracts due for execution only after the end of work are not eligible.

If the granting authority does not receive the report within the deadline, only costs and contributions

which are included in an approved periodic report will be taken into account (no costs/contributions if no periodic report was ever approved).

Improper termination may lead to a grant reduction (see Article 28).

After termination, the beneficiaries' obligations (in particular Articles 13 (confidentiality and security), 16 (IPR), 17 (communication, dissemination and visibility), 21 (reporting), 25 (checks, reviews, audits and investigations), 26 (impact evaluation), 27 (rejections), 28 (grant reduction) and 42 (assignment of claims)) continue to apply.

32.2 Consortium-requested beneficiary termination

32.2.1 Conditions and procedure

The coordinator may request the termination of the participation of one or more beneficiaries, on request of the beneficiary concerned or on behalf of the other beneficiaries.

The coordinator must submit a request for **amendment** (see Article 39), with:

- the reasons why
- the opinion of the beneficiary concerned (or proof that this opinion has been requested in writing)
- the date the beneficiary ends work on the action ('end of work date')
- the date the termination takes effect ('termination date'); this date must be after the date of the submission of the amendment request.

If the termination concerns the coordinator and is done without its agreement, the amendment request must be submitted by another beneficiary (acting on behalf of the consortium).

The termination will **take effect** on the termination date specified in the amendment.

If no information is given or if the granting authority considers that the reasons do not justify termination, it may consider the beneficiary to have been terminated improperly.

32.2.2 Effects

The coordinator must — within 60 days from when termination takes effect — submit:

(i) a **report on the distribution of payments** to the beneficiary concerned

(ii) a **termination report** from the beneficiary concerned, for the open reporting period until termination, containing an overview of the progress of the work, the financial statement, the explanation on the use of resources, and, if applicable, the certificate on the financial statement (CFS; see Articles 21 and 24.2 and Data Sheet, Point 4.3)

(iii) a second **request for amendment** (see Article 39) with other amendments needed (e.g. reallocation of the tasks and the estimated budget of the terminated beneficiary; addition of a new beneficiary to replace the terminated beneficiary; change of coordinator, etc.).

The granting authority will calculate the amount due to the beneficiary on the basis of the report

submitted and taking into account the costs incurred and contributions for activities implemented before the end of work date (see Article 22). Costs relating to contracts due for execution only after the end of work are not eligible.

The information in the termination report must also be included in the periodic report for the next reporting period (see Article 21).

If the granting authority does not receive the termination report within the deadline, only costs and contributions which are included in an approved periodic report will be taken into account (no costs/ contributions if no periodic report was ever approved).

If the granting authority does not receive the report on the distribution of payments within the deadline, it will consider that:

- the coordinator did not distribute any payment to the beneficiary concerned and that
- the beneficiary concerned must not repay any amount to the coordinator.

If the second request for amendment is accepted by the granting authority, the Agreement is **amended** to introduce the necessary changes (see Article 39).

If the second request for amendment is rejected by the granting authority (because it calls into question the decision awarding the grant or breaches the principle of equal treatment of applicants), the grant may be terminated (see Article 32).

Improper termination may lead to a reduction of the grant (see Article 31) or grant termination (see Article 32).

After termination, the concerned beneficiary's obligations (in particular Articles 13 (confidentiality and security), 16 (IPR), 17 (communication, dissemination and visibility), 21 (reporting), 25 (checks, reviews, audits and investigations), 26 (impact evaluation), 27 (rejections), 28 (grant reduction) and 42 (assignment of claims)) continue to apply.

32.3 EU-initiated GA or beneficiary termination

32.3.1 Conditions

The granting authority may terminate the grant or the participation of one or more beneficiaries, if:

(a) one or more beneficiaries do not accede to the Agreement (see Article 40)

(b) a change to the action or the legal, financial, technical, organisational or ownership situation of a beneficiary is likely to substantially affect the implementation of the action or calls into question the decision to award the grant (including changes linked to one of the exclusion grounds listed in the declaration of honour)

(c) following termination of one or more beneficiaries, the necessary changes to the Agreement (and their impact on the action) would call into question the decision awarding the grant or breach the principle of equal treatment of applicants

(d) implementation of the action has become impossible or the changes necessary for its

continuation would call into question the decision awarding the grant or breach the principle of equal treatment of applicants

(e) a beneficiary (or person with unlimited liability for its debts) is subject to bankruptcy proceedings or similar (including insolvency, winding-up, administration by a liquidator or court, arrangement with creditors, suspension of business activities, etc.)

(f) a beneficiary (or person with unlimited liability for its debts) is in breach of social security or tax obligations

(g) a beneficiary (or person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has been found guilty of grave professional misconduct

(h) a beneficiary (or person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed fraud, corruption, or is involved in a criminal organisation, money laundering, terrorism-related crimes (including terrorism financing), child labour or human trafficking

(i) a beneficiary (or person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) was created under a different jurisdiction with the intent to circumvent fiscal, social or other legal obligations in the country of origin (or created another entity with this purpose)

(j) a beneficiary (or person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed:

  (i) substantial errors, irregularities or fraud or

  (ii) serious breach of obligations under this Agreement or during its award (including improper implementation of the action, non-compliance with the call conditions, submission of false information, failure to provide required information, breach of ethics or security rules (if applicable), etc.)

(k) a beneficiary (or person having powers of representation, decision-making or control, or person essential for the award/implementation of the grant) has committed — in other EU grants awarded to it under similar conditions — systemic or recurrent errors, irregularities, fraud or serious breach of obligations that have a material impact on this grant (extension of findings from other grants to this grant; see Article 25)

(l) despite a specific request by the granting authority, a beneficiary does not request — through the coordinator — an amendment to the Agreement to end the participation of one of its affiliated entities or associated partners that is in one of the situations under points (d), (f), (e), (g), (h), (i) or (j) and to reallocate its tasks, or

(m) other:

  (i) linked action issues: not applicable

  (ii) the action has lost its scientific or technological relevance, for EIC Accelerator actions: the action has lost its economic relevance, for challenge-based EIC Pathfinder actions

and Horizon Europe Missions: the action has lost its relevance as part of the Portfolio for which it has been initially selected

32.3.2 Procedure

Before terminating the grant or participation of one or more beneficiaries, the granting authority will send **a pre-information letter** to the coordinator or beneficiary concerned:

- formally notifying the intention to terminate and the reasons why and
- requesting observations within 30 days of receiving notification.

If the granting authority does not receive observations or decides to pursue the procedure despite the observations it has received, it will confirm the termination and the date it will take effect (**confirmation letter**). Otherwise, it will formally notify that the procedure is discontinued.

For beneficiary terminations, the granting authority will — at the end of the procedure — also inform the coordinator.

The termination will **take effect** the day after the confirmation notification is sent (or on a later date specified in the notification; 'termination date').

32.3.3 Effects

(a) for **GA termination**:

The coordinator must — within 60 days from when termination takes effect — submit a **periodic report** (for the last open reporting period until termination).

The granting authority will calculate the final grant amount and final payment on the basis of the report submitted and taking into account the costs incurred and contributions for activities implemented before termination takes effect (see Article 22). Costs relating to contracts due for execution only after termination are not eligible.

If the grant is terminated for breach of the obligation to submit reports, the coordinator may not submit any report after termination.

If the granting authority does not receive the report within the deadline, only costs and contributions which are included in an approved periodic report will be taken into account (no costs/contributions if no periodic report was ever approved).

Termination does not affect the granting authority's right to reduce the grant (see Article 28) or to impose administrative sanctions (see Article 34).

The beneficiaries may not claim damages due to termination by the granting authority (see Article 33).

After termination, the beneficiaries' obligations (in particular Articles 13 (confidentiality and security), 16 (IPR), 17 (communication, dissemination and visibility), 21 (reporting), 25 (checks, reviews, audits and investigations), 26 (impact evaluation), 27 (rejections), 28 (grant reduction) and 42 (assignment of claims)) continue to apply.

(b) for **beneficiary termination**:

The coordinator must — within 60 days from when termination takes effect — submit:

(i) a **report on the distribution of payments** to the beneficiary concerned

(ii) a **termination report** from the beneficiary concerned, for the open reporting period until termination, containing an overview of the progress of the work, the financial statement, the explanation on the use of resources, and, if applicable, the certificate on the financial statement (CFS; see Articles 21 and 24.2 and Data Sheet, Point 4.3)

(iii) a **request for amendment** (see Article 39) with any amendments needed (e.g. reallocation of the tasks and the estimated budget of the terminated beneficiary; addition of a new beneficiary to replace the terminated beneficiary; change of coordinator, etc.).

The granting authority will calculate the amount due to the beneficiary on the basis of the report submitted and taking into account the costs incurred and contributions for activities implemented before termination takes effect (see Article 22). Costs relating to contracts due for execution only after termination are not eligible.

The information in the termination report must also be included in the periodic report for the next reporting period (see Article 21).

If the granting authority does not receive the termination report within the deadline, only costs and contributions included in an approved periodic report will be taken into account (no costs/ contributions if no periodic report was ever approved).

If the granting authority does not receive the report on the distribution of payments within the deadline, it will consider that:

- the coordinator did not distribute any payment to the beneficiary concerned and that
- the beneficiary concerned must not repay any amount to the coordinator.

If the request for amendment is accepted by the granting authority, the Agreement is **amended** to introduce the necessary changes (see Article 39).

If the request for amendment is rejected by the granting authority (because it calls into question the decision awarding the grant or breaches the principle of equal treatment of applicants), the grant may be terminated (see Article 32).

After termination, the concerned beneficiary's obligations (in particular Articles 13 (confidentiality and security), 16 (IPR), 17 (communication, dissemination and visibility), 21 (reporting), 25 (checks, reviews, audits and investigations), 26 (impact evaluation), 27 (rejections), 28 (grant reduction) and 42 (assignment of claims)) continue to apply.

SECTION 3 OTHER CONSEQUENCES: DAMAGES AND ADMINISTRATIVE SANCTIONS

ARTICLE 33 — DAMAGES

33.1 Liability of the granting authority

The granting authority cannot be held liable for any damage caused to the beneficiaries or to third parties as a consequence of the implementation of the Agreement, including for gross negligence.

The granting authority cannot be held liable for any damage caused by any of the beneficiaries or other participants involved in the action, as a consequence of the implementation of the Agreement.

33.2 Liability of the beneficiaries

The beneficiaries must compensate the granting authority for any damage it sustains as a result of the implementation of the action or because the action was not implemented in full compliance with the Agreement, provided that it was caused by gross negligence or wilful act.

The liability does not extend to indirect or consequential losses or similar damage (such as loss of profit, loss of revenue or loss of contracts), provided such damage was not caused by wilful act or by a breach of confidentiality.

ARTICLE 34 — ADMINISTRATIVE SANCTIONS AND OTHER MEASURES

Nothing in this Agreement may be construed as preventing the adoption of administrative sanctions (i.e. exclusion from EU award procedures and/or financial penalties) or other public law measures, in addition or as an alternative to the contractual measures provided under this Agreement (see, for instance, Articles 135 to 145 EU Financial Regulation 2018/1046 and Articles 4 and 7 of Regulation 2988/95[21]).

SECTION 4 FORCE MAJEURE

ARTICLE 35 — FORCE MAJEURE

A party prevented by force majeure from fulfilling its obligations under the Agreement cannot be considered in breach of them.

‘Force majeure’ means any situation or event that:

- prevents either party from fulfilling their obligations under the Agreement,
- was unforeseeable, exceptional situation and beyond the parties’ control,
- was not due to error or negligence on their part (or on the part of other participants involved in the action), and
- proves to be inevitable in spite of exercising all due diligence.

Any situation constituting force majeure must be formally notified to the other party without delay, stating the nature, likely duration and foreseeable effects.

[21] Council Regulation (EC, Euratom) No 2988/95 of 18 December 1995 on the protection of the European Communities financial interests (OJ L 312, 23.12.1995, p. 1).

The parties must immediately take all the necessary steps to limit any damage due to force majeure and do their best to resume implementation of the action as soon as possible.

CHAPTER 6 FINAL PROVISIONS

ARTICLE 36 — COMMUNICATION BETWEEN THE PARTIES

36.1 Forms and means of communication — Electronic management

EU grants are managed fully electronically through the EU Funding & Tenders Portal ('Portal').

All communications must be made electronically through the Portal, in accordance with the Portal Terms and Conditions and using the forms and templates provided there (except if explicitly instructed otherwise by the granting authority).

Communications must be made in writing and clearly identify the grant agreement (project number and acronym).

Communications must be made by persons authorised according to the Portal Terms and Conditions. For naming the authorised persons, each beneficiary must have designated — before the signature of this Agreement — a 'legal entity appointed representative (LEAR)'. The role and tasks of the LEAR are stipulated in their appointment letter (see Portal Terms and Conditions).

If the electronic exchange system is temporarily unavailable, instructions will be given on the Portal.

36.2 Date of communication

The sending date for communications made through the Portal will be the date and time of sending, as indicated by the time logs.

The receiving date for communications made through the Portal will be the date and time the communication is accessed, as indicated by the time logs. Formal notifications that have not been accessed within 10 days after sending, will be considered to have been accessed (see Portal Terms and Conditions).

If a communication is exceptionally made on paper (by e-mail or postal service), general principles apply (i.e. date of sending/receipt). Formal notifications by registered post with proof of delivery will be considered to have been received either on the delivery date registered by the postal service or the deadline for collection at the post office.

If the electronic exchange system is temporarily unavailable, the sending party cannot be considered in breach of its obligation to send a communication within a specified deadline.

36.3 Addresses for communication

The Portal can be accessed via the Europa website.

The address for paper communications to the granting authority (if exceptionally allowed) is the official mailing address indicated on its website.

For beneficiaries, it is the legal address specified in the Portal Participant Register.

ARTICLE 37 — INTERPRETATION OF THE AGREEMENT

The provisions in the Data Sheet take precedence over the rest of the Terms and Conditions of the Agreement.

Annex 5 takes precedence over the Terms and Conditions; the Terms and Conditions take precedence over the Annexes other than Annex 5.

Annex 2 takes precedence over Annex 1.

ARTICLE 38 — CALCULATION OF PERIODS AND DEADLINES

In accordance with Regulation No 1182/71[22], periods expressed in days, months or years are calculated from the moment the triggering event occurs.

The day during which that event occurs is not considered as falling within the period.

'Days' means calendar days, not working days.

ARTICLE 39 — AMENDMENTS

39.1 Conditions

The Agreement may be amended, unless the amendment entails changes to the Agreement which would call into question the decision awarding the grant or breach the principle of equal treatment of applicants.

Amendments may be requested by any of the parties.

39.2 Procedure

The party requesting an amendment must submit a request for amendment signed directly in the Portal Amendment tool.

The coordinator submits and receives requests for amendment on behalf of the beneficiaries (see Annex 3). If a change of coordinator is requested without its agreement, the submission must be done by another beneficiary (acting on behalf of the other beneficiaries).

The request for amendment must include:

- the reasons why
- the appropriate supporting documents and
- for a change of coordinator without its agreement: the opinion of the coordinator (or proof that this opinion has been requested in writing).

The granting authority may request additional information.

[22] Regulation (EEC, Euratom) No 1182/71 of the Council of 3 June 1971 determining the rules applicable to periods, dates and time-limits (OJ L 124, 8/6/1971, p. 1).

If the party receiving the request agrees, it must sign the amendment in the tool within 45 days of receiving notification (or any additional information the granting authority has requested). If it does not agree, it must formally notify its disagreement within the same deadline. The deadline may be extended, if necessary for the assessment of the request. If no notification is received within the deadline, the request is considered to have been rejected.

An amendment **enters into force** on the day of the signature of the receiving party.

An amendment **takes effect** on the date of entry into force or other date specified in the amendment.

ARTICLE 40 — ACCESSION AND ADDITION OF NEW BENEFICIARIES

40.1 Accession of the beneficiaries mentioned in the Preamble

The beneficiaries which are not coordinator must accede to the grant by signing the accession form (see Annex 3) directly in the Portal Grant Preparation tool, within 30 days after the entry into force of the Agreement (see Article 44).

They will assume the rights and obligations under the Agreement with effect from the date of its entry into force (see Article 44).

If a beneficiary does not accede to the grant within the above deadline, the coordinator must — within 30 days — request an amendment (see Article 39) to terminate the beneficiary and make any changes necessary to ensure proper implementation of the action. This does not affect the granting authority's right to terminate the grant (see Article 32).

40.2 Addition of new beneficiaries

In justified cases, the beneficiaries may request the addition of a new beneficiary.

For this purpose, the coordinator must submit a request for amendment in accordance with Article 39. It must include an accession form (see Annex 3) signed by the new beneficiary directly in the Portal Amendment tool.

New beneficiaries will assume the rights and obligations under the Agreement with effect from the date of their accession specified in the accession form (see Annex 3).

Additions are also possible in mono-beneficiary grants.

ARTICLE 41 — TRANSFER OF THE AGREEMENT

In justified cases, the beneficiary of a mono-beneficiary grant may request the transfer of the grant to a new beneficiary, provided that this would not call into question the decision awarding the grant or breach the principle of equal treatment of applicants.

The beneficiary must submit a request for **amendment** (see Article 39), with

- the reasons why
- the accession form (see Annex 3) signed by the new beneficiary directly in the Portal Amendment tool and

- additional supporting documents (if required by the granting authority).

The new beneficiary will assume the rights and obligations under the Agreement with effect from the date of accession specified in the accession form (see Annex 3).

ARTICLE 42 — ASSIGNMENTS OF CLAIMS FOR PAYMENT AGAINST THE GRANTING AUTHORITY

The beneficiaries may not assign any of their claims for payment against the granting authority to any third party, except if expressly approved in writing by the granting authority on the basis of a reasoned, written request by the coordinator (on behalf of the beneficiary concerned).

If the granting authority has not accepted the assignment or if the terms of it are not observed, the assignment will have no effect on it.

In no circumstances will an assignment release the beneficiaries from their obligations towards the granting authority.

ARTICLE 43 — APPLICABLE LAW AND SETTLEMENT OF DISPUTES

43.1 Applicable law

The Agreement is governed by the applicable EU law, supplemented if necessary by the law of Belgium.

Special rules may apply for beneficiaries which are international organisations (if any; see Data Sheet, Point 5).

43.2 Dispute settlement

If a dispute concerns the interpretation, application or validity of the Agreement, the parties must bring action before the EU General Court — or, on appeal, the EU Court of Justice — under Article 272 of the Treaty on the Functioning of the EU (TFEU).

For non-EU beneficiaries (if any), such disputes must be brought before the courts of Brussels, Belgium — unless an international agreement provides for the enforceability of EU court judgements.

For beneficiaries with arbitration as special dispute settlement forum (if any; see Data Sheet, Point 5), the dispute will — in the absence of an amicable settlement — be settled in accordance with the Rules for Arbitration published on the Portal.

If a dispute concerns administrative sanctions, offsetting or an enforceable decision under Article 299 TFEU (see Articles 22 and 34), the beneficiaries must bring action before the General Court — or, on appeal, the Court of Justice — under Article 263 TFEU.

For grants where the granting authority is an EU executive agency (see Preamble), actions against offsetting and enforceable decisions must be brought against the European Commission (not against the granting authority; see also Article 22).

ARTICLE 44 — ENTRY INTO FORCE

The Agreement will enter into force on the day of signature by the granting authority or the coordinator, depending on which is later.

SIGNATURES

For the coordinator	For the granting authority

ANNEX 1

Horizon Europe (HORIZON)

Description of the action (DoA)

Part A

Part B

DESCRIPTION OF THE ACTION (PART A)

COVER PAGE

Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.

PROJECT *Grant Preparation (General Information screen) — Enter the info.*	
Project number:	101135927
Project name:	A catalyst for EuropeaN ClOUd Services in the era of data spaces, high-performance and edge computing
Project acronym:	NOUS
Call:	HORIZON-CL4-2023-DATA-01
Topic:	HORIZON-CL4-2023-DATA-01-02
Type of action:	HORIZON-RIA
Service:	CNECT/G/01
Project starting date:	fixed date: 1 January 2024
Project duration:	36 months

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PROJECT SUMMARY

Project summary

Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.

Use the project summary from your proposal.

NOUS will develop the architecture of a European Cloud Service that allows computational and data storage resources to be used from edge devices as well as supercomputers, through the HPC network, and Quantum Computers. NOUS will be an Infrastructure-as-a-Service (IaaS)/Platform-as-a-Service (PaaS) cloud provider, harnessing edge computing and decentralisation paradigms to incorporate a wide array of devices and machines in its computational flow to provide leaps in Europe's capability to process vast amounts of data.
The pipeline of the NOUS in the project will include three types of components: i) computational components that are responsible for executing computations, ii) edge components that are responsible for communicating with edge devices (such as IoT sensors/ actuators/ devices), iii) data storage components that are responsible for data storage and storage management. Components are researched individually, expecting to yield breakthroughs, and jointly, to create the architecture and cloud-level services such as syndication with other platforms and virtual labs. The project has defined 4 use-cases that will allow the testing of the developed technologies in real-world scenarios that industry leaders face.
The NOUS architecture will be made open source to allow the capitalization by companies and organisations. Furthermore, a set of workshops and collaboration activities is envisioned with Data Spaces Support Centre, Gaia-X, FIWARE and EOSC powered by a strong consortium of 21 partners from 11 European countries.

LIST OF PARTICIPANTS

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
1	COO	AIR	FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION	ES	904614338
2	BEN	AETHON	AETHON ENGINEERING SINGLE MEMBER PC	EL	918590292
3	BEN	USAL	UNIVERSIDAD DE SALAMANCA	ES	999846610
4	BEN	NCSR "D"	NATIONAL CENTER FOR SCIENTIFIC RESEARCH "DEMOKRITOS"	EL	999978239
5	BEN	ICOOR	CONSORZIO INTERUNIVERSITARIO PER L'OTTIMIZZAZIONE E LA RICERCA OPERATIVA	IT	996385068
5.1	AE	POLITO	POLITECNICO DI TORINO	IT	999977754
5.2	AE	UNIMORE	UNIVERSITA DEGLI STUDI DI MODENA E REGGIO EMILIA	IT	999840887
6	BEN	TIM	TELECOM ITALIA SPA	IT	999908496
7	BEN	NET-INTRA	NETCOMPANY-INTRASOFT SA	LU	999702371

PARTICIPANTS

Grant Preparation (Beneficiaries screen) — Enter the info.

Number	Role	Short name	Legal name	Country	PIC
8	BEN	ITML CY	IOTAM INTERNET OF THINGS APPLICATIONS AND MULTI LAYER DEVELOPMENT LTD	CY	912549326
9	BEN	UNIPI	UNIVERSITA DI PISA	IT	999862712
10	BEN	UNPARALLEL	UNPARALLEL INNOVATION LDA	PT	953318426
11	BEN	KU Leuven	KATHOLIEKE UNIVERSITEIT LEUVEN	BE	999991334
12	BEN	HPE Hellas	HEWLETT - PACKARD HELLAS ETAIREIA PERIORISMENIS EFTHYNIS	EL	913711871
13	BEN	ECL	ECLIPSE FOUNDATION EUROPE GMBH	DE	944758564
14	BEN	F6S IE	F6S NETWORK IRELAND LIMITED	IE	900885658
15	BEN	PPC	DIMOSIA EPICHEIRISI ILEKTRISMOU ANONYMI ETAIREIA	EL	999938954
16	BEN	ARCTUR d.o.o.	ARCTUR RACUNALNISKI INZENIRING DOO	SI	991568048
17	BEN	AEGIS	AEGIS IT RESEARCH GMBH	DE	911867804
18	BEN	CS GROUP-FRANCE	CS GROUP-FRANCE	FR	999938275
19	BEN	INESC TEC	INESC TEC - INSTITUTO DE ENGENHARIADE SISTEMAS E COMPUTADORES, TECNOLOGIA E CIENCIA	PT	999513706

LIST OF WORK PACKAGES

Work packages *Grant Preparation (Work Packages screen) — Enter the info.*						
Work Package No	**Work Package name**	**Lead Beneficiary**	**Effort (Person-Months)**	**Start Month**	**End Month**	**Deliverables**
WP1	Project management and coordination	1 - AIR	84.00	1	36	D1.1 – Interim report D1.2 – Final report D1.3 – Data Management Plan
WP2	Gap analysis and requirements	1 - AIR	109.00	1	36	D2.1 – Status quo, gap analysis, requirements and project evaluation framework D2.2 – A data –life-cycle framework
WP3	#compute - The computational component of NOUS	4 - NCSR "D"	120.00	6	33	D3.1 – Connecting to the HPC Network D3.2 – An interface between Quantum Computers and HPC
WP4	#edge - The edge component of NOUS	8 - ITML CY	125.00	6	36	D4.1 – Distributed architectures D4.2 – The NOUS Federated Inference framework, Interoperability in the Fog and Structured parallel environments
WP5	#data - The data component of NOUS	7 - NET-INTRA	137.00	1	36	D5.1 – Legal assessment and recommendations on blockchain technologies D5.2 – Blockchain-based database storage component of NOUS and the Auto-Standardiser and Data services
WP6	Towards the creation of the NOUS architecture	2 - AETHON	185.00	6	36	D6.1 – The NOUS Architecture and Open-Source development platform (1st version) D6.2 – The NOUS Architecture and Open-Source development platform (2nd version)

Work packages

Grant Preparation (Work Packages screen) — Enter the info.

Work Package No	Work Package name	Lead Beneficiary	Effort (Person-Months)	Start Month	End Month	Deliverables
WP7	The data economy and use-cases	1 - POLITO	220.00	18	36	D7.1 – Use cases testing and project assessment D7.2 – Project evaluation, Roadmap and Best practices for data economy
WP8	Communication, dissemination, exploitation and liaison	14 - F6S IE	132.00	1	36	D8.1 – Plan for dissemination and exploitation including communication activities D8.2 – Collaboration and synergies with other projects & initiatives

Work package WP1 – Project management and coordination

Work Package Number	WP1	Lead Beneficiary	1. AIR
Work Package Name	Project management and coordination		
Start Month	1	End Month	36

Objectives
The specific objectives of this WP are i) to ensure the achievement of all project's objectives on time, quality and costs, ii) to manage and coordinate smoothly and efficiently the project's partners, ensuring the achievement of the specific objectives from each of the WPs, iii) to communicate a high-quality project reporting to the EC and iv) to define and monitor the protocols and those responsible for the management of the data used in the project.

Description
WP1 comprises the following tasks: T1.1. Administrative and financial management [M1 - M36 | L: AIR, P: All]:This project-wide task will aim to ensure project compliance, budget transparency and corporate responsibility from the consortium's point of view, managing the financial resources and addressing the EC's administrative, legal, and financial requirements; by defining the project compliance and operational procedures in administrative, financial, and legal terms, establishing effective communication channels among partners and with the EC. A predictive control of possible deviations of the technical, human, and financial resources will be conducted and preventive mitigation measures will be implemented. A project management book will be produced, detailing compliance, administrative forms, financial aspects, and legal commitments. T1.2. Technical management [M1 - M36 | L: AIR, P: All]: A Technical Committee (TC) will be set up under the Steering Committee (SC), as well as three Advisory Boards made up of external experts: the Technical Advisory Board (TAB), the Innovation Advisory Board (IAB), and the Ethics Advisory Board (EAB). A detailed project plan will be defined, a schedule of regular meetings (giving preference to virtual meetings to reduce the carbon footprint), as well as the internal communication procedures and tools for sharing documentation, data, and source code necessary for the smooth running of the project. A Project Balanced Scorecard will be designed with SMART objectives, milestones, and KPIs to monitor the development of the project. The different technical reports will be validated by the EC. T1.3. Quality and risk management [M1 - M36 | L: AIR]: This task will aim to monitor the overall progress of the project to ensure compliance with the minimum acceptance criteria foreseen for the different deliverables, analysing, and recording potential risks and providing tools for the effective control of the quality of the results. Objective methodologies will be defined for the quality assurance and monitoring of deliverables by at least two partners who have not participated in the elaboration of the associated results and with the validation of the TAB. An online dashboard shared by all partners will be designed to monitor risks by means of a matrix including probability, impact and mitigation measures foreseen for each risk (technical, budgetary, stakeholders). The planned mitigation measures will be implemented in advance. T1.4. Ethics, legal and data management [M1 - M36 | L: AETHON, P: AIR, NCSRD, TIM, NET-INTRA, KUL, HPE, PPC, CS]: This task will be aimed at defining and monitoring protocols to ensure the quality and privacy of the data collected, stored, processed, analysed and published during the project, as well as for the life cycle of the data of the NOUS platform itself resulting from the project. A Data Management Plan will be developed in accordance with the European GDPR and FAIR principles in order to ensure the appropriate and secure treatment of the information. Furthermore, this task will define an ethical framework based on the guidelines of a Trustworthy AI from the EC to guide all data analysis and ML model training processes on NOUS. Both the DMP and the ethical guidelines as well as the periodic monitoring reports on ethical and data privacy issues will be validated by the Ethics Advisory Board.

Work package WP2 – Gap analysis and requirements

Work Package Number	WP2	Lead Beneficiary	1. AIR
Work Package Name	Gap analysis and requirements		
Start Month	1	End Month	36

Objectives
The specific objectives of this WP are i) to determine the current state in data standards, technologies of the project and architectures, ii) to elicit requirements for the NOUS architecture and for each technological component of the project, iii) to create an evaluation framework of the project to be used for assessing its impact and iv) to develop a data life cycle framework that will govern the data flows within NOUS and update it after testing.

Description
WP2 comprises the following tasks: T2.1. Status quo and gap analysis of existing and emerging standards, models and architectures [M1 - M6 | L: NCSRD, P: AIR, AETHON, UNIMORE, NET-INTRA, ITML, UNIPI, UNP, KUL, HPE, ARCTUR, INESC TEC]: This task involves the mapping of commercial and SotA technologies and the review of existing IaaS/PaaS/MLaaS architectures. This will include neural-evolutionary algorithms for Cloud-HPC-Quantum computing resource balance optimisation; cybersecurity in Zero Trust environments; federated learning and Edge/Fog computing, energy-efficient DLTs, NLP for cross-language conversational interfaces; legal implications of federated networks and multi-tenant cloud services in countries with different legislations; and breakthroughs of other initiatives, such as, Gaia-X and EOSC. The incorporation of DSSC workings into NOUS will be reviewed (e.g., DSSC Glossary, Starter Kit). This will determine the gaps between current implementations leading to business definition of the NOUS model that will be expressed using the "Working Backwards" methodology which defines that the product creation (i.e., the NOUS architecture) should start at the end. This creates a common vision for the consortium while working towards NOUS architecture and the various technological components. The inputs of the gap analysis leading to the implementation of Working Backwards method will be retrieved through workshops (at least 2 – first with project partners and second will be extended to also include non-partners) and desk research. T2.2. Requirements elicitation and prioritisation for the NOUS architecture and NOUS components [M4 - M10 | L: AETHON, P: AIR, USAL, NCSRD, UNIMORE, TIM, NET-INTRA, ITML, UNIPI, UNP, HPE, F6S, PPC, ARCTUR, AEGIS, CS, INESC TEC]: The task will generate the functional and non-functional requirements for NOUS architecture and each technology created in WP3, WP4, WP5, WP6 with the methodology defined in Part 1 of ch. 1.2.2 incorporating human-centred practices This activity will be co-created with consortium partners and especially use-case's providers through a series of surveys and workshops (at least 1 – can be merged with a T2.1 workshop), considering the gaps determined in T2.1. The data collected from surveys/workshops will assist in determining needs and expectations of the NOUS' partners and community, which, in combination with desk research, will lead to the creation of user stories, functional requirements (e.g., variety of computing and virtualisation services, types of SQL, conversational assistant functionalities facilitating user interaction) and technical requirements (e.g., service latencies, storage capacities, computing power, built-in authentication and cybersecurity measures, legal aspects of data privacy) considering the human dimension of the technology being developed. T2.3. Identify, adapt and implement a suitable evaluation framework [M4 - M10 | L: USAL, P: AIR, NCSRD, UNIMORE, ITML, KUL]: This task will identify the most suitable evaluation framework and methodology for measuring project KPIs. KPIs will extend those determined in ch. 2.1 and will represent impacts related to, e.g., acceptance, accessibility, operational cooperation structures, safety, security, costs, energy, and externalities. The evaluation framework will be assessed and validated by at least eight external advisors from European universities, research centres and data industry. This task will also design and prepare the surveys needed to collect users' perceptions during the testing, in T7.2. The evaluation methodology will be implemented in T7.3. T2.4. A novel data life cycle framework [M4 - M36 | L: AIR, P: AETHON, USAL, UNIMORE, TIM, HPE, PPC, CS, INESC TEC]: The task aims to break down methodologies for data life cycle handling. Methodologies currently used by consortium partners and the use-cases' providers will be extracted through targeted interviews. Following the above, the task will determine current issues in data life cycle management and opportunities to enable correct allocation and enforcement of data-related rights, obligations and responsibilities ranging through all phases (Data creation, ingestion, or capture, Data processing, Data analysis, Data sharing or publication, Archiving). This part must consider the DSSC Glossary and Starter Kit, as well as EIF principles and recommendations. Human language and context awareness issues for multimodal (images, voice and text) AI models will be included to an emerging framework that the task will deliver for data life-cycle handling. This task will also address concerns for privacy and the implementation of GDPR across the life cycle. Data standardisation is conducted in two ways: (1) Perform an overall review of the standards that the project associates with (see O4 and T5.4, in Data Spaces of Mobility, Energy, Green Deal), followed by recommendations towards harmonisation and updating the viewpoints of EIRA and the EIF Implementation Strategy in scope of facilitating

the creation and deployment of tools such as NOUS and NOUS components. (2) The inclusion of standardisation activities in the data life cycle. This task will provide an updated version of the framework following the testing of T7.2

Work package WP3 – #compute - The computational component of NOUS

Work Package Number	WP3	**Lead Beneficiary**	4. NCSR "D"
Work Package Name	#compute - The computational component of NOUS		
Start Month	6	**End Month**	33

Objectives
The specific objectives of this WP are i) to design the optimum of I/O communication architecture between HPC and NOUS, ii) to achieve high-speed, error-free and seamless classical large data exchange between HPC and NOUS, iii) to realize a software environment for encoding of classical large datasets to qubit-ready format for processing in a quantum computer and iv) to develop unconventional and novel cybersecurity architectures.

Description
WP3 comprises the following tasks: T3.1. Leveraging the HPC Network and its resources for the NOUS [M6 - M12 | L: ARCTUR, P: NCSRD, TIM, HPE, PPC, AEGIS, CS]: This task will identify system-level policy and infrastructure requirements to enable the design and implementation of seamless communication and data transfer between the NOUS and HPC systems. This can include access control, networking, queueing systems, containers, data repositories, software and similar. Defining key services like network and resource management, which can be virtualised, and the communication protocols between them will enable integration of various HPC systems, regardless of the underlying hardware and setup. ARCTUR has the knowledge to manage HPC with different architectures and deploy them to the participants. In this, task also experts will help for the development or improvement of the available resources to the participants. T3.2. Seamless interaction between Quantum Computers and HPC [M12 - M28 | L: NCSRD, P: AIR, USAL, HPE, PPC, ARCTUR, AEGIS]: The task will focus on a continuous software stack that enables a user to harness the combined computational capabilities of both system components in a seamless fashion. On the QC side, this task will start with the standard programming packages available to QC researchers, such as Cirq and Qiskit, and link them to the HPC resources management as well as augment their back-ends to efficiently and transparently onload compute intensive tasks back to the HPC system, ideally using the HPC workload manager to identify otherwise unused resources. QC systems only target single-user operations and offer limited support for multiuser operations. Therefore, integration requires a careful redesign of the runtime component to enable efficient overlap of QC executions from multiple users and to mitigate idle times caused by complex preparation tasks. On the HPC side, this task will build on top of widely available technology forming the software stacks as they are now available on most HPC systems. This requires the ability to specify quantum computation from within HPC programs, which is challenging due to the different programming approaches. To solve this challenge, T3.2 will develop new techniques to enable Python-driven front ends and integrate them into existing offload approaches such as OpenMP and use noninterpreted approaches directly in C/C++. Finally, this task will explore the capability of using Quantum Machine Learning in collaboration with the Federated Learning system in collaboration with T4.2. T3.3. Cybersecurity of the NOUS - A darknet approach and Zero Trust architecture [M18 - M33 | L: AEGIS, P: AIR, USAL, NCSRD, TIM, ITML, UNIPI, UNP, HPE, PPC, ARCTUR, CS, INESC TEC]: The aim of this task is to design and develop the cybersecurity component of NOUS as a standalone software to be also used in cloud networking environments. This component will be based on a combination of two pillars (i) darknets and (ii) Zero Trust. On one hand, a darknet (a distinct secure network across the Internet) can be used to segregate the services offered by NOUS, improving trust and efficiency along the way, by minimising the requirements for additional cybersecurity inside the darknet. On the other hand, the Zero Trust security model can be leveraged, which dictates that access to a protected resource (e.g., a NOUS service) is granted to a requestor (user/system) only after verifying the identity, context, and policy adherence of each specific request.

Work package WP4 – #edge - The edge component of NOUS

Work Package Number	WP4	Lead Beneficiary	8. ITML CY
Work Package Name	#edge - The edge component of NOUS		
Start Month	6	End Month	36

Objectives
The specific objectives of this WP are i) to design and deliver innovative distributed architectures working with a traditional cloud service, ii) to facilitate a new paradigm in the edge computing exploiting structured parallel programming environments, based on FastFlow, which will enable operators to instantiate parallel applications as compositions of pre-defined, configurable parallel skeletons, iii) to deliver a technological breakthrough in FL approaches driven by first, a suite of algorithms for decentralised ML for user- and device-centric analytics and second, an FL-based hierarchical, algorithmic data and computation distribution layer and iv) to develop advanced interoperability mechanisms for seamless data sharing and exchange at the edge.

Description
WP4 comprises the following tasks: T4.1. Distributed architectures working with a traditional cloud service [M6 - M12 | L: ITML, P: NCSRD, TIM, NET-INTRA, UNIPI, UNP, HPE, PPC, ARCTUR, AEGIS]: This task will provide an in-depth analysis of available approaches for distributed architectures to (co-)operate in an optimal and most efficient manner with traditional cloud services. It will identify the best practices to address the challenges of edge-to-cloud interoperability and collaboration given the requirements from T2.2, and it will deliver a set of concrete architectural designs for distributed environments, that will be used as a basis for the deployment of FL technologies that will take place in T4.3 as well as for the computing mechanisms to be delivered by T4.2. T4.2. Computing on the edge, the far edge and the cloud [M13 - M36 | L: UNIPI, P: AIR, NET-INTRA, ITML, UNP, ARCTUR, INESC TEC]: This task will deliver structured parallel programming environments, based on FastFlow, which will enable the operators to instantiate parallel applications as compositions of both pre-defined, configurable parallel primitive components and user-provided parallel patterns. The software developed on top of FastFlow will enable users of NOUS to instantiate efficient parallel applications. By means of this software will be also provided efficient mechanisms for thread cooperation (through lock-free queues), and a set of architectural optimization strategies for getting the best performance from multicores (e.g., pinning/sticking of threads onto cores, blocking/non-blocking concurrency control, NUMA-aware mapping strategies of threads onto cores) will be delivered to address scenarios with strict throughput and latency requirements in services related to executing edge computing operations. Moreover, in the context of this task, FastFlow will leverage the high-level interface of the Multi-Transport Communication Library (MTCL), developed at UNIPI, to best fit the needs and protocol requirements of the NOUS project by integrating an extensive set of communication protocols and libraries (e.g., TCP/IP, MPI, UCX, MQTT, ZeroMQ, Kafka, etc.) to target different deployment scenarios transparently. In addition, the MTCL library will support communication fault tolerance and resiliency. T4.3. Federated Learning and Federated (Distributed) Inference framework [M18 - M36 | L: ITML, P: AIR, NCSRD, TIM, UNIPI, HPE, PPC, ARCTUR, AEGIS, INESC TEC]: This task will be responsible for delivering a technological breakthrough in FL approaches. It will build on ITML's "analytics-as-a-service" technology and will deploy a suite of algorithms for decentralised ML for user- and device-centric analytics. Algorithms to be deployed will run, both at training and inference time, on edge and beyond-edge devices (e.g., user personal devices and small size edge nodes), and generate models based on local data, which are then progressively refined through tailored collaborations between nodes. Local learning processes will be optimised to minimise the resource footprint on resource-constrained devices, particularly at learning time. FL will also be deployed to realize a hierarchical, algorithmic data and computation distribution layer that will offer intelligent resource allocation optimization recommendations between edge devices and cloud/fog data centres, and the respective decision-making systems (real time alerts and short-term decisions). The data and computation distribution will be able to instruct how to move data where needed satisfying application constraints. In case of non-real time constraints, it will target maximum guaranteed delay. Moreover, data distribution will optimise relevant device resource usage (e.g., energy consumption) with respect to current SotA solutions. T4.4. Interoperability in the Fog using a delegated network of edge mediators [M22 - M36 | L: UNP, P: USAL, NCSRD, ITML, UNIPI, ARCTUR, INESC TEC]: This task will develop an advanced interoperability solution for

seamless data sharing and exchange at the edge. The task will implement the mechanisms for dynamically enabling interoperability based on the Plug'n'Play Interoperability (Plug'n'Interoperate) concept to automate as much as possible, the configuration and participation of systems into an Interoperability environment. The task will provide the necessary interoperability specifications to enable interoperations between different heterogeneous systems (IoT and edge) in the NOUS environment, provide an interoperability mediator component to execute data translations directly at-the-edge (close to the point of origin of data) also using tools from T5.4 and information from T2.4, and implement a delegated network of edge mediators (considering human-in-the-loop as well) for distributed (Fog) interoperability.

Work package WP5 – #data - The data component of NOUS

Work Package Number	WP5	**Lead Beneficiary**	7. NET-INTRA
Work Package Name	#data - The data component of NOUS		
Start Month	1	**End Month**	36

Objectives
The specific objectives of this WP are i) to assess the sustainability of a Blockchain-based database in comparison with other existing solutions and provide a Blockchain Sustainability Index that will be usable in the future to qualify sustainability and scalability of blockchain-based databases, ii) to develop data services to enable efficient management and integration of existing and new Data Spaces, iii) to analyse the legal challenges related to blockchain used in the project and iv) to develop the auto-standardiser and data lake for translating data to specific standards in Mobility (≥1 standard), Energy (≥1), Green Deal (≥1).

Description
WP5 comprises the following tasks: T5.1. Legal evaluation of Blockchain technologies [M1 - M16 | L: KUL, P: AIR, USAL, POLITO, NET-INTRA, ARCTUR, INESC TEC]: Considering the objectives envisaged by the European Commission's blockchain strategy, this task will conduct a legal assessment of technical possibilities combining blockchain technology and the cloud. It will examine the relevant legal frameworks that apply to such technologies (e.g., GDPR, the Directive on Free Flow of Non-Personal Data, the NIS2 Directive on cybersecurity and the DA proposal impacting smart contracts), considering relevant sectors and investigating the potential frictions between those frameworks and DLTs (e.g., under the GDPR as set out in section 1.2.3). Moreover, EU-relevant initiatives will be taken into consideration, such as the EU Blockchain Observatory and Forum and the Regulatory sandbox for blockchain. In cases where consumer protection is in question, EU consumer law on digital fairness will be consulted. Overall, this task will aim at providing a clear legal and regulatory framework for blockchain technology in the context of Data Spaces. T5.2. A Blockchain-based framework for cloud and edge storage [M6 - M36 | L: POLITO, P: AIR, AETHON, USAL, TIM, NET-INTRA, KUL, PPC, CS]: The objective of this task is to establish a technological layer (PoC) utilizing a digital ledger/blockchain to track all the data throughout its entire lifecycle within a dataset. The layer will be based on the framework of T2.4 created to ensure transparency and monitoring the entire data's lifecycle, including adherence to legal requirements (especially GDPR), connection with data spaces (DSSC), permissions for internal and external use, economic characteristics, and compatibility with licenses and other agreements. Initially the task will determine the suitability of using a low-energy distributed ledger (e.g., Ethereum 2.0, tangles/IOTA) to accomplish these goals and the issues identified in the use-cases leading to proper parameterization of the deployed technology. defining consensus algorithms (e.g., PoW, PoS), smart contract languages (e.g., Solidity) and oracles / sidechains to interact with data external to the ledger. The solution will be tested in use-cases, compared to existing ones and assess its sustainability and scalability, including environmental and cost. The aim is to ensure free flow of data to customers while maintaining complete control over it, guaranteeing data's high quality and safety. A Sustainability Index for assessing the sustainability and scalability of databases based on blockchain technology will be presented. T5.3. Data spaces, data pools and data warehouses interconnection over an operational NOUS [M8 - M36 | L: NET-INTRA, P: AIR, TIM, ITML, UNIPI, KUL, HPE, PPC, ARCTUR, INESC TEC]: The main objective of this task is to create the necessary data services to enable efficient management and integration of existing and new Data Spaces supported by the exchange of data assets and applications. It will employ existing open-source solutions and technologies, as well as the respective standards (e.g., semantic languages such as OWL and RDF triples for semantic ontologies

modelling data; binary formats such as Apache Avro, Parquet and ORC; ETL tools such as Apache Camel; open linked data publication via CKAN, etc.), to offer capabilities such as data/application sharing, discovery, and data-relative functionalities such as data processing, data governance, and orchestration. More specifically, the data services will support the implementation of a set of mechanisms to allow different modalities of data access (e.g., online streaming, batch download, etc.) and transfer (bulk transfer and data streaming) and integration capabilities to merge data coming from various sources or Data Spaces. Additionally, the task will define the methodology and process of integration of Data Spaces, including the procedure that will be followed to reach a consensus on common needs and required features of each data space. This task, in collaboration with T7.2, will provide common data storage places for use by all partners in the project.

T5.4. The Auto-Standardiser tool for translation to data standards [M20 - M36 | L: AETHON, P: TIM, NET-INTRA, ITML, UNP, HPE, PPC, CS, INESC TEC]: The task aim is to create the Auto-Standardiser (AS). The AS is created for users to switch to data standards by translating their existing databases to a data standard. Database fields/column names should be translated to standards fields' names through an algorithm that takes into consideration linguistics matters such as grammar (e.g., synonyms, similar words) and language (i.e., translation). Furthermore, the AS will allow the incorporation of human-in-the-loop with a supervised ML component. When the main algorithms cannot match fields between input database and standard, human input will be requested. The interface for input request will allow the incorporation of additional information (e.g., the best matches of fields will be displayed with information on the data standard's field). While the user goes through all the issues with matching fields, a supervised learning component will monitor user inputs to improve its matching sequence. The software component will be built with common programming languages, e.g., Python, Java, Go, it will be published as open-source and it will be connected to a data lake component that will be used as a staging area for uploading data in files to be translated by the AS.

Work package WP6 – Towards the creation of the NOUS architecture

Work Package Number	WP6	**Lead Beneficiary**	2. AETHON
Work Package Name	Towards the creation of the NOUS architecture		
Start Month	6	**End Month**	36

Objectives
The specific objectives of this WP are i) to design the architecture of the NOUS platform in such a way that it harmoniously integrates the components of cloud computing, edge and distributed storage, ii) to investigate and design a framework based on Trustworthy AI guidelines for data and model management at the Edge and Cloud layers, iii) to publish the platform architecture and the different components in an open way so that they can be reused by researchers, developers and integrators and iv) to assess legal mechanisms impacting cloud and edge technologies and iv) to develop the collaboration platform (Virtual Lab) for exchanging insights, knowledge and interests on data analytics and science.

Description
WP6 comprises the following tasks: T6.1. Design and develop a high level architecture that brings #compute, #edge and #data together [M6 - M36 | L: NET-INTRA, P: AIR, AETHON, NCSRD, UNIMORE, TIM, ITML, UNIPI, UNP, KUL, HPE, ECL, PPC, ARCTUR, AEGIS, CS]: The primary objective of this task is to provide the main layout of the NOUS functional blocks, interfaces, and operations, creating an IaaS/PaaS/MLaaS architecture that follows the European Interoperability Reference Architecture (EIRA) viewpoints and building blocks and the European Interoperability Framework (EIF) suggestions, incorporating the #compute, #edge, and #data NOUS components that WP3, WP4, WP5 will deliver. The task will generate the main principles that the NOUS architecture will be built upon, using input from T2.1, T2.2 and T2.4. To efficiently follow and support the identified principles, the envisioned architecture will be composed of multiple layers (data services layer, administration services layer, infrastructure and connectors layer, governance layer, etc.), combining the various NOUS building blocks with existing open-source solutions and cloud orchestration and automation tools. Consequently, the task will develop effective and efficient mechanisms following standard DevSecOps practices and technologies to enable the development, integration, and operation of the individual NOUS components, and support the automated setup and runtime management of the multi-layer #compute, #edge, and #data services. A first release of the architecture is expected in M18, facilitating the interoperability of the NOUS components and updated following the testing in T7.2.

T6.2. Utilising and delivering open-source software and open architecture with and for the NOUS [M6 - M36 | L: ECL, P: AIR, AETHON, NCSRD, NET-INTRA, ITML, UNIPI, UNP, PPC, ARCTUR]: This task will set up the framework and guidelines for good open-source practices to effectively manage NOUS contributions to the European Open-Source Cloud to Edge continuum. This includes contributions to relevant existing open-source projects as well as the contribution of new NOUS assets to the open-source community. The task will define IP and license management guidelines for inbound and outbound IP to ensure that the open-source project results will be sustainable, open and exploitable. Further, this task will implement open-source governance processes and infrastructure (e.g., Eclipse Research Labs) to foster commitment and provide the basis for the creation of a vendor-neutral NOUS open-source community from the very beginning of the project. This task is responsible for managing project's repositories where open-source code and documentation will reside.

T6.3. Cloud and edge legal challenges [M16 - M36 | L: KUL, P: AETHON, NET-INTRA, PPC, INESC TEC]: This task will explore the current and prospective legal and regulatory challenges arising from the cloud and edge deployment and operation. By doing so, the task seeks to assist the European Commission towards its work for the creation of the 'EU Cloud rulebook' and the work of the EU Alliance on data, cloud and edge. It will analyse the foreseen role of the NOUS in the cloud-data ecosystem, based on the notion of 'neutral facilitator' where it operates as a 'digital utility'. It will also contour its role in the Data Spaces ecosystem considering the new EU Data Laws (DGA, DA, DSA, DMA as analysed under 1.2.3) and in strengthening the EU's digital sovereignty. The impact of satellite legal frameworks (e.g., the GDPR, Act on Artificial Intelligence) will also be examined. Potential data access obligations for IoT data will be considered, in light also of the use cases. The task will give emphasis on privacy and how to ensure data-related rights, such as data portability. Its connection to interoperability will be analysed, not only for personal data but also industrial, IoT and public sector data. New rights such as 'a right to switching' between cloud and edge computing service providers will also be explored. Key self-regulatory initiatives such as the EU Cloud Code of Conduct for Data Protection and the Code of Conduct on data portability and cloud switching will be considered.

T6.4. A Virtual Lab for data collaboration [M20 - M36 | L: AETHON, P: USAL, UNIMORE, TIM, HPE, F6S, PPC, INESC TEC]: The task will deliver a web application that will have the functionalities of: posting questions and answers (like in Stack Overflow), attaching link to data (and authentication information) and proper categorization of posts. Functionalities for the platform will be collected during T2.2 and correlated in this task with existing research on Virtual Labs (e.g., see MobiDataLab in ch. 1.2.3). The platform will be implemented in AETHON's Virtual Lab which will be extended to facilitate the new requirements leading either to an extension of the existing platform's functionalities. One important extension is the creation of a plugin that implements a query interface. The interface will include a chatbot/voicebot interface. This interface will facilitate interaction with users and will require the creation and training of Natural Language Processing (NLP) model using data from existing Virtual Lab implementation. The model will consider language specificities and cultural background of users. In collaboration with T7.1, the initial content required to attract traffic to the collaboration platform will be created.

T6.5. Tools for Syndication between NOUS instances and other platforms [M24 - M36 | L: AIR, P: AETHON, USAL, NCSRD, TIM, NET-INTRA, UNP, KUL, ECL, ARCTUR]: The NOUS architecture will be open and accessible to everyone through source code repositories such as GitHub. As such, it is expected that various organisations will replicate and implement it. This is warranted and the project aims to facilitate in many ways. First, consider how and why NOUS instances will communicate to each other and generate tools to facilitate it, also considering the EU Cloud Rulebook. Second, by means of the establishment of cooperation agreements to other European projects (T8.4) and the creation of a technological framework for the interactivity between NOUS and other cloud-related initiatives such as Gaia-X, FIWARE and EOSC (as defined in T2.1). The requirements for the connections will be reviewed (from T2.2), e.g., APIs and what architecture (REST, GraphQL etc.), open and linked data exchange (data virtualisation, ingesting, queueing, use of open-source technologies such as CKAN, etc.), user authentication (OAuth 2.0, eIDAS etc.), security concerns. This will extend the architecture of T6.1 and provide a set of tools that will be published openly.

Work package WP7 – The data economy and use-cases

Work Package Number	WP7	**Lead Beneficiary**	1. POLITO
Work Package Name	The data economy and use-cases		
Start Month	18	**End Month**	36

Objectives

The specific objectives of this WP are i) to create a quantification framework for data economy assessment that will act as a tool for policy makers and professionals working in Data Spaces, ii) to perform the evaluation of the project given the framework of T2.3, iii) to execute the testing of the technologies of the project through the use-cases, iv) to develop the roadmap for the extension of the NOUS and the guidelines for NOUS adoption.

Description
WP7 comprises the following tasks: T7.1. The Economy of Data - A quantification framework for data economy assessment [M18 - M36 | L: POLITO, P: USAL, NCSRD, TIM, NET-INTRA, ITML, UNIPI, UNP, KUL, HPE, PPC, ARCTUR, CS, INESC TEC]: The task will deliver a methodology based on the Lean Business GUEST Methodology by POLITO (POLITO) for assessing the value of data in terms of storing, sharing, and usage. We adopt the GUEST to guide the entire process. GUEST is already proven to be easy-to-understand by people with limited knowledge of innovation processes and to let them to be involved in the process without the need of a long learning period. We want to overcome the issues of the state of the current methodologies by specializing the GUEST for incorporating the business development of the economy of Data and understanding the sustainability (operational, economic, social, and environmental) of the NOUS' outcomes on users and stakeholder. Thus, the task will provide a specialized GUEST-based methodology able to make the process of data economy quantification measurable and replicable, as already done for the Blockchain-based use cases. Through a harmonization of the results of T6.4 and the use cases, the task will request NOUS partners to pose a question or a question/solution or some other form of content (e.g., articles) in the collaboration platform of T6.4 about a data analytics/science subject in the Data Spaces of Energy, Mobility, Green Deal, Tourism, Cultural Heritage, and Sciences. In particular, the results of T6.4 will provide a viable interface for the identification of data-patterns that occurs in the different applications in order to reduce costs and risks connected to the data-usage. T7.2. Setting up the use-cases and testing the developed technologies [M18 - M33 | L: NCSRD, P: AIR, AETHON, POLITO, UNIMORE, TIM, NET-INTRA, ITML, UNIPI, UNP, HPE, F6S, PPC, ARCTUR, AEGIS, CS, INESC TEC]: This task is responsible for coordinating testing of technologies, achieving the objectives of the use-cases as expressed in Tables 3, 4, 5, 6, aggregate data from the use cases for the testing of technologies and updating of requirements following the testing. The task will start with a plan that will elaborate the use case descriptions presented in ch. 1.2.2 and provide detailed descriptions for them as well as plan out how to achieve the objectives of the use-cases considering the technologies developed in WP3, WP4, WP5, WP6. This task will also be a liaison between use case providers and other consortium partners for collecting the data required by technology providers. The task participants (which are also technology developers in said WPs) will adjust and adapt the developed technologies to address the objectives of the use-cases. This adaptation as well as the demonstration of the end-result constitutes the testing of the technologies. Each technology provider is responsible for adapting their technologies to the requirements of the use-case as discussed in 1.2.2. For usage of positioning data, earth observations with satellites, navigation and timing data, usage of Copernicus and Galileo/EGNOS is envisioned by use case providers. The start of the testing will be signified by a workshop with all participants to discuss the expectations for the testing. A set of questionnaires (created in T2.3 and improved here) and interviews will be conducted with testing participants to test the acceptance of developed technologies, determine new requirements arisen and measure KPIs, leading to updating D2.1. Technical feedback for tested technologies (e.g., bugs) will be collected and provided to T2.4, T3.3, T4.2, T4.3, T4.4, T5.2, T5.3, T5.4, T6.1 and T6.4 for improvements. T7.3. Evaluation and assessment [M27 - M36 | L: UNIMORE, P: USAL, POLITO, NET-INTRA, F6S]: This task will implement the evaluation methodology defined in T2.3 by using the data gathered during the testing in T7.2. Derived data will be used to measure KPIs that cannot be measured with data coming from testing. As these KPIs are at least as important as the directly measurable ones, effort must be taken to calculate these values as exact and reproducible as possible. Finally, the economic impact will be addressed through a Cost-Benefit Analyses. T7.4. Roadmap and guidelines enabling the future of European Cloud that fosters data sharing [M30 - M36 | L: AIR, P: AETHON, NCSRD, POLITO, KUL, HPE, F6S, CS]: This task aims to generate roadmap for the further development of NOUS and generate a set of guidelines addressed towards the EuroHPC network for utilising the NOUS architecture in their offering. The roadmap will receive input from T2.4, T6.1, T6.5 (this three together represent the high-level architectural outcome of the project), and the testing of the technologies executed in T7.2 as well the assessment of T7.3. The roadmap will provide an overview of new services and technologies to be added in NOUS and new user groups to address, thus, constituting a holistic overview of the next steps for NOUS. The guidelines aim to increase the uptake of the NOUS solution by demonstrating how the NOUS outcomes fit into the overall map of European tools and initiatives regarding cloud technologies, data sharing and how they can fit to the business model of EuroHPC. The roadmap is an input for guidelines since adoption will be executed within a timeframe of NOUS evolution.

T7.5. Fostering human-centred practices for a data-agile economy [M18 – M36 | L: INESC TEC, P: AIR, AETHON, NCSRD, POLITO, UNIMORE, NET-INTRA, ITML, KUL, F6S, AEGIS, CS, INESC TEC]: This task aims to establish human-centred practices within a data-agile economy. This involves recognizing the vital role of humans in the loop, ensuring that their needs and perspectives are considered when designing and implementing data-driven solutions as for defining the high-level architecture (T6.1). Special emphasis will be placed on the social sciences and humanities, which provide important insights into human experience and can help ensure that data solutions are ethically responsible and equitable. The novel data life cycle framework to be established at T2.4 will foster human-centred practices, resulting from a deep understanding of how people interact with technology and data, gathered from T7.2. This involves engaging experts in the social sciences and humanities, with practitioners in data science, engineering, and related fields. As a result, a human-centred approach to data-driven innovation is expected, that empowers individuals and fosters placing humans at the centre of the generation of data. This task also expects to provide input for the data life cycle framework (T2.4).

Work package WP8 – Communication, dissemination, exploitation and liaison

Work Package Number	WP8	**Lead Beneficiary**	14. F6S IE
Work Package Name	Communication, dissemination, exploitation and liaison		
Start Month	1	**End Month**	36

Objectives
The specific objectives of this WP are i) to disseminate NOUS technologies to the scientific community, the edge computing communities & groups, and the general stakeholders, raise awareness about the project and boost the exploitation activities, ii) to perform the appropriate communication activities to spread knowledge in various stakeholder communities and maximize the project's visibility, iii) to manage the preparation, planning, and execution of exploitation of the project's results and achievements, iv) to draft a sustainability plan for the maintenance of the project's results following its completion and v) to monitor the market environment relevant to the work performed in the project in order to develop an efficient exploitation plan for the NOUS technologies

Description
WP8 comprises the following tasks: T8.1. Communication, dissemination, exploitation, innovation plans & strategies [M1 - M6 | L: F6S, P: ECL]: This task will identify, define, coordinate, and organize the activities aiming to deliver the NOUS Communication, Dissemination, Exploitation and Innovation Plans & Strategies. It will deal with the following issues in particular: (i) Drafting of a Dissemination & Communication Plan, which will be updated in regular intervals, identifying events of interest and detailing the appropriate communication mechanisms and procedures to be followed by the consortium; (ii) Exploitation strategy: establish a continuous "market watch" and develop a process that enable key consortium members to analyse & explore the market needs and opportunities; (iii) Innovation management: conducting studies of market trends, opportunities and new technological achievements, and update the NOUS innovation strategy in the annual management reports; (iv) Collaborate with T8.4 to kick-start project collaboration with relevant initiatives such as Eclipse CloudDevTools, EdgeNative, IoT Working Groups, to place NOUS in the European Open-Source Cloud to Edge continuum landscape. T8.2. Communication and dissemination implementation and monitoring [M6 - M36 | L: F6S, P: All]: This task will implement and monitor the activities associated with the promotion of the NOUS project and its results/ community. The task will deal with first, Project Communication and Promotion: conducting promotional activities for the dissemination of project work and achievements (examples include, but are not limited to, the production of project brochures, project videos, the project web presence, social media accounts, etc.). Second, Dissemination and participation in project-related events: the consortium will be informed regularly about events that may be worth attending (workshops, national and international events/ conferences, coordination with external organizations, special interest groups and communities, technology driven conferences, etc.). Then, individual organizations, a combination of partners or the entire consortium will participate, representing the project or individual parts thereof, demonstrating the project's innovations and seeking feedback from the presentations to advance the project's results. Third, Assisting and organising project workshops: the task will assist with the organisation of the (second) workshop of T2.1, T7.2 and T8.4 (the first workshop of T2.1 and the workshop in T2.2 are internal and will be handled by respective task leaders). This includes invitation of non-

project partners to the workshops, monitoring and feedback management. Fourth, Deployment of monitoring processes: ensure the achievement of proposed KPIs and identification of best practices and improvement measures. This task is responsible for organising the final event that will take place at around a major cloud event as discussed in ch. 2.2.1 by the coordinator.

T8.3. Exploitation and innovation management, implementation and monitoring [M8 - M36 | L: F6S, P: All]: This task will generate individual and joint plans for the exploitation of the project's results, white papers about technology evolution in the field and tackle IPR management. Business cases will be monitored continuously and used as input to draft the Business Models, providing useful quantitative and qualitative evaluation elements and considering the scientific and industrial perspective of results. A standard methodology, such as Canvas Model, will be used (determination of target groups, value proposition, partners, key resources). A strategy will be decided to exploit NOUS' results a whole but plans for individual exploitable products will also be drawn up for enabling individual partners to utilize their part of the work as separate products, services, or potential plug-in modules to existing systems. This task will collaborate with T7.4 in real time; while T8.4 views adoption from the EuroHPC perspective (among others), this task takes a much broader view considering many user groups. During this task, the Innovation Manager will interact with the partners engaged in techno-economic analysis as well as with the scientific and technical manager, end-users and industry stakeholders to ensure the project's innovation and long-term sustainability. Regular reviews of each partner's innovation activities and updates of the business plan are activities within this task. Apart from the internal innovation monitoring, NCSRD will undertake the task to monitor the European and International innovation created in the topics of the proposal (i.e., annual patent reports; strategies to minimize the risk of a patent infringing; findings that may support future further develop plans and business concepts). NCSRD will provide two white papers (at the mid and end of NOUS, within updates of D8.1) with an analysis and summary of all potential technologies in the field. These documents will be presented during the events organized in T8.4 and will influence the business plan as well as the roadmap (T7.4).

T8.4. Clustering, networking, synergies with other projects & international cooperation [M1 - M36 | L: UNIMORE, P: AIR, NCSRD, TIM, NET-INTRA, ITML, UNP, HPE, ECL, F6S, PPC, ARCTUR, CS]: The purpose of this task is to build links with other projects relevant initiatives and partnerships within the AI, Data & Robotics domain to (1) maximise synergies, (2) offer high visibility for the NOUS results, (3) develop collaborations as well as (4) exchange knowledge and expertise with similar research initiatives. The activities covered are: (i) networking activities, (ii) joint research meetings, including two online workshops, one with project partners and beyond to discuss preliminary findings leading to the generation of a press release (see Milestone 22) and another with many stakeholders to present project results (see Milestone 23) and 2 live or online workshops with project partners and DSSC, (iii) exchange of knowledge and results, (iv) improvement of technical and socio-environmental analysis of the results as well as new possibilities for scalability, (v) collaboration on data collection, (vi) joint research papers, (vii) creation of a common web site, (viii) invitations of researchers from other activities to give talks, (ix) launch of common special issues and conference sessions, (x) launch of a common brochures and newsletters, and (xi) alignment with commercial & business accelerator programs to support uptake of project outcomes from NOUS, the engagement of early-adopters and attract next step investors. Initial targets for synergising activities are Gaia-X, FIWARE, DSSC, EOSC, DOME (revised in T2.2 and T8.1).

STAFF EFFORT

Staff effort per participant

Grant Preparation (Work packages - Effort screen) — Enter the info.

Participant	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	Total Person-Months
1 - AIR	29.00	23.00	4.00	4.00	7.00	20.00	14.00	12.00	113.00
2 - AETHON	12.00	12.00			24.00	30.00	16.00	5.00	99.00
3 - USAL	2.00	12.00	4.00	5.00	2.00	9.00	5.00	2.00	41.00
4 - NCSR "D"	3.00	12.00	25.00	9.00		6.00	25.00	10.00	90.00
5 - ICOOR	2.00							3.00	5.00
5.1 - POLITO	1.00				20.00		29.00	1.00	51.00
5.2 - UNIMORE	1.00	8.00				7.00	12.00	9.00	37.00
6 - TIM	3.00	5.00	6.00	3.00	6.00	3.00	10.00	5.00	41.00
7 - NET-INTRA	3.00	2.00		2.00	23.00	29.00	8.00	4.00	71.00
8 - ITML CY	2.00	3.00	4.00	35.00	5.00	6.00	11.00	3.00	69.00
9 - UNIPI	2.00	2.00	1.00	24.00	3.00	3.00	6.00	2.00	43.00
10 - UNPARALLEL	2.00	2.00	1.00	18.00	4.00	5.00	5.00	8.00	45.00
11 - KU Leuven	3.00	2.00			17.00	17.00	4.00	2.00	45.00
12 - HPE Hellas	3.00	6.00	17.00	3.00	6.00	5.00	12.00	3.00	55.00
13 - ECL	2.00					13.00		14.00	29.00
14 - F6S IE	2.00	2.00				6.00	5.00	35.00	50.00
15 - PPC	3.00	5.00	10.00	5.00	6.00	10.00	10.00	4.00	53.00
16 - ARCTUR d.o.o.	2.00	2.00	19.00	8.00	6.00	7.00	6.00	3.00	53.00
17 - AEGIS	2.00	1.00	22.00	3.00		4.00	7.00	2.00	41.00

Staff effort per participant

Grant Preparation (Work packages - Effort screen) — Enter the info.

Participant	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	Total Person-Months
18 - CS GROUP-FRANCE	3.00	5.00	4.00		3.00	1.00	12.00	3.00	31.00
19 - INESC TEC	2.00	5.00	3.00	6.00	5.00	4.00	23.00	2.00	50.00
Total Person-Months	84.00	109.00	120.00	125.00	137.00	185.00	220.00	132.00	1112.00

LIST OF DELIVERABLES

Deliverables

Grant Preparation (Deliverables screen) — Enter the info.

The labels used mean:

Public — fully open (⚠ automatically posted online)
Sensitive — limited under the conditions of the Grant Agreement
EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444

Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
D1.1	Interim report	WP1	1 - AIR	R — Document, report	SEN - Sensitive	18
D1.2	Final report	WP1	1 - AIR	R — Document, report	SEN - Sensitive	36
D1.3	Data Management Plan	WP1	2 - AETHON	DMP — Data Management Plan	SEN - Sensitive	5
D2.1	Status quo, gap analysis, requirements and project evaluation framework	WP2	4 - NCSR "D"	R — Document, report	PU - Public	9
D2.2	A data –life-cycle framework	WP2	1 - AIR	R — Document, report	PU - Public	34
D3.1	Connecting to the HPC Network	WP3	16 - ARCTUR d.o.o.	R — Document, report	PU - Public	12
D3.2	An interface between Quantum Computers and HPC	WP3	4 - NCSR "D"	OTHER	PU - Public	28
D4.1	Distributed architectures	WP4	8 - ITML CY	R — Document, report	PU - Public	12
D4.2	The NOUS Federated Inference framework, Interoperability in the Fog and Structured parallel environments	WP4	8 - ITML CY	OTHER	PU - Public	30
D5.1	Legal assessment and recommendations on blockchain technologies	WP5	11 - KU Leuven	R — Document, report	PU - Public	16
D5.2	Blockchain-based database storage	WP5	1 - POLITO	OTHER	PU - Public	34

Deliverables

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Deliverable No	Deliverable Name	Work Package No	Lead Beneficiary	Type	Dissemination Level	Due Date (month)
	component of NOUS and the Auto-Standardiser and Data services					
D6.1	The NOUS Architecture and Open-Source development platform (1st version)	WP6	7 - NET-INTRA	OTHER	PU - Public	18
D6.2	The NOUS Architecture and Open-Source development platform (2nd version)	WP6	13 - ECL	OTHER	PU - Public	36
D7.1	Use cases testing and project assessment	WP7	4 - NCSR "D"	R — Document, report	SEN - Sensitive	33
D7.2	Project evaluation, Roadmap and Best practices for data economy	WP7	2 - UNIMORE	R — Document, report	PU - Public	36
D8.1	Plan for dissemination and exploitation including communication activities	WP8	14 - F6S IE	R — Document, report	SEN - Sensitive	6
D8.2	Collaboration and synergies with other projects & initiatives	WP8	1 - AIR	R — Document, report	PU - Public	20

Deliverable D1.1 – Interim report

Deliverable Number	D1.1	**Lead Beneficiary**	1. AIR
Deliverable Name	Interim report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	18	**Work Package No**	WP1

Description
Detailed technical and financial execution (including risk management, prevention / corrective mitigation measures applied) of the project (T1.1, T1.2, T1.3)

Deliverable D1.2 – Final report

Deliverable Number	D1.2	**Lead Beneficiary**	1. AIR
Deliverable Name	Final report		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	36	**Work Package No**	WP1

Description
Detailed technical and financial execution (including risk management, prevention / corrective mitigation measures applied) of the project (T1.1, T1.2, T1.3)

Deliverable D1.3 – Data Management Plan

Deliverable Number	D1.3	**Lead Beneficiary**	2. AETHON
Deliverable Name	Data Management Plan		
Type	DMP — Data Management Plan	**Dissemination Level**	SEN - Sensitive
Due Date (month)	5	**Work Package No**	WP1

Description
Periodically updated protocols, formats, and roles responsible for the data lifecycle (T1.4).

Deliverable D2.1 – Status quo, gap analysis, requirements and project evaluation framework

Deliverable Number	D2.1	**Lead Beneficiary**	4. NCSR "D"
Deliverable Name	Status quo, gap analysis, requirements and project evaluation framework		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	9	**Work Package No**	WP2

Description
Report on the status and gaps in current implementations (T2.1). Elicited requirements, results of data collection organised for the NOUS architecture (T2.2). The evaluation framework (T2.3) and the surveys will be presented.

Deliverable D2.2 – A data –life-cycle framework

Deliverable Number	D2.2	**Lead Beneficiary**	1. AIR
Deliverable Name	A data –life-cycle framework		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	34	**Work Package No**	WP2

Description
A framework for data life-cycle management with standardisation aspects (T2.4).

Deliverable D3.1 – Connecting to the HPC Network

Deliverable Number	D3.1	**Lead Beneficiary**	16. ARCTUR d.o.o.
Deliverable Name	Connecting to the HPC Network		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP3

Description
A report demonstrating how the interactivity between HPC and NOUS can take place (T3.1).

Deliverable D3.2 – An interface between Quantum Computers and HPC

Deliverable Number	D3.2	**Lead Beneficiary**	4. NCSR "D"
Deliverable Name	An interface between Quantum Computers and HPC		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	28	**Work Package No**	WP3

Description
Demonstration of the component for QC and HPC interfacing and an investigation on implementation of Quantum Machine Learning (T3.2).

Deliverable D4.1 – Distributed architectures

Deliverable Number	D4.1	**Lead Beneficiary**	8. ITML CY
Deliverable Name	Distributed architectures		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	12	**Work Package No**	WP4

Description
A report on the distributed architectures created in T4.1.

Deliverable D4.2 – The NOUS Federated Inference framework, Interoperability in the Fog and Structured parallel environments

Deliverable Number	D4.2	**Lead Beneficiary**	8. ITML CY
Deliverable Name	The NOUS Federated Inference framework, Interoperability in the Fog and Structured parallel environments		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	30	**Work Package No**	WP4

Description
The deliverable will present methods and tools for initiating structured parallel programming environments (T4.2), the Federated Learning tool of NOUS (T4.3) and the outcomes of T4.4 being interoperability at the edge and presentation of the mediator component.

Deliverable D5.1 – Legal assessment and recommendations on blockchain technologies

Deliverable Number	D5.1	**Lead Beneficiary**	11. KU Leuven
Deliverable Name	Legal assessment and recommendations on blockchain technologies		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	16	**Work Package No**	WP5

Description
A report with the legal challenges and recommendations for blockchain technologies from T5.1.

Deliverable D5.2 – Blockchain-based database storage component of NOUS and the Auto-Standardiser and Data services

Deliverable Number	D5.2	**Lead Beneficiary**	1. POLITO
Deliverable Name	Blockchain-based database storage component of NOUS and the Auto-Standardiser and Data services		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	34	**Work Package No**	WP5

Description
The blockchain storage component and the Sustainability Index, will be presented (T5.2) as well as the tool that allows automatic translation to data standards and its Machine Learning component (T5.4). Report on the created services for Data Spaces interconnectivity (T5.3).

Deliverable D6.1 – The NOUS Architecture and Open-Source development platform (1st version)

Deliverable Number	D6.1	**Lead Beneficiary**	7. NET-INTRA
Deliverable Name	The NOUS Architecture and Open-Source development platform (1st version)		
Type	OTHER	**Dissemination Level**	PU - Public

Due Date (month)	18	**Work Package No**	WP6

Description
The NOUS architecture presenting elements to create an IaaS/PaaS architecture as developed in T6.1. The last version includes guidelines and tools to assist NOUS interoperate with other EU platforms (see T6.5). The framework and guidelines manage NOUS contributions as open-source updated through time (see T6.2).

Deliverable D6.2 – The NOUS Architecture and Open-Source development platform (2nd version)

Deliverable Number	D6.2	**Lead Beneficiary**	13. ECL
Deliverable Name	The NOUS Architecture and Open-Source development platform (2nd version)		
Type	OTHER	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP6

Description
The NOUS architecture presenting elements to create an IaaS/PaaS architecture as developed in T6.1. The last version includes guidelines and tools to assist NOUS interoperate with other EU platforms (see T6.5). The framework and guidelines manage NOUS contributions as open-source updated through time (see T6.2).

Deliverable D7.1 – Use cases testing and project assessment

Deliverable Number	D7.1	**Lead Beneficiary**	4. NCSR "D"
Deliverable Name	Use cases testing and project assessment		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	33	**Work Package No**	WP7

Description
The report will provide an overview of the testing executed and collected feedback from T7.2.

Deliverable D7.2 – Project evaluation, Roadmap and Best practices for data economy

Deliverable Number	D7.2	**Lead Beneficiary**	2. UNIMORE
Deliverable Name	Project evaluation, Roadmap and Best practices for data economy		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	36	**Work Package No**	WP7

Description
Methodology for data of value assessment per T7.1 workings. Results on the project evaluation from T7.3 and the roadmap and guidelines developed in T7.4. Report in the loop best practices for human in the loop for data-agile economy.

Deliverable D8.1 – Plan for dissemination and exploitation including communication activities

Deliverable Number	D8.1	**Lead Beneficiary**	14. F6S IE
Deliverable Name	Plan for dissemination and exploitation including communication activities		
Type	R — Document, report	**Dissemination Level**	SEN - Sensitive
Due Date (month)	6	**Work Package No**	WP8

Description
Dissemination & Communication plans, strategy and implemented activities (update) (T8.1, T8.2). Report on the exploitation plan, business models and innovation management (including IPR) in order to maximize impact including the analysis (white papers, M24/36) of potential technologies (SWOT) (T8.3)

Deliverable D8.2 – Collaboration and synergies with other projects & initiatives

Deliverable Number	D8.2	**Lead Beneficiary**	1. AIR
Deliverable Name	Collaboration and synergies with other projects & initiatives		
Type	R — Document, report	**Dissemination Level**	PU - Public
Due Date (month)	20	**Work Package No**	WP8

Description
Proceedings of joint research meetings and conference sessions, online networking workshop presentations and video, common website, joint publications and common data repositories. (T8.4) Updated on M36.

LIST OF MILESTONES

Milestones *Grant Preparation (Milestones screen) — Enter the info.*					
Milestone No	**Milestone Name**	**Work Package No**	**Lead Beneficiary**	**Means of Verification**	**Due Date (month)**
1	Constitution of Steering and Technical Committees	WP1	1-AIR	KOM minutes	1
2	Yearly General Assembly conducted (I)	WP1	1-AIR	Progress meeting minutes and presentations	12
3	Yearly General Assembly conducted (II)	WP1	1-AIR	Progress meeting minutes and presentations	24
4	Workshop with project partners to analyse technologies, gaps and requirements	WP2	1-AIR	Meeting minutes, presentations, pictures	3
5	Workshop with project partners and other stakeholders to analyse technologies, gaps and requirements	WP2	1-AIR	Meeting minutes, presentations, pictures	5
6	Status quo, gap analysis, requirements and project evaluation framework validated	WP2	1-AIR	D2.1 submitted (including report on the status and gaps in current implementations; elicited requirements, results of data collection organised for the NOUS architecture; evaluation framework and surveys). Consortium meeting held. Validation by at least eight external experts (Technical Advisory Board).	11
7	Preliminary data life cycle framework developed	WP2	1-AIR	Preliminary version of the D2.2 (data –life-cycle framework) developed and validated by at least 8 external experts (Technical Advisory Board).	20
8	Access to HPC network mapped	WP3	4-NCSR "D"	Report demonstrating how the interactivity between HPC and NOUS can take place (D3.1) published openly in the project website and repositories.	12
9	Interface between HPC and QC delivered	WP3	4-NCSR "D"	D3.2 submitted, including (i) Demonstration of the component for Quantum computing and	28

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
				HPC interfacing (code published in repositories; video available publicly in social media); (ii) Investigation on implementation of Quantum Machine Learning conducted (at least 2 research papers published).	
10	NOUS Cybersecurity component delivered	WP3	4-NCSR "D"	Report on the cybersecurity in NOUS using a combination of tools with darknets and Zero Trust models published openly in the project website and repositories (and at least 2 research papers published).	30
11	Analysis of distributed network and start of WP4 technology development	WP4	8-ITML CY	D4.1 submitted: report on the analysis on distributed architectures developed, validated by at least 8 external experts (Technical Advisory Board) and presented to the consortium.	12
12	Federated (Distributed) Inference framework developed	WP4	8-ITML CY	Federated Learning tool of NOUS (T4.3) developed over ITML's "analytics-as-a-service" technology, including a suite of algorithms for decentralised ML for user- and device- centric analytics (at least 2 research papers published reflecting these algorithms and experimental results).	30
13	Legal analysis and recommendations developed	WP5	7-NET-INTRA	A report with the legal challenges and recommendations for blockchain technologies in the context of Data Spaces from T5.1 published openly in project website and repositories (at least 1 review paper published).	16
14	Connectors with existing Data Spaces developed and tested	WP5	7-NET-INTRA	Data services to enable efficient management and integration of existing Data Spaces supported by the exchange of data assets and applications	15

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
				developed: connectors developed and validated, gathering and publishing data with at least 2 existing Data Spaces.	
15	First version of the open-source development platform openly published	WP6	2-AETHON	Framework and guidelines for good open-source practices to effectively manage NOUS contributions to the European Open-Source Cloud to Edge continuum validated by at least 8 external experts (Technical Advisory Board), presented to the consortium and published openly in the project website and repositories.	18
16	Virtual Lab online and available	WP6	2-AETHON	Virtual Lab accessible online and validated by at least 8 external experts (Technical Advisory Board). A report presenting the collaboration and the process leading to its creation openly published in the project website and repositories.	30
17	Data collected from use-cases providers and testing plan ready	WP7	1-POLITO	First version of report including an overview of the testing executed and collected feedback from T7.2 elaborated. Non-sensitive data will be published openly in project website and repositories. Further update will be done at M33.	26
18	Start of testing of the developed technologies in the use cases (Workshop)	WP7	1-POLITO	Meeting minutes, presentations, pictures.	27
19	End of testing of the developed technologies in the use cases (Meeting)	WP7	1-POLITO	Meeting minutes, presentations, pictures.	33
20	Project website publication, kick off social media campaigns, social influencing	WP8	14-F6S IE	Website accessible including partners, objectives and workplan. Social media profiles public. First newsletter and posts in social media delivered.	3

Milestones

Grant Preparation (Milestones screen) — Enter the info.

Milestone No	Milestone Name	Work Package No	Lead Beneficiary	Means of Verification	Due Date (month)
21	Online networking workshop with other projects and initiatives within the AI, Data & Robotics celebrated	WP8	14-F6S IE	Presentations, videorecording of meetings, posts on social media.	18
22	Joint research meeting with consortium partners, experts and other projects and initiatives to discuss preliminary findings and results celebrated	WP8	14-F6S IE	Meeting minutes, presentations, pictures.	20
23	Joint research meeting with stakeholders and other projects and initiatives to present final results celebrated	WP8	14-F6S IE	Meeting minutes, presentations, pictures.	35

LIST OF CRITICAL RISKS

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
1	Difficulties managing a wide and heterogenous consortium.	WP1	Consortium partners have proven experience coordinating and participating in European (FP7, H2020, HE, PRIMA) projects which ensure a smooth cooperation. A Steering and a Technical Committee will conduct online monitoring meetings twice a month. A project Balanced Scorecard will be shared with all the partners to monitor SMART objectives and KPIs.
2	A partner leaves the consortium.	WP1	Strong commitment of the partners in the project. Partners have a well-proven wide cross-sectoral expertise that allows for the redistribution of tasks should the need arise.

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
			Partner networks (e.g., EDIHs, associations) would guarantee that all tasks are conducted successfully
3	Low acceptance of proposed NOUS frameworks by industrial and research users.	WP2	Both the evaluation and the data life-cycle frameworks will be assessed and validated by at least 8 external advisors from European universities, research centres and data industry forming the Technical Advisory Board. Perform additional iterations of proposed frameworks
4	Difficulties to develop a scalable HPC-QC data encoding interface software.	WP3	Algorithm optimization will be performed on the HPC. Offload approaches such as OpenMP will be adopted and noninterpreted approaches directly in C/C++ will be followed.
5	A darknet overlay designed on a zero-trust security model can hinder productivity and efficiency.	WP3	Execute a variety of Zero Trust trials and security evaluations before releasing the framework to production environments. Establish secure endpoints among the involved tools/modules. Design and follow a gradual approach; identify the most sensitive and critical workflows and then scale the deployment. Use CI/CD pipelines to ensure data and user permissions and controlled fine-tuning.
6	Lack of accuracy of ML and FL models at the edge.	WP4	Verified expertise of partners in several edge intelligence-driven R&I initiatives. Partners have tested and verified a number of algorithms at the edge, ensuring thus a solid starting point for the project. Enhance the algorithm development with new data, open or paid.
7	Blockchain PoC is complicated to implement is scale.	WP5	Proper Blockchain optimisation will be executed through partners that have significant expertise in the field. The existing installation of Hyperledger by AETHON will be considered. Blockchain-as-a-Service will be considered.
8	Lack of standardization during the interconnection of new data spaces.	WP5	Partners responsible for the interconnection of the identified data spaces will ensure that the provided data comply with well-established industry standards. Common data vocabularies will be employed to harmonize and curate the identified data sources.
9	Potential changes in legislation to cause delays in analyses.	WP5, WP6	Continuous monitoring of changes in legislation. Monitoring of proposals for regulations as well and incorporation in the study. Amendment and delays in deliverables' submission to account for all changes in legislation.
10	Interoperability/integration challenges during the platform architecture design and implementation.	WP6	The platform architecture will be designed with interoperability in mind from the outset. NET-INTRA will utilize their CI/CD integration stack to thoroughly test, validate in multiple iterations, and further develop where needed the individual integration points

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
			between different components before the overall platform deployment.Custom integration and middleware systems will be developed.
11	Tools for syndication do not cover all connections with other platforms.	WP6	Given the amount of platforms that the project can be syndicated with, it is possible that some requirements of certain platforms are not covered. Connections will be prioritised as follows, between NOUS instances, with Gaia-X, EOSC, FIWARE and others. Utilise external tools (open-source code) for the connection.
12	Use cases cannot provide data due to privacy or business concerns.	WP7	Early planning will allow the determination of most issue related to data. Amendment to the objectives of the use-case without changing the final results (testing of defined technologies). Synthetic data will be used.
13	Different requested timelines of partners for testing.	WP7	Partners creating technologies need to coordinate with T7.2 leader to execute testing in a timely manner so that use-cases' providers have high availability. Testing will be executed in bunches risking unavailability of use-cases' providers personnel.
14	Low participation in multi stakeholders workshops.	WP8	F6S will utilise their digital platform for promoting the workshops. Project partners with significant communities will be requested to assist in promotion of events (e.g., ECL, HPE). The events will be rescheduled to achieve the expected amount of participants.
15	Data (partially) unavailable for testing phase.	WP7	Public datasets will be identified within T5.3 from the M8 to train ML models and simulate use cases scenarios in the first stages of setup. Data connectors with European Data Spaces will be available in M15 (MS14). The partners (e.g. AIR) have expertise in synthetic data generation techniques for use in different scenarios.
16	Difficulties to deal with large-scale testing due to simultaneous constraints.	WP7	First version of the NOUS Architecture and Open-Source development platform available from M18. "Working Backwards" methodology. Coordination of testing phase in a specific workshop involving partners and end-users. Use of synthetic data and emulated data/services connectors in case some of the components are delayed or some data sources are not yet available.
17	Risk of not reaching a financial equilibrium with the solution developed	WP8	"Working Backwards" methodology. Stakeholder involvement from the very beginning (T2.1, T2.2). Auto-Standardiser approach to reduce integration costs. Developments strongly based on open-source technologies (e.g., IOTA, CKAN). The business plan will continuously feed back into T6.4 (Virtual laboratory for data collaboration), T7.1 (Economic data

Critical risks & risk management strategy

Grant Preparation (Critical Risks screen) — Enter the info.

Risk number	Description	Work Package No(s)	Proposed Mitigation Measures
			framework) and T7.3 (Cost-benefit analysis) and identify the least cost-effective components that may be discarded for the initial business plan.

PROJECT REVIEWS

Project Reviews *Grant Preparation (Reviews screen) — Enter the info.*			
Review No	**Timing** (month)	**Location**	**Comments**
RV1	19	Luxembourg or Videoconference	
RV2	36	Luxembourg or Videoconference	

A catalyst for European Cloud Services in the era of data spaces, high-performance and edge computing

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1. History of changes

HISTORY OF CHANGES		
VERSION	PUBLICATION DATE	CHANGE
1.0	29.03.2023	NOUS proposal submitted to the European Commission for evaluation.
1.1	07.09.2023	Formatting issues: removing the cover page, deleting the header, replacing the footer, removing the list of participants, and adding a table of contents with page numbers.
1.2	07.09.2023	Removing tables 3.1a (list of work packages), 3.1b (work packages descriptions), 3.1c (list of deliverables), 3.1d (list of milestones), 3.1e (critical risks for implementation) and 3.1f (summary of staff effort) from section 3.1.
1.3	07.09.2023	Creating the section 'ethics self-assessment'.
1.4	07.09.2023	Reducing the number of deliverables and balancing them between the two reporting periods (M1-M18 and M19-M36).
1.5	07.09.2023	Updating the milestone list (information uploaded in the F&T Portal).
1.6	08.09.2023	Following the comment in the ESR *"The proposal fails to consider appropriate dependencies between work packages and tasks. For example, the testing phase of the project (T7.2) is planned to last for two months at the very end of the project (M30-M32). But the proposal does not explain in sufficient detail how to ensure implementation compliance at this later stage"*. The testing phase has been extended to span M27-M33 (MS18 and MS19, respectively). Moreover, further explanation has been included in the methodology, deliverables and milestones description.
1.7	08.09.2023	Following the comment in the ESR *"The proposal fails to clarify when individual architectural components are developed with respect to their use in the implementation of the initial use cases, e.g. the description of the safe component ensuring computation capabilities before cybersecurity is delivered in T3.3 from M18 to M33"*, further explanation has been included in the methodology (Section 1.2).
1.8	08.09.2023	Three new risks identified and included along with mitigation measures and contingency plans in the project risk table (information uploaded in the F&T Portal, #15, #16, #17).
2.0	11.09.2023	New version of the document with clarifications required by the Project Officer uploaded in the F&T Portal.
2.1	13.09.2023	New version of the document with minor corrections required by the Project Officer uploaded in the F&T Portal.

2. Glossary

BC	Blockchain	**EIRA**	European Interoperability Reference Architecture
DLT	Distributed Ledger Technologies	**EIF**	European Interoperability Framework
IaaS	Infrastructure as a Service	**MLaaS**	Machine Learning as a Service
IoT	Internet of Things	**PoC**	Proof-of-Concept
PaaS	Platform as a Service	**PaaS**	Platform as a Service
DFD	Data Flow Diagram	**SaaS**	Software as a Service
ML	Machine Learning	**HPC**	High-performance Computing
NLP	Natural Language Processing	**SSH**	Social Sciences and Humanities

1 Excellence

1.1 Objectives and ambition

European Cloud is at a negative where 75% of the European cloud market is dominated by non-EU players which are bound to laws and regulations of foreign powers, such as PR China and USA, generating legitimate concerns for the EU, its businesses and citizens. These players shape the way that Europe is deploying and using cloud services. However, EU's digital future resides on having secure, high-quality data processing capacity. This can only be offered by cloud services both centrally and at the edge. This project's ambition is completely in line with the European Strategy for data as it aims to create the foundations for a European Cloud Service which exploits the HPC network and tackles specific-to-the-EU-economy requirements:

1. It must **research and develop revolutionary cloud technologies**.
2. It must **facilitate the cloud-edge computing continuum and provision of edge computing services**.
3. It must **be conducive to data sharing and exchange**.

4. It must work towards **trustworthy data infrastructure, empowering EU's digital sovereignty**.

The project does not envision to create the cloud service with all the infrastructure expected by a cloud service such as hyper-verticalized providers (Microsoft Azure, Amazon AWS, Google GCP). It envisions to create a comprehensive blueprint for implementing a cloud service where each implementation is interconnected with other EU platforms. The project **aspires to be the technological leg of Europe's Next Generation Cloud**.

1.1.1 Detailed project objectives

The NOUS project objectives can be found in Table 1, followed by the explanation of pertinence to the call.

Table 1: NOUS Project Objectives

O1: To develop a technological connectivity framework between Europe's High-Performance Computing network and NOUS. To extend computational capabilities of NOUS by studying the seamless interaction of Quantum computers with the HPC network based on the NOUS use-case. [#compute]			
Key Achievements: 1 Architecture design and implementation of high-speed, error-free and secure data exchange and I/O communication between HPC with NOUS. | **2** Design and deploy an environment allowing the fast data retrieval from cloud, formatting them to be accessible by the quantum computer and implement the interface between NOUS and the quantum computer. More precisely regarding the interface, after organizing the classical data describing the case under study, these should be transformed to quantum states, and this is achieved by utilizing quantum circuits and a quantum channel. | **3** Design a quantum data encoding scheme according to the number of qubits available by the quantum computer, taking advantage of quantum superposition, classical large datasets will be simplified to a few superimposed qubits for processing in a quantum computer. | **4** Design the I/O interface of NOUS with a quantum computer. | **5** Design and develop a cybersecurity module for use by NOUS (and other cloud environments) for creating a secure network by researching the topics of darknets and Zero Trust security model.			
Indicators of Achievements: Achieve operational high-speed, secure and error-free data exchange between HPC and a (simulated) cloud environment (NOUS). Determine requirements for using HPC nodes as the computational component of a cloud infrastructure using information from interviews with HPC nodes. Development of an environment allowing easy, fast and secure exchange of (big) data between NOUS and a quantum computer. Develop the quantum data encoding scheme and demonstrate loading of data and encoding to qubits. Determine and explore the capability of using Quantum Machine Learning in collaboration with the Federated Learning system of Objective. Design the I/O interface and define requirements for the NOUS architecture. Design and develop the NOUS cyber-security component as a standalone software to be used in cloud networking environments. Demonstrate the cybersecurity software's capabilities using a cloud environment.			
Partners	**NCSRD, ARCTUR, AEGIS, HPE, PPC, ITML**	**WPs**	**WP3, WP2, WP7**
O2: To develop a data processing module used for fusion of data taken from physical ubiquitous assets and digital models while tackling computation distribution among the available pervasive devices in the Edge-to-Fog-to-Cloud (E2F2C) continuum. [#edge]			
Key Achievements: 1 Design and deploy a suite of SSH-aware algorithms for decentralised ML user- and device-centric. Algorithms run, both at training and inference time, on edge and beyond-edge devices (e.g., user personal devices and small size edge nodes), and generate models based on local data, which are then progressively refined through tailored collaborations between nodes. The objective of NOUS is to ensure that local learning processes will be optimised to minimise the resource footprint on resource-constrained devices, particularly at learning time. | **2** Deploy a hierarchical, algorithmic data and computation distribution layer that offers intelligent resource allocation optimization recommendations between edge devices and cloud/fog data centres, and the respective decision-making systems (real time alerts and short-term decisions). | **3** Deliver structured parallel programming environments, based on FastFlow, which will enable the operators to instantiate parallel applications as compositions of pre-defined, configurable parallel skeletons.			
Indicators of Achievements: A ML-as-a-Service framework comprising a suite of algorithms for decentralized ML that reduce resource footprint by 50% with respect to existing implementations and reduce learning time by 70%. The above will be verified through testing with the use cases. A load balancer optimising the operation of edge devices achieving 100% identification of non-real time constraints and ensuring maximum guaranteed delay at 95% of the cases.			
Partners	**ITML, UNIPI, UNP, USAL, ARCTUR, INESC TEC, PPC**	**WPs**	**WP4, WP2, WP7**
O3: To consider the implementation of Blockchain as a decentralised database that is connected to the #edge and compare it to classical storage models for the use-case of NOUS for standard applications (cloud-client) and IoT-powered applications. [#data]			
Key Achievements: 1 Analyse the impact of Blockchain-based solutions on trust among actors of the system and			

on (reduction of) risks connected to cyber threats. | 2 Compare the Blockchain-based solution to other centralised and decentralised solutions giving the technical and managerial insights on how to ensure the sustainability of the blockchain-based database (from the operational, economic, and environmental points of view especially including the way legislation, e.g., GDPR, is addressed by said solutions). | 3 Demonstrate the impact of the blockchain as a sustainable technological mechanism for data life-cycle management through testing of a proof-of-concept (PoC) for monitoring data accessibility, ownership and data lifecycle processes. 4 Create a framework for assessing Blockchain sustainability. | 5 Assess the legal factors that influence blockchain technologies on the cloud

Indicators of Achievements: Determine trends in tools and processes used in data life cycle management and determine the usability of Blockchain in data life cycle management. Determine the impact of Blockchain on trust of users through targeted interviews and/or questionnaires with professionals. Determine a list of actions for ensuring the sustainability of blockchain-based databases. Implement a Blockchain-based PoC for data life cycle (definition of ledger entries, smart contracts-based data policies, and interconnection of the blockchain with external data sources such as IoT sensors networks). Develop a full assessment framework as a specialization of one of the main Blockchain use-case methodologies. Compare the use case before/after the Blockchain solution introduction using Blockchain Sustainability index. Evaluate at least two legal frameworks, including the GDPR.

Partners	**POLITO, AETHON, NET-INTRA, AIR, TIM, CS**	**WPs**	**WP5, WP2, WP7**

O4: To design an extendable translation-to-standards concept using data lake and develop it as a software component for standards in the Mobility (NeTEx, GTFS), Energy (IEC, SAREF) and Green Deal (various CEN/CENELEC standards) Data Spaces. [#data]

Key Achievements: 1 Determine a methodological structure and develop algorithms (including an active Supervised Machine Learning component) used for the translation of diverse data sources to data standards. | 2 Develop the software for the translation of any dataset to a data standard, demonstrate its usage using specific standards across three Data Spaces and publish it as open-source. | 3 Consider and design the implementation of the software as a microservice published in a cloud interface (as the NOUS will be, this achievement will allow the NOUS architecture to use data lakes given the auto-standardiser practical example).

Indicators of Achievements: A state of the art review of data lake concepts, data standards for Mobility/Energy/Green Deal Data Spaces and harmonisation between standards in said domains. Generate one algorithm for the translation of a dataset to any of four selected standards (one per data space). Develop an implementation of a Supervised ML to allow learning during usage. Create a software that tackles the deployment and management of a local data lake, file handling and algorithm deployment. Test the software using data from use-cases, open or synthetic datasets. Design the architecture of connecting the software to NOUS as a core service (e.g., as MySQL is preinstalled and universally available for GCP, AWS, Azure and others). Define extensions to the NOUS architecture to support data lakes as a core service (e.g., as Blobs in GCP). Publish the translation algorithms as open-source.

Partners	**AETHON, NET-INTRA, UNP, UNIMORE, INESC TEC, TIM, CS**	**WPs**	**WP5, WP2, WP7**

O5: To develop the overall architecture of a European Cloud Service suite, combining #compute, #edge and #data and that can increase EU's digital sovereignty.

Key Achievements: 1 Evaluate applicable regimes such as EU data laws and the GDPR and analyse selected legal frameworks that impact cloud and edge technologies, as well as the specific necessities to protect the seven key requirements for solutions based on EU's Ethics Guidelines for Trustworthy AI (New Futurium, Futurium).: (i) human agency and oversight, (ii) technical robustness and safety, (iii) privacy and data governance, (iv) transparency, (v) diversity, non-discrimination and fairness, (vi) environmental and societal well-being and (vii) accountability. | 2 Analyse the current status and gaps in current implementation involving Cloud and Edge Computing, HPC, Quantum Computing, Internet of Things, Federated Learning, Human in the Loop, and Distributed Ledger Technologies, considering the differences between Europe and the rest of the world. | 3 A syndicated approach to NOUS deployment, focusing on interoperability between potential NOUS implementations (coming from the implementation of the open architecture) and with other initiatives (e.g., Gaia-X, EOSC, FIWARE). | 4 Design and develop an open architecture that integrates #compute, #edge and #data layers by means of open-source HPC, Cloud, Edge, IoT and ML frameworks. | 5 Develop and implement a strategy for open-source software and architecture, guiding the publication of the technological and technical project outcomes.

Indicators of Achievements: Cloud and legal challenges analysis and recommendations. Report on the current status and gaps in current implementations published openly. Functional and technical requirements along with results of data collection organised and classified for the NOUS architecture and per component. Project evaluation framework validated by at least eight external European advisors and including surveys used to assess project KPIs and impact. NOUS data life-cycle management framework following European regulations and ethical guidelines including interoperability with European Data Spaces and standardisation aspects. Novel open-source

IaaS/PaaS/MLaaS architecture presenting the functional blocks, interfaces, and operations. Tools and guidelines for the interoperation of the architecture with other EU platforms and between multiple NOUS implementations (with publication of tools). A large repository of open-source software. Public deliverables for the NOUS architecture. Engagement of the open-source community through newsletters, open workshops and guidelines.			
Partners	**ALL**	**WPs**	**WP6, WP2, WP7**
O6: To develop a collaboration platform, based on Living Labs concept and acting as a "Stack Overflow for data sharing", that will allow users to ask questions, exchange data analysis results, information on data analysis especially using cross-domain data.			
Key Achievements: 1 Implement and deploy a digital platform based on a Virtual Lab platform. This will utilise an existing platform (https://labs.mobidatalab.eu/) and extend its functionalities to allow a question-response type system (as in Stack Overflow or Quora) with the capability to relate the question to a dataset. | 2 Design query interfaces through chatbots and voicebots that facilitate interaction with users using Natural Language Processing (NLP) models that consider the language specificities and cultural background of the users. | 3 Identify specific use-cases with synergies in data analytics in the Data Spaces where NOUS project and its partners are working on (project – Energy, Mobility, Green Deal Data Spaces / partners – Tourism, Cultural Heritage). | 4 Elaborate a roadmap for the implementation of Virtual Labs in data sharing, especially cross-domain.			
Indicators of Achievements: More than 50 registered users in the platform. Requirements collected from >500 questionnaires and turned to functionalities in the Virtual Lab. Identification of at least 10 use-cases of synergies on cross-domain data sharing. Validation through complementary pilots using the use case providers. Increase of data sharing (predicted) using a Virtual Lab as a data sharing conductor. Measure the impact on the data economy through implementation and testing of the Virtual Lab with the use-cases' partners. Connect different databases (e.g., from other European projects such as MobiDataLab) and execute real-life use-cases of co-development and co-experimentation using the Virtual Lab as a catalyst.			
Partners	**AETHON, USAL, NCSRD, F6S, TIM, PPC, CS, HPE**	**WPs**	**WP6, WP2, WP7**
O7: To collaborate and find synergies with other initiatives and organisations working on European Data Spaces in order to facilitate the sharing of data within NOUS focusing on converging ideas around a common (or a set of common) European Cloud Services.			
Key Achievements: 1 Implement data analysis services running on top of NOUS and ingesting data from European Data Spaces (e.g., via DSSC), such as the Mobility Data Space or the future Language Data Space or the Cultural Heritage Data Space. New Data Spaces deployed partially or fully on top of NOUS storage and analysis services. | 2 Establish cooperation agreements between NOUS and organisations such as Gaia-X or Alastria (AIR Institute is a member of the Spanish data hub of Gaia-X; both AIR and USAL are members of Alastria). European and national research and development projects in different sectors (smart cities and territories, mobility, precision agriculture, language learning, heritage monitoring, Industry 4.0/5.0, climate change forecast, omics data analysis) making use of NOUS services for the deployment of data-driven use cases employing different digital enabling technologies (AI, Big Data, Robotics, IoT, Digital Twins and metaverses). | 3 Develop testbeds and proofs of concept developed on NOUS by European entrepreneurs, SMEs, research centres and universities as part of the services provided by the network of European Digital Innovation Hubs (AIR is the coordinator of the EDIH DIGIS3, with agreement with a network of more than 50 EDIHs).			
Indicators of Achievements: At least 4 European Data Spaces (Mobility, Energy, Green deal, Manufacturing) linked to NOUS (e.g., data consumed by services deployed via NOUS, use of NOUS as storage service provider by some data space, etc.). Collaboration with at least 6 Horizon Europe / LIFE / PRIMA projects and at least 18 national projects within the European Union to validate proofs of concept involving the use of services deployed via NOUS. Collaboration agreements with at least 24 EDIHs for the implementation of proofs of concept on NOUS by linked SMEs in at least 12 European countries.			
Partners	**ALL**	**WPs**	**WP8, WP2**

Pertinence to the scope of the call: The call HORIZON-CL4-2023-DATA-01-02, requests to *"address the entire data life cycle from data generation/collection to the final use and disposal/deletion of data"*. Addressing this will require first a good overview of existing data life cycle management techniques, which is tackled by studying how use cases' partners handle this and by literature review. The project will look into practical solutions for data life-cycle management (Blockchain and comparison with existing technologies – **O3**), standardisation (**O4**) and a communication channel for using, exploiting and sharing information on data (**O6**). The call also requests to build *"on existing standards, models and architectures"*. This is tackled by the project through the creation of the NOUS architecture (**O5**), the use of FastFlow in edge computing research (**O2**), the utilisation of the HPC network for the

computational aspect of NOUS (**O1**) while the creation of the auto-standardiser as open-source (**O6**) provides an extendable tool for others to build on standards easier while looking into standards harmonisation. The call also requests the *"consideration of cybersecurity issues and analysing the use and re-use potential, especially in view of use of data across sectors"*, with cybersecurity tackled directly in **O1** with the creation of a tool and (re-)use potential of data supported fully by the collaboration platform of **O6**. For the request to *"address relevant human language issues at all stages of data life cycle, addressing the social and cultural factors as necessary"*, the project tackles this directly in the auto-standardiser (**O4**) where language issues are truly relevant during fields matching and in the collaboration platform (**O6**) where the project will research query interfaces facilitate interaction with users using NLP models. The project's cloud backbone will be defined to be able to tackle *"human-generated and human-related data"* while *"seamless integration of 'human in the loop'"* is tackled in the process of the auto-standardiser that includes a supervised learning module (**O4**), the structured parallel programming environments that can be controlled and initiated by operators (**O2**), while the collaboration platform promotes human-centric approaches to problem solving enhanced with algorithms and data reusability (**O6**). **O7** of the project is dedicated to achieving "collaboration with existing projects and develop synergies with other relevant European, national or regional initiatives, funding programmes and platforms". The request to *"include use cases or pilots addressing or involving at least three different common European Data Spaces and/or related ecosystems"* is tackled in the project since it includes 4 use-cases each tied with a different data space (Mobility, Energy, Green Deal and Manufacturing). The project is also linked *"with the Data Spaces support centre"* through its partners being members to BDVA (NET-INTRA, ITML, NCSRD), Gaia-X (AIR, TIM, ECLIPSE, CS GROUP) and directly through KUL who is a member of DSSC while the values and models of DSSC will be included in the analysis for the data economy and the project will provide to DSSC new tools for achieving its objectives. For the request on addressing interoperability for data sharing *"in line with the European Interoperability Framework (EIF)"*, the project will create a study, as part of **O3** and **O4**, for determining the principles and guidelines of EIF and European Interoperability Reference Architecture (EIRA). The project will also tackle the call's request for *"enabling cross-domain data sharing and data markets"* through the collaboration platform (**O6**), the auto-standardiser that eases the exchange of data through standards (**O4**), the #edge component that will enable federated learning across edge nodes (**O2**) and the Blockchain-based data life cycle component that facilitates transparency being the precursor to data sharing. The call's demand that *"envisaged architectures and systems should enable correct allocation and enforcement of data-related rights, obligations and responsibilities across the life cycle*" will also be tackled through identifying and analysing those rights, obligations and responsibilities from a legal perspective.

1.1.2 Vision, Ambition and Main Challenges

The vision of NOUS is to provide a uniquely suitable infrastructure and services to scientists and professionals for running computationally intensive and complex algorithms on a commercial cloud interface. According to the Destination challenge of the call, "75% of the European cloud market is dominated by non-EU players". This argument indirectly pushes for innovations that will assist European economy to move away from non-EU players' technologies. This project works towards this direction. However, the core of the NOUS idea is that instead of calling for an end to utilisation of non-EU based cloud technologies e.g., by French President Emmanuel Macron and Euclidia initiative among many, a new paradigm for cloud infrastructure must be put forward. The vision of NOUS came to be given three main challenges:

3. **Utilize the network of existing HPCs and Quantum computers from different providers for high-performance computations that allow solving extraordinarily complex problems** (e.g., climate change forecasting, the synthesis of new drugs or the creation of new, more energy-efficient materials) The importance of this challenge lies in delivering a cloud service built for the European economy; a cloud service that foster excellence in science and the development of highly performant digital services.
   Our European economy is focused on "Excellence in Science". The HPC network was created in that direction however, the major issue is how to assist scientists convert their ideas from a research project to a commercial product. This is an endeavour for any scientist and various organisations in Europe already provide non-technological tools to assist in this transition. The NOUS project will tackle the technological aspect for the above while providing an HPC-QC interface that can greatly enhance computational capabilities while assisting programmers who need to consider the unique architecture of any specific QC operating during the NISQ era.
4. **Foster and promote edge computing services.** The NOUS project will build towards the next generation Cloud-Edge-IoT in Europe's technological base. This is in line with the destination challenges being the strengthening/transition to a computation (and storage) continuum that is better distributed between central points (i.e., cloud) and near-edge/far-edge nodes.
   The edge computing challenge is well founded in the economy and society and how it progresses. Devices, from smartwatches to cars, are as well-equipped as Personal Computers. Their computational capabilities are increasing yet they are only used for basic functions. Organisations need to have better tools to utilise these devices for complex, computationally intensive and storage related tasks (e.g., ML model training and

executing). This is a resource-allocation problem, nodes must collaborate efficiently to each other and with the cloud, while respecting users' data privacy The NOUS tackles the entire spectrum in that regard.

5. **Connect with dataspaces and provide a seamless data flow within, towards and from a European Cloud Service.** As with previous challenge, a main goal for the EU economy is the creation of a single data market ensured through common Data Spaces (see Staff working document on data spaces). It comes as a natural challenge working towards increasing the EU capacity to process vast amounts of data, to connect and/or facilitate the creation of Data Spaces. NOUS will be a conductor for the establishment of the single data market. While Data Spaces are in their infancy and not well defined from the technological aspect, Data Spaces "An infrastructure that enables data transactions between different data ecosystem parties based on the governance framework of that data space. Data space should be generic enough to support the implementation of multiple use cases" (DSSC Glossary). NOUS project will make strides towards this direction both in the technological as well as the business scopes, assisting in the convergence of dataspaces. A vision that can be manifested during the project is that NOUS can become the technological scope of EU vision on dataspaces.

The challenges were brought forth by the multidisciplinary team of the project and drove the proposal of NOUS. One question that the consortium tackled with early on, is why create an architecture when other initiatives are also attempting to do the same. This question refers to Gaia-X, FIWARE and EOSC and it is central to the NOUS idea.

First, NOUS project argues that a common cloud architecture is missing. Each of the aforementioned initiatives are making strides, some more than others, in bringing data sharing (Gaia-X), open standards (FIWARE) and scientific collaboration (EOSC) to the European economy. None of these initiatives aims to assist European (and other) cloud providers in the core of their business which is offering cloud services. **A cloud architecture on the other hand, can and will assist cloud providers to progress their services and will assist emerging providers.**

Second, there are distinct differences with each initiative. Firstly, both EOSC and Gaia-X are platforms for data sharing offering SaaS/PaaS and IaaS (Gaia-X). Moreover, Gaia-X distinctly states that the "outcome will not be a cloud.". EOSC on the other hand is a platform with a specific scope to assist researchers to share and use data and services. Similar context-specific cloud initiatives can be found such as, ECCCH. FIWARE, on the other hand, drives the definition of key open standards for delivering extendable, transferable and interoperable solutions. NOUS is complementary to the above aiming to function as the connecting infrastructure for all said initiatives.

1.1.3 Advances beyond the state-of-the-art

HPC cloud computing and Edge cloud computing: Cloud computing has become a means of consolidating computing functions, storage and network management centrally and, as result, private and public platforms have undergone a rapid growth [1]. This is because this paradigm facilitates the development of distributed computing systems, data management and computing resources through a scalable network, data processing centres and web services [2]. Unfortunately, architectures based exclusively on a centralized cloud experience many difficulties because they cannot meet some requirements, like the Internet of Things (IoT) paradigm, of current mobile Internet applications, nor can they support the massive volume of data these applications generate. Over the last few years there has been a huge increase in solutions that decentralise communications, data collection and processing, bringing all these computing and storage tasks from the Cloud to the Edge [3]. This trend has led to the emergence of the Edge Computing paradigm [4], which is based on technologies that enable computing at the edge of the network. In this way, computing and network resources are closer to the data source than to the data centres in the Cloud [5]. The Edge Computing paradigm allows to improve the performance of computing systems by reducing latency, reducing the cost of resources and increasing responsiveness, scalability, reliability, security or privacy [6].

Cloud providers such as Amazon Web Services, Microsoft Azure or Google Cloud offer pricing plans based on the amount of customer resources for a certain period of time [7]. Cloud pricing plans are quite complex to describe, but, without going into too much detail, cloud providers charge for computing resources (CPU usage in terms of the number of cores multiplied by the clock frequency in GHz multiplied by the usage time in s, as well as RAM in terms of GB-s), storage and database resources (in terms of GB-month, as well as total read and write operations per month), as well as network resources (in terms of amount of incoming and outgoing traffic), among others that currently include advanced services, such as Machine Learning as a Service (e.g. Natural Language Processing APIs) or highly specialised services (e.g. tensor processing units or TPUs). In this way, Edge Computing can help reduce the amount of data traffic moving to and the amount of data stored in the Cloud and, more importantly in terms of cost, the computing resources and data traffic moving out of the Cloud [8]. NOUS will go beyond the state-of-the-art advancing the following topics.

Distributed Registration Technologies: This type of technology has shown some of its potential, materialising in some of the most revolutionary applications of recent years, Bitcoin and Ethereum. Distributed log technologies or DLTs (blockchain and directed acyclic graphs (DAGs) as the most widespread examples, as well as other implementations) base their reliability and robustness on the intelligent use of different combined technologies (encryption, timestamping, P2P, PoW/PoS), designed in such a way that, when well implemented, they are seamless.

This provides a technological infrastructure for the exchange of digital assets with the following characteristics: security, reduced operating costs, immediacy, transparency, and transparency. Although there are multiple options and their focus of interest is on cryptocurrencies, their application has spread to many other sectors and has given rise to a wide variety of businesses and implementations, which base their competitive advantages on this type of technology: marketplace transactions [9], distributed storage [10], property registration, traceability and logistics [11], etc.). In the same way that there are multiple end applications, these technologies are also offered as a service, to be able to implement a solution of this type, without the need to develop the entire infrastructure, i.e., they provide the capabilities of a distributed registry, on which to design an API to build the desired applications. This type of proposal reduces costs and development time but means that the registry is hosted on external nodes, limiting the possibilities for research into this type of technology, as well as the ad-hoc design of the internal mechanisms and characteristics that define the chain or network.

Federated Learning: The emphasis on privacy has become a force majeure around the world forcing large companies to compromise on data security and confidentiality. Federated Machine Learning proposes a new approach, based on a decentralised artificial intelligence model that allows machine learning to take place on local devices in real time, without compromising their functionality and without having to share their data [12]. It is collaborative learning that consists of creating a global model trained with computed updates on local devices while maintaining users' data and thus ensuring their security and privacy. Local models are aggregated and contribute to a global model, but as there is no need to share the raw data, confidentiality is guaranteed. Moreover, by training locally on user data, it offers a high degree of personalisation, while reducing latencies and the cost of continuously exchanging data with a server. Privacy is one of the essential properties of federated learning. This requires security and analytics models that provide meaningful privacy guarantees. Among the privacy techniques for federated learning is Secure Multiparty Computation (SMC) which aims to guarantee complete zero-knowledge, i.e., each party knows only its input and output, but this desired property often requires complicated computational protocols and may not be efficiently achieved. In some cases, partial knowledge disclosure may be considered acceptable if security guarantees are provided. It is possible to build a security model with SMC with lower security requirements in exchange for efficiency, one of the most advanced SMC frameworks being Sharemind, although 3PC [13] models have also been proposed with an honest majority considering security under both semi-honest and malicious assumptions. On the other hand, there are differential privacy [14], k-anonymisation [15] and diversification [16] methods for privacy protection, which consist of adding noise to the data or using generalisation methods to hide certain sensitive attributes until the third party cannot distinguish the individual, thus making the data impossible to restore in order to protect the user's privacy. Additive homomorphic encryption [17] is widely used and polynomial approximations are needed to evaluate non-linear functions in machine learning algorithms resulting in a trade-off between accuracy and privacy.

Natural Language Processing: Natural language processing (NLP) employs computational techniques for the purpose of learning, understanding, and producing human language content. Computational linguistic systems can have multiple purposes: The goal can be aiding human-human communication, such as in machine translation; aiding human-machine communication, such as with conversational agents; or benefiting both humans and machines by analysing and learning from the enormous quantity of human language content that is now available online. Early computational approaches to language research focused on automating the analysis of the linguistic structure of language and developing basic technologies such as machine translation, speech recognition, and speech synthesis [18]. Today's researchers refine and make use of such tools in real-world applications, creating spoken dialogue systems and speech-to-speech translation engines, mining social media for information about health or finance, and identifying sentiment and emotion toward products and services. The field has increasingly relied on data-driven computation involving statistics, probability, and machine learning. Recent increases in computational power and parallelization, harnessed by graphical processing units (GPUs) [19], now allow for Deep Learning (DL) which utilizes artificial neural networks (ANNs), sometimes with billions of trainable parameters. In addition, the contemporary availability of large data sets, facilitated by sophisticated data collection processes, enables the training of such deep architectures [20]. Deep learning has been successfully applied to image retrieval (also known as text to image), in which query and image are first transformed into vector representations with CNNs, the representations are matched with DNN and the relevance of the image to the query is calculate. DL is also employed in generation-based natural language dialogue, in which, given an utterance, the system automatically generates a response, and the model is trained in sequence-to-sequence learning [21].

AI software algorithms across the edge continuum: Edge computing allows to shift part of the computation to the local resources, providing better privacy, latency, and power efficiency. However, running AI on the edge poses many challenges. Training of a DL-based model at the edge usually takes advantage of several software-related approaches that accelerate training, such as transfer learning and distributed/federated learning. Knowledge distillation techniques can also be used to derive a smaller model from a larger pre-trained model. The design and training of DL models that can provide optimized Explainable Artificial Intelligence (XAI) and human-interpretable decisions is crucial for the widespread use of AI in edge-enabled scenarios. For inference at the edge, the key

challenges to come to more efficient software paradigms that apply smart, selective processing are twofold: 1) designing SoC and hardware accelerators that support event-based processing of data elements (e.g., spiking neural networks), or 2) designing alternative model architectures that better support smart, selective processing of data elements during inference on current streaming-based processing hardware (CPU and GPU). For this, exploration of transformed-based architectures is still in its infancy and requires research and exploration of realistic use cases. The potential benefit of transformer-based architectures is that they do not consider the spatial dependency between data elements in their processing paradigm. Where the kernels of a CNN encode a spatial dependency between nearby data elements, i.e., a grid, transformers encode spatial dependency by adding an extra dimension to the data elements. This allows transformers, contrary to CNNs, to process data elements in arbitrary spatial orders. Thus, the usage of transformer-based architectures at the edge can potentially lead to 1) being able to run larger deep AI models efficiently on edge devices, or 2) running deep AI models with higher frequencies or higher resolutions on edge devices. Moreover, the design of algorithms for automatically optimizing the trained deep models, based on network pruning and quantization, is crucial to deploying DL-based models at the edge, using low-cost devices with limited computational power.

Training at the edge: Edge training is remarkably different from traditional training procedures which typically are positioned on powerful servers, computing clusters or in the cloud. The traditional approach is faster by having access to high computing resources, however, faces multiple challenges with the level of security, the risks of data confidentiality, reliability, latency, and even so-called backdoor attacks [22]. Edge training typically relies on strictly limited power resources, therefore training architecture, acceleration and optimisation are the main challenges of edge training. Currently, two training architectures have been designed: solo training and collaborative training. It is challenging to build upon the collaborative training approaches, towards distributed collaboration approaches, with great potential for communication and distributed learning across edge devices, thus improving the training and communication efficiency, privacy, security and by expanding the approach to multi-agent platforms. The limited computing capability needs to be especially tackled by training acceleration through a multi-agent platform and the efficiency by which the cross-modal transfer through agents is performed. Currently, in state-of-the-art systems the communication efficiency (communication frequency and communication cost) is limiting the full exploitation potential of training at the edge. For one of the most typical edge training techniques, federative learning, communication frequency is the most important operation, which needs to be improved not only by addressing the stated issues but also by exploring and developing new edge AI training architectures, and acceleration/optimisation methods. Training of deep AI models in the industry is mainly performed using fully supervised training requiring many manual annotations of data samples. There are many disadvantages to this and using weak or self-supervision has many benefits to improving or adopting a model that is pre-trained using labelled data. Specifically, techniques like unsupervised domain adaption (UDA) [23], leveraging self-training on self-generated pseudo labels and student-teacher training, can significantly increase task performance for a specific target domain without requiring extra human-labelling effort. A challenge to be addressed when deploying state of the art UDA models at the edge is that such models are extremely compute-intensive and currently are only applicable to server-grade hardware. The goal is thus to come to significantly less compute intensive solutions to UDA that can run at the edge. In federated learning, the need to share raw data is prevented but edge devices receive manual input for fully supervised learning.

AI middleware to support AI-based data analytics at the edge: Edge applications real-time decision-making exploiting information from distributed components in an IoT ecosystem (e.g., deployed sensors, gateway devices, HPC/cloud). The capability of orchestrating computation and resources in a distributed environment needs support from a middleware layer capable of abstracting low-level hardware and OS details while providing applications with a set of common APIs and functionalities. In edge computing scenarios, middleware solutions such as Eurotech's Everyware Software Framework and similar, provide simplified data analytics and data management on field-deployed IoT gateways and computers. From the robotics viewpoint, middleware such as ROS is largely used to communicate and orchestrate among components. From a general AI perspective, middleware software has been developed in different contexts such as distributed linear algebra, multi-agent systems (MAS), games AI, dialogue systems, and robotics. Examples of distributed engines for linear algebra are Apache Mahout and Weka. OpenCog is a more general open-source software for AGI (Artificial General Intelligence) that is for cognitive systems in general, with strong applications also on robotics. The AI middleware are focused on easing application development, with little emphasis on runtime resource optimisation considering the complexity of heterogeneous architectures that may be equipped with different flavours of AI accelerators. Despite recent advances, such as DECI, there is still a gap to fill to make existing AI middleware efficiently exploit edge computing resources.

HPC-QC interface: Quantum computing (QC) systems will not replace existing computer architectures; they more likely augment them by accelerating certain suitable tasks or kernels. The computation of new kernels and work needed for I/O and workflow management will (at least for the foreseeable future) remain bound to the existing computer approaches. Additionally, quantum computing relies on several compute-intensive tasks, which require support from HPC systems. Consequently, QC must seamlessly become part of HPC, enabling common user access and experience. To enable the needed integration, it will be essential to not only develop QC hardware and physically

connect it to HPC systems, which is currently the focus for several groups with approaches ranging from loosely coupled cloud or modular access to near quantum compute options [24] to actual deep integration [25] targeting low-latency access. Combining these two worlds, however, is not trivial, as they work on radically different assumptions. The HPC world is trimmed for low latency and pre-compilation; on the other hand, QC systems require a much larger runtime component. The latter is caused by compiling at runtime as input arguments are likely to change the intended quantum program (i.e., the generated circuits). Further, the computation needed to compile and map to the final topology of the QC system is substantial. Currently, enabling HPC software stacks exist and developed in the framework of EU supporting instruments. Examples for this are HPC enabled stacks, such as OpenHPC, the E4S initiative, or the DEEP-SEA stack developed for the European exascale systems, as they are used in most large-scale HPC centres. These include state-of-the-art components for compilers, runtimes, parallel programming abstractions including MPI and OpenMP support for accelerators following the evolution of system architectures, as well as a wide range of libraries to support efficient application development. The most natural way of integration is to allow HPC systems to push workloads to the QC system, referred as "offloading mechanisms" in congruence with mechanisms in other accelerators. Such an approach requires intensive computations to achieve the needed optimizations and mapping to the underlying qubit topologies on the quantum devices. Ultimately, this will form the needed bridge between the two worlds and enable efficient HPCQC operations. Recently, Finnish research and technology organisation VTT announced a milestone when, in collaboration with CSC – IT Center for Science and Aalto University, it connected the quantum Finnish 5-qubit QC HELMI with the pan-European supercomputer LUMI. NOUS will push the HPC-QC interface beyond the current SotA targeting to realize an interface that will be scalable meaning that the interface software is independent on the number of qubits of the QC. Furthermore, the proposed HPC-QC interface software package is inspired by the fact that supercomputers could help optimise quantum algorithms and they can post process measurement data to mitigate errors in the computation. As quantum computers get larger, these optimisation problems get more challenging. NOUS will address this issue as well as the data encoding before send to QC.

1.1.4 Positioning of the project in terms of R&I maturity

Project results are expected to reach between TRL 4/6. Various components will be researched individually allowing them to reach a higher TRL level while the incorporation of all components into the architecture, which is the spearhead of project's outcomes, allows for an almost hands on consideration of technology incorporation: the NOUS architecture is not only a thought exercise but a practical methodology to interconnect disparate technological components into a singular offering.

Table 2: Assets of partners directly used and advanced in NOUS (Asset - A, Partner - P, Technology Readiness Level - TRL)

A	Distributed Ledger Technologies for transparency and monitoring of the data lifecycle	P	USAL	TRL	4 → 6
Technology description: Distributed Ledger Technologies or DLTs (e.g., blockchain) base their reliability and robustness on the intelligent use of different combined technologies (encryption, timestamping, P2P, PoW/PoS), designed in such a way that, when well implemented, they are seamless. This provides a technological infrastructure for the exchange of digital assets with the following characteristics: security, reduced operating costs, immediacy, transparency, and transparency. However, legal factors, such as the correct application of accountability, and of data subject rights are difficult to implement in blockchains existing legal essential requirements for them for data sharing and interoperability, **Use/Improvement/Adaptation:** The project will use distributed ledger technologies, such as tangles and green blockchains, for transparency and monitoring of the data lifecycle, from creation to end use, through proofs of concept to assess the sustainability of blockchain. Legal factors influencing cloud and blockchain technologies will be assessed in relation to the proposed Data Act and the GDPR to understand the construction of data protection-friendly data space infrastructures. The blockchain decentralised database will allow data sharing while ensuring privacy and compliance with the GDPR.				
A	**Federated Learning for the training of a DL-based models at the edge**	**P**	**AIR**	**TRL**	**3 → 4**
Technology description: Federated learning is a new approach with a decentralised AI model that performs learning locally, and then allows local models to be aggregated and contribute to a global model without having to share the raw data, ensuring greater confidentiality, offering high personalisation, reducing latencies and costs of exchanging data continuously with a server, and facilitating fast and easy integration with Edge Computing. The communication efficiency (frequency and cost) is limiting the full exploitation potential of training at the edge., **Use/Improvement/Adaptation:** A common layer and repository of federated Artificial Intelligence algorithms to be run on the Edge will be designed and developed. NOUS will incorporate a layer for federated learning between Edge nodes in the network, enabling model training in a distributed manner while protecting user privacy. Privacy techniques for federated learning include Secure Multiparty Computation (SMC), 3PC models, differential privacy methods, k-anonymisation, diversification, or homomorphic encryption.				
A	**HPC-QC interface**	**P**	**NCSRD**	**TRL**	**3 → 4**
Technology description: The ecosystem around quantum computing is not mature enough to offer a good set of tools to program quantum computers, to execute programs and to read the output. The most natural way of integration Quantum computing with HPC systems is to allow HPC to push workloads to the QC system, referred as "offloading mechanisms". Such an approach requires intensive computations to achieve the needed optimizations and mapping to the underlying qubit topologies on the quantum devices, **Use/Improvement/Adaptation:** NOUS will realize a scalable HPC-QC interface					

software (independent on the number of qubits of the QC), design a quantum data encoding scheme taking advantage of quantum superposition. NOUS will explore the capability of using Quantum Machine Learning in collaboration with the Federated Learning NOUS will design and deploy an environment allowing the fast data retrieval from cloud, formatting them to be accessible by the quantum computer and implement the interface between NOUS and the quantum computer.

A	Natural Language Processing (NLP) models for query interfaces through chatbots and voicebots	P	AIR/USAL	TRL	4 → 5

Technology description: Early computational approaches to language research focused on automating the analysis of the linguistic structure of language and developing basic technologies such as machine translation, speech recognition, and speech synthesis developing spoken dialogue systems and speech-to-speech translation engines. The contemporary availability of large data sets and the increase in computational power and parallelization, facilitated enables the training of Deep Learning architectures. **Use/Improvement/Adaptation:** Query interfaces through chatbots and voicebots that facilitate interaction with users using Natural Language Processing (NLP) models will be implemented in the Virtual Lab platform that will allow a question-response type with the capability to relate the question to a dataset. The language specificities and cultural background of the users will be considered.

A	TransiTool (www.transitool.com)	P	AETHON	TRL	7 → 8

Technology description: TransiTool is a web app for Transport Service providers to digitize their service data using standards such as GTFS and NeTEx/SIRI. **Use/Improvement/Adaptation:** The auto-standardiser (AS) and data lake will be incorporated in TransiTool (with respect to transport data). Standardisation techniques employed in TransiTool will be used in the AS. Web app resources of TransiTool will be used to deliver the AS.

A	Hyperledger Implementation	P	AETHON	TRL	2 → 4

Technology description: AETHON has created a local PoC using Hyperledger technologies for a transportation ticketing implementation. **Use/Improvement/Adaptation:** AETHON will employ its experience from the PoC to requirements elicitation and implement it to deliver a DLT for monitoring data accessibility, ownership and data processes.

A	Virtual Lab (https://labs.mobidatalab.eu/)	P	AETHON	TRL	4 → 5

Technology description: AETHON developed the Virtual Lab for the MobiDataLab project. The tool contains functionalities to assist the co-creation/design/ experimentation between different stakeholders, **Use/Improvement/Adaptation:** The Virtual Lab platform will be the basis for the creation of the collaboration platform, using and extending current functionalities especially with regards to Q&A, forums. Connection with F6S will also be studied.

A	Methodology for the deployment of Blockchain-based use-cases	P	POLITO	TRL	4 → 5

Technology description: POLITO developed one of the leading frameworks for assessing Blockchain-based solutions in non-crypto applications (cite, see comment). **Use/Improvement/Adaptation:** The framework will be the basis for the Distributed Ledger/Blockchain infrastructure of the project, extending it to consider additional aspects, including the environmental ones.

A	GUEST	P	POLITO	TRL	7 → 8

Technology description: GUEST is Lean Business methodology already in use in more than 50 companies and 10 European and National projects. It is already proven to be easy-to-understand by people with limited knowledge of innovation processes and to let them to be involved in the process without the need of a long learning period. **Use/Improvement/Adaptation:** We will specialize the GUEST to perform an analysis to understand the sustainability (operational, economic, social, and environmental) of the economy of Data fostered by the project on users and stakeholders.

A	Arctur-2 HPC infrastructure	P	ARCTUR	TRL	9

Technology description: Arctur-2 is the flagship hyperconverged HPC & Cloud computing infrastructure. It has been designed and built to fit various and diverse usage scenarios, from general use Cloud Computing to HPC computing as well as High Performance Data Analytics. By converging diverse types of servers into single entity the highest performance, energy efficiency and seamless management are achieved thus guaranteeing excellent and cost-effective services to users. **Use/Improvement/Adaptation:** Arctur-2 will be used for testing and development of the computational components of the NOUS. By gathering data from real HPC usage scenarios, we will be able to better assess the needs of the NOUS, in terms of computational resources, but also regarding the interface to other parts of the NOUS.

A	Cybersecurity component based on a Darknet overlay and the Zero Trust model.	P	AEGIS	TRL	3 → 4

Technology description: A Darknet is a completely secure subset of the Internet, which can protect enclosed services, enhancing trust and efficiency throughout the network. The Zero Trust security model assumes no trust among the assets of a network (devices, users, services, etc.) and monitors access to a protected resource verifying the identity, context, and policy adherence of all requests. **Use/Improvement/Adaptation:** The Cybersecurity component of NOUS will be based on a combination of the two models. A consortium-wide decision will be made on which one will be leveraged (one or both) in the development of the framework, based on the added benefits of each model towards securing the NOUS platform.

A	Plug'n'Play Interoperability in the Fog	P	UNP	TRL	3 → 6

Technology description: Advanced Interoperability Solution for seamless data sharing and exchange at the edge implementing a delegated network of edge mediators for distributed (fog) interoperability. **Use/Improvement/Adaptation:** Implement mechanisms for dynamic interoperability based on the Plug'n'Play concept so to automate, as much as possible, the configuration and exchange of data between constituency systems in the interoperability environment.

A	3ACES	P	ITML	TRL	4 → 6

Technology description: A proprietary technology that provides big data infrastructure and AI-based data analytics as a service. The service provides encapsulated functionality for analysing heterogeneous data, providing insights with short or no training period. It provides the necessary cloud infrastructure for the data sets to be classified with high accuracy, delivering

tailor-made insights through open, extensible and scalable deployments that host any kind of adapters, from traditional relational databases, No-SQL databases or Kafka streams. **Use/Improvement/Adaptation:** 3ACES will be further developed to cover the need for delivering Federated Learning as a service and the further facilitation of edge analytics for the project.					
A	**Streamhandler Platform**	**P**	**NET-INTRA**	**TRL**	**6 → 7**
Technology description: A high-performance (low latency and high throughput) distributed streaming platform for managing real-time data based on Apache Kafka. It can efficiently ingest and manage massive amounts of data into processing pipelines, for both real-time and batch processing, and create a set of integrated connectors for managing cross-streaming architectures from various data spaces. **Use/Improvement/Adaptation:** The streamhandler platform will be used for creating the necessary connectors that will be used for integrating multiple data spaces into the project platform. It will be further developed to enable advanced stream handling service optimised and evaluated for the domains addressed by the projects.					

1.2 Concept and Methodology

1.2.1 Overall Concept

NOUS is envisioned to be the cloud computing suite in the European Union that is connected to the HPC network and designed to offer IaaS and PaaS components to companies and organisations. The pipeline of a NOUS will include three types of components:

- **Computational components**: The components that are responsible for executing computations. This is the standard cloud computation service and the sandbox environment for researchers to test their applications (as is the HPC network today).
- **Edge components**: The components that are responsible for communicating with edge devices such as smartphones and IoT sensors/actuators/devices.
- **Data components**: The components that are responsible for data storage, storage management, data sharing, standardisation and exchange.

NOUS will allow users to develop applications and services built for and operated largely with the use of edge computing devices while offering unprecedented computational power for backend services. At the same time, the NOUS will utilize connection with Data Spaces to foster data sharing and provide data analytics related services especially focusing on knowledge sharing and collaboration.

The **computational components** of the NOUS must allow unprecedented computational power to its users. Europe has to harness the power of its research industry; the NOUS will be connected with the HPC network and utilize its resources for executing computations, while providing a seamless interface to quantum computers. Cybersecurity is also an issue in this regard which will be solved by the design of a new overlay network of the internet (darknet) to offer encrypted and traceable connections between NOUS and the computational endpoints. Given its encryption, it can provide a physical-like division with the internet thus, providing paramount security. NOUS will also employ self-sovereign identity techniques for computational nodes, encouraging distributed ownership and management of cloud resources by citizens.

For the **edge components** the project will employ distributed methodologies with on-premises (IoT) resources to implement data analysis exploiting not only global cloud services, but also local resources available at the devices of the operators and data owners. In such environments, some (potentially mobile and/or resource constrained) devices are available, and typically their computation and storage resources are underutilised. Instead of relying exclusively on storage and computation services provided by a global cloud provider, the project's approach optimises the distribution of storage and computation tasks among those devices which will be eventually viewed as a local mobile cloud. NOUS will also incorporate a layer for federated learning between Edge nodes in the network, enabling model training in a distributed manner while protecting user privacy.

The **data components** of NOUS address the data life cycle and various of its aspects. NOUS aims to provide a comprehensive framework for data life cycle management, that incorporates, governance, accessibility and privacy concerns. NOUS will research the implementation of low-power distributed ledger technologies to guarantee the integrity of the stored data (expanding upon the concept of directed acyclic graphs or *tangles*, and registering only cryptographic summaries / *hashes* of the stored data to respect the citizens' right to be forgotten). In addition, the data storage components will function as a place to combine data from different Data Spaces and platform, using tools for syndication that the project will generate. NOUS can provide practical tools to developers and engineers of data spaces. First, an auto standardiser will be created to standardise data from any format to a known standard (in Mobility, Energy and Green Deal data spaces) and second, a Virtual Lab for sharing knowledge on data analytics.

All three components will be studied individually and together in order to create the envisioned NOUS architecture and lay the foundations for a first European cloud suite. The architecture will be made open-source to allow the capitalization of it by companies and organisations, public and private. In addition, NOUS will develop the components for testing and demonstration purposes which can become spin-offs of the project in and of themselves. The components will be tested with specific use-cases to measure the KPIs of the project and assess the expected

impact of the project.

1.2.2 Methodology

The methodology is devised to cover certain requirements laid out by the objectives and the concept of the project:

1. various installations of NOUS managed by different tenants can take place, however, with different implementation (partly or entirely) of its components;
2. the project should be geared towards the creation of the NOUS product thus, provide deliverables that are close to the market and its requirements;
3. the research executed in the project should allow innovations beyond NOUS; each component discussed in ch. 1.2.1 covers a different part of cloud architecture while delivering exploitable and self-sufficient innovations;
4. the methodology should allow the achievements of the objectives as discussed in ch. 1.1.1.

Requirements 1 and 2 demand a generalisable methodology to be leveraged by future projects. To match these requirements, the methodology of the project is two-layered. **The upper (generalisable) layer consists of a more widely implemented approach** that, while more abstract, it can be implemented easily by others. **The lower (specific) layer consists of a specific-to-the-project methodology**, that tackles the entirety of the Objectives defined in 1.1.1. The two layers are executed in parallel, with the lower layer being a more detailed version of the upper layer.

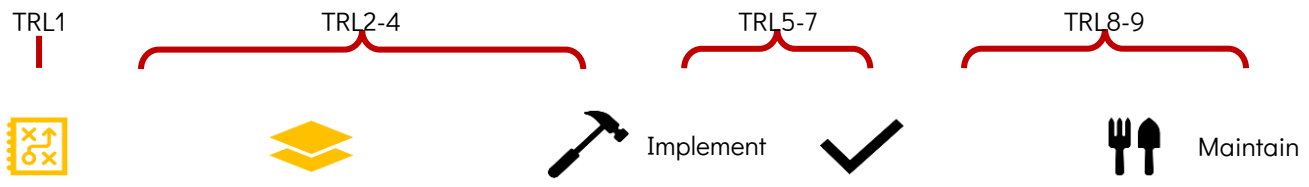

*Figure SEQ Figure * ARABIC1: The generalisable (upper layer) methodology of NOUS - The SDLC process*

On the **upper layer**, the development of the project will follow the Software Development Life Cycle (SDLC) approach but not in its entirety. The SDLC defines steps from idea to product full release. The project will reach TRL 5, thus, only some steps will be executed. The methodology of the SDLC that will be implemented is shown in Figure 1. NOUS will follow the SDLC process until implementation and verification steps which will be partly executed.

On the **lower layer** lies the core, specific methodology of NOUS project. This part of methodology specifically covers requirements 3 and 4 and is derived from the SDLC approach. Figure 2 demonstrates the lower layer methodology. NOUS methodology is designed in a way it is not significantly harming (**'Do no significant harm'**) any of the six environmental objectives of the EU Taxonomy Regulation.

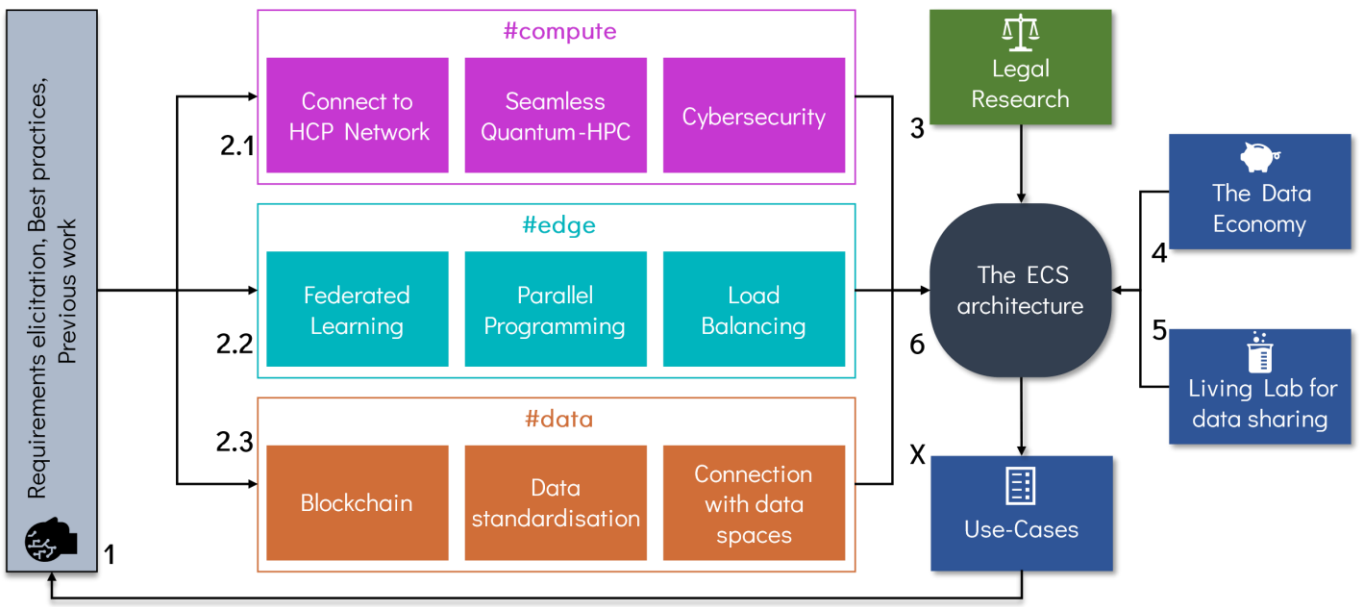

Figure 2: The specific (lower layer) methodology of NOUS (colours correspond coarsely to WPs, see 3.1.1)

Part 1 – Requirements Elicitation: The requirements elicitation is the first step of the project. The requirements elicitation in the project aims to collect functional and non-functional requirements for the NOUS product and for each technological component individually. The end-result of requirements elicitation is a clear objective for the creation of each component of the project shown in Figure 2. First, requirements will be retrieved from the Use Cases (**1st function of use cases**). Second, the project will extend its research to other commercial cloud providers (e.g., Google, Microsoft, Amazon and others, such as consortium partner HPE), as well as to public or private open-source

clouds (e.g., OpenStack and OpenShift). Third, the project will collect requirements from partners of the project that already use cloud infrastructure. Finally, requirements will be related and collected based on Data Spaces needs, existing infrastructure, i.e., Gaia-X, EOSC, ECCCH, as well as other initiatives identified through the literature review that this part will execute (e.g., distributed ledger technologies such as IOTA or Ethereum 2.0) and the European Cloud Marketplace (DOME). This step will also perform a review of the European Interoperability Framework and determine how its principles and recommendations can be used in the NOUS, as (non-)functional requirements. This step is also responsible for refining and reviewing the business objectives of the NOUS as defined in this proposal text (also see Part 6 in this chapter).

Part 2.1 – #compute: The computational component of the project aims to define how the project and the envisioned NOUS cloud can use resources of the HPC node. This is not only practical (e.g., how to authenticate, how to send data to the HPCs) but also theoretical since the project aims to determine the optimal usage methodology of the HPC network for the NOUS. To this end the project will assume some configurations being, first, a Kubernetes powered application, a VM powered application and an application that has significant requirements for HPC resources (e.g., simulation or digital twins). Similarly for the interaction between HPC and Quantum computers the project aims to determine the following: when a difficult/highly computational problem arises, which actions can be executed on the HPC network and which on the Quantum computers based on multi-criteria evolutionary algorithms that optimise load scheduling based on computational requirements, infrastructure costs and energy consumption. The project aims to determine which actions will be executed on which computer and why and create a software application that handles this problem when the NOUS is operational given the problems derived in the use-cases (see Part X below). Finally, this component will look into the Cybersecurity of the NOUS with two principles. First, the project will research darknets which can create a completely secure network that acts as a sub-network of the internet. Within the darknet, the NOUS could offer digital services that are completely cut-off from the internet thus improving trust and efficiency (the latter because on an inherently secure compartmentalized environment, the requirements for additional cybersecurity and security applications diminishes). Second, the project will research Zero Trust Security Model that can work in addition or separately from the darknet security model. In both cases, the question to be solved concerns the added benefits of each model (darknet and Zero Trust) towards securing the NOUS (and other cloud services) and their configurations.

A report demonstrator how the interactivity of between HPC and NOUS can take place will be delivered in **M12** (**D3.1**), and a demonstrator of the interface between QC and HPC will be delivered in **M28**. The NOUS Cybersecurity component using a combination of tools with darknets and Zero Trust models will be published in **M30** (**MS10**).

Part 2.2 – #edge: The edge component of the project will be researching aspects regarding edge computing as part of the NOUS. As with most technological components developed in the project, the Federated Learning and Inference framework, parallel computation programming and load balancing components, aim to be services that a user could activate in the NOUS one operational. The federated learning and inference framework will consist of a suite of SSH-aware algorithms for decentralised ML in user- and device-centric analytics. The project will research and develop the framework as well as optimise local learning processes to minimise the resource footprint on resource-constrained devices, particularly at learning time. Regarding the parallel computation programming, the project will develop a software based on FastFlow to enable users of NOUS to instantiate parallel applications as compositions of pre-defined, configurable parallel skeletons. Efficient mechanisms for thread cooperation (through lock-free queues), and a set of architectural optimization strategies for getting the best performance from multicores (e.g., pinning/sticking of threads onto cores, blocking/non-blocking concurrency control, NUMA-aware mapping strategies of threads onto cores) will be delivered to address scenarios with strict throughput and latency requirements in services related to executing edge computing operations. Finally, and regarding Load Balancing, the project will develop a hierarchical, algorithmic data and computation distribution layer that offers intelligent resource allocation optimization recommendations between edge devices and cloud/fog data centres, and the respective decision-making systems (real time alerts and short-term decisions). The design of this component will consider the utilisation of edge devices for distributed and redundant storage; an aspect that will be researched is how edge devices storage and its distribution can cooperate with cloud storage while protecting users' privacy and data sovereignty.

An in-depth analysis of available approaches for distributed architectures to (co-)operate in an optimal and most efficient manner with traditional cloud services will be published in **M12** (**D4.1**) and a package of key components will be published in **M30** under **D4.2**: a set methods and tools for initiating structured parallel programming environments, the Federated Learning tool of NOUS and the interoperability specifications to enable interoperations between different heterogeneous systems (IoT and edge) in the NOUS environment.

Part 2.3 – #data: The data component of project will be focused on delivering a blockchain implementation, research standardisation processes and connecting with Data Spaces. Distributed Ledger Technologies such as *tangles* and green blockchains will be used in the project for data life cycle transparency and monitoring. While this is mentioned as a blockchain part, the project is more interested into the distributed ledger and the way it is implemented in the Blockchain. Thus, the project will determine if and how the distributed ledger can be used for the aforementioned

goals, designed and validated given the use cases' defined problems. Furthermore, the use-cases will be used extract information on how data life cycle is currently handled in real-world scenarios (**2nd function of use cases**). The standardisation processes will be researched for the domains of mobility, energy and green deal in the scope of first, determining harmonisation possibilities between the standards per data space and with EIRA and second, creating an auto-standardiser (AS). Research on harmonisation of standards should consider the Architectural Building Blocks of EIRA as well as the European Interoperability Framework - Implementation Strategy. The AS aims to assist in a highly practical issue; organisations and companies that want to standardise their data, need to spend a significant amount of money to do, since not all domains have standardisation tools. The AS is a data-lake concept where data are "thrown" in the "buckets" and the AS attempts to relate the data elements of the provided dataset to data elements of a standard. When it fails, it will make specific requests to humans to manually correlate the data elements. This algorithm is expected to be heuristic and recursive, making use of imitation learning techniques to learn from human manual correlations and progressively improve their performance. Lastly, the connection with Data Spaces aims to address the call request for the successful deployment of Data Spaces. NOUS is a conductor of data analysis. It is where the actual data analytics take place because the cloud suites have computational resources. Within the NOUS data should flow freely in two ways, first, data can be uploaded by users or created by users' applications (e.g., a web blog that allows registrations and has videos) or data can flow from other data pools/warehouses. The latter part is what is of interest for the project while relating it with data standardisation and data sovereignty (see blockchain above). Thus, the latter part of #data aims to address the entirety of the data life cycle in the NOUS considering that data can flow in from external data warehouses.

A set of connectors with existing Data Spaces developed and tested with at least 2 existing Data Spaces will be published in **M15** (**MS14**). A report with the legal challenges and recommendations for blockchain technologies will be openly published in **M16** (**D5.1**). The blockchain-based database storage component will be delivered in **M34**, along with the Auto-Standardiser component and a report with the final services created for Data Spaces interconnectivity, all included in (**D5.2**).

Part 3 – Legal Research: In a 2020 Joint declaration on "Building the next generation cloud in the EU", Member States agreed that the next generation EU cloud offering needs to comply with the highest standards in terms of data protection, cybersecurity, data portability, interoperability, and consumer protection. The legal research will seek to shed light on the legal challenges cloud technologies raise and how to address them. For example, the centralized cloud computing systems, enabling data controllers to access the data processed, raise concerns in relation to data protection and privacy [26]. While edge computing provides solutions to some of these problems, such as microdata storage aligning with the data minimization principle (Art 5 GDPR), these offer low computing power at the time being. The problems raised by blockchain and the cloud in isolation are essential to understand to build data-protection friendly data space infrastructures. Legal scholars have also highlighted a general clash between blockchain technology and the GDPR that should be considered [27] [28]. Among others, the correct application of accountability, and of data subject rights are difficult to implement in blockchains [29]. The Data Act proposalThe Data Act proposal (DA proposal) also raises questions for blockchain as it explicitly mentions smart contracts and introduces essential requirements for them for data sharing and interoperability. At the same time, operators need to navigate a complex EU legal landscape. The Digital Services Act (DSA), the Digital Markets Act (DMA) and the Data Governance Act (DGA) have changed the applicable regime on platforms/intermediaries imposing significant obligations on the operation of such services. Classifying the NOUS is a challenge in itself as the notion of "intermediary" is defined differently in each of the said legislation.[1] Furthermore, a common definition and terminology for actors active in the Data Spaces framework (e.g., "data provider", "data recipient" and "data user") is only currently being developed (see DSSC Glossary) and it is not evident how each actor in the NOUS ecosystem would be classified in relation to the data transactions taking place and the role they would play in preserving digital sovereignty. Competition law considerations also come into play if market power can be attributed to the cloud operator based on the accumulation of data. Privacy protection poses significant challenges. For example, some national authorities (CNIL, Slovenian Supervisory Authority) have identified cloud service providers as joint controllers, rather than mere data processors (see CNIL Recommendations and Slovenian Supreme Court Decision). The NOUS may also host sensitive data coming from the public sector, such as 'data concerning health' (e.g., electronic health records, patient registries) that require elevated protection under the GDPR. Finally, the development of the data economy necessitates better access to data but also more control for users on what they can do with their data. Rights provided under the DA proposal – such as an access right to IoT product data – need to be examined on whether they can be claimed against cloud services. The same applies for the data portability right[2] and

[1] The DSA refers to "providers of intermediary services" and imposes obligations on platforms according to their size. The DMA regulates "gatekeepers" defined as "platforms that have a significant impact on the internal market, serve as an important gateway for business users to reach their end users, and which enjoy, or will foreseeably enjoy, an entrenched and durable position" while the DGA refers to "data intermediation services". At the same time, the DA proposal adds an extra layer of complexity when introducing "data processing services".

[2] Under the DA proposal, data portability applies to IoT products, and is extended to legal persons and to observed data. Data portability

a 'right to switching' from one cloud and edge computing service provider to another. But data portability and 'switching' will only deliver on expectations – if at all – provided that interoperability is guaranteed.[3]

Part 4 – The Data Economy: The intersection of technology and data presents a huge opportunity to address various social, environmental and economic issues. Efforts are ongoing to leverage the use of data for the well-being of society in several domains, such as agriculture, financial services, logistics, health and mobility, among others. Data is a valuable resource. Just as traditional economies based on the production and consumption of goods and services, the data economy will fund on generation and use, re-use of data, moving the economy scheme from an economy of repeatable processes (Economy of Kronos) to an economy based on events and data flows (Economy of Kairos). This process is evident in digital businesses and technologies, such as Web 3.0 and Metaverses (to cite some). Thus, there is the need to assess the value of Data in all their lifecycle, from their creation to their (eventual) destruction, by measuring with a repeatable and measurable methodology all the aspects of the Economy of Data (economic, operational, social, and environmental). Thanks to the GUEST, an established Lean Business methodology, NOUS will design a methodological framework that will become a standard for data clouds and data lakes applications.

Part 5 – Living Lab for data sharing: The Living Lab that the project envisions to create is a virtual location that will allows different users to co-create, co-design and co-develop data analytics for a specific and across domains. It is a tool for data experimentation, usability and even exploitation. While Data Spaces aim to facilitate flow of data, develop infrastructure to access/share/process/use data, clear governance models and more, the Living Lab for data sharing aims to cover the "why" part of data sharing and exchanging. It will be a "Stack-Overflow"-like system, based on Living Lab principles and AETHON's Virtual Lab (https://labs.mobidatalab.eu/) where researchers and professionals can exchange ideas and questions on data analytics, usability and exploitation. This tool is a companion of the NOUS and, while the NOUS is expected to be replicated many times, this Living Lab will be a singular application.

Part 6 – The NOUS architecture: The NOUS architecture is the main contribution of the project towards the scope of the call, the scientific and professional communities.

The NOUS architecture is an IaaS/PaaS/MLaaS architecture that incorporates the connection with HPC network and seamless computing between HPC Quantum computers in a (cyber) secure environment, provides a federated learning framework as a service with a proper load balancing based on AI (i.e., neural-evolutionary algorithms) and capability for parallel applications, utilises low-consumption DLTs for ensuring data life cycle handling; while assisting in data standardisation using semantic data lakes and lakehouses, as well as recursive algorithms, and promoting self-sovereign data exchange through its connection with a Virtual Lab. The above should also incorporate connection with Data Spaces and address legal and ethical issues (e.g., following the EC's Ethics Guidelines for Trustworthy AI).

The architecture should be comprehensive, for the matters addressed, and should be ready to be implemented by project partners or beyond, thus, there are high requirements for documentation. The architecture should tackle all aspects of a cloud suite including networking (e.g., VPNs), traditional storage (i.e., SQL, NoSQL, NewSQL, object repositories), virtualization (e.g., containerisation), general-purpose processing (e.g., Big Data processing, serverless computation), as well as machine learning model training (e.g., AutoML approaches, computer vision and natural language APIs). The methodology of designing the architecture is a stepwise process:

1. **Business Objectives Definition**, including stakeholders' identification (see Part 1 in this chapter);
2. **Requirements elicitation**, functional/non-functional, constraints and limitations (see Part 1 in this chapter);
3. **Architectural drafts with iterations for different views**, following established models (e.g., Rozanski and Woods [30] or Kruchten [31]) and which should be presented to all partners in various workshops or other formats (e.g., live document, one-on-one interviews);
4. **Documentation**, should be executed for the final drafts per view as determined in previous step;
5. **Evaluation**, using Architecture Trade-off Analysis Method or Cost-Benefit Analysis Method or similar (for a review of methods see [32].

A first version of the NOUS Architecture and Open-Source development platform will be published in **M18** (**D6.1**). This will include elements to create an IaaS/PaaS architecture. The last version of them (**M6.2**, **M36**) will include guidelines and tools to assist NOUS interoperate with other EU platforms, as the framework and guidelines to manage NOUS contributions as open-source updated through time. In the meantime, the Virtual Lab will be online and available in **M30** (**MS16**).

Part X – Use Cases: The use-cases are a part of methodology that contributes to many other parts, hence the signification as "X". The functions of the use-cases have been designated in the text above. One additional function

provisions also emerge in the context of European Data Spaces Commission, 'Proposal for a Regulation of the European Parliament and of the Council on the European Health Data Space' COM/2022/197 final (EHDS proposal).

[3] This essential lesson learnt from the application of the data portability right under the GDPR, shifts the focus from portability to the regulation of interoperability as a key factor. White Paper on the Data Act Proposal.

is that each use case presents a specific challenge that will be tackled by NOUS and the technologies that will be created. Thus, use cases will also implement some technologies in an industrially relevant environment (**3rd function of use cases**) which are described, along with the use case description, in the following tables. For usage of satellite data, use of Copernicus and Galileo/EGNOS is envisioned in use case #1 (and #3, depending on available data).

Use Cases selection criteria: The use cases in the project are defined by organisations which are undergoing digital transformation (PPC, TIM), have high requirements for AI technologies (PPC, TIM, NCSRD), have high requirements to analyse and improve their data life cycle (PPC, TIM, NCSRD) and/or have requirements regarding the analysis of the data life cycle in their products (CS). Each of the use-cases' organisations have a specific domain of operations that is connected to a different data space (energy – PPC, mobility – TIM, green deal – CS, science data - NCSRD). Each organisation has defined a different challenge in their data life cycle and a requirement for an AI algorithm that require processing of vast data.

Table 3: Use Case #1 – Perception of Connected Vehicles using camera data

IDENTITY, ACTIVITIES & OBJECTIVES: TIM is an Italian telecommunications company operating in Italy and Brazil offering mobile and fixed telephony, cloud and IoT end-to-end services, operating the largest data centre network in Italy as well as many other activities. The goal of this use case is to apply NOUS services to improve connected vehicles perception relying on roadside cameras and other cars' cameras. Relying on the combination of a distributed network of heterogeneous IoT devices (e.g., smart cameras), a widespread connectivity infrastructure, and NOUS' edge servers, the system will be able to precisely locate moving physical assets (with a special focus on pedestrians and vulnerable users) and provide the obtained data to cars and road users in real-time. This will allow connected cars and road users to perceive the environment from multiple point-of-view (depending on installed cameras and road-side units) and provide increased awareness of the nearby environment. To achieve the above, the use case will collect data from multiple cameras using #edge component firstly, analysing requirements to deploy the required service on edge devices with a special focus on 5G MEC architectures and secondarily, to assess and measure the performance of different communication protocols to enable interaction between devices, car, cameras and NOUS. AI models will be created also considering contextualisation of information. A special concern for privacy aspects during collection and processing will be given, also considering the NOUS data life cycle framework. The obtained solution will be tested end-to-end in a real urban environment, using "standard" communication infrastructure and production edge cloud computational nodes.
EXISTING WORK IN THE DOMAIN OF THE PROJECT: This use case will rely on the existing MODENA AUTOMOTIVE SMART AREA (MASA - https://www.automotivesmartarea.it): a project, created through a public-private partnership, aiming to develop a living lab to test, evaluate, and implement Cooperative Connected and Autonomous Mobility (CCAM), on different environments (a proving ground and a real-life urban area).
DATA, DATA MANAGEMENT PROCESSES & DATA TO BE USED IN NOUS TASKS: The MASA area is already provided with: 1. The MASA area has deployed sensors, cameras, on an area of 3 km² inside the town of Modena, with an already plan to extend and interconnect it to a provincial 15 km high-speed road. We have deployed a network of 20+ cameras and a cluster of GPU servers running object detection (and anonymization) on the camera continuously. The stream of result of the object detection will be made available to be used in the use case. 2. The use case area is covered by the commercial cell phone network featuring 5G and 4G antennas. The MEC edge server located in Bologna can serve as the deployment infrastructure for the services.
PARTNERS INVOLVED: POLITO (Leading), TIM (Use Case Provider), ITML, UNIPI, AIR, KUL, UNP

Table 4: Use Case #2 – Energy Prediction and Energy Data Lifecycle Management

IDENTITY, ACTIVITIES & OBJECTIVES: PPC constitutes one of the largest power suppliers in North-eastern Europe, with ~10 million customers in Greece, North Macedonia, and Romania, while maintaining a total installed generation capacity of 10.5 GW in Greece and 0.5 GW in Romania, through a series of thermal, hydroelectric, solar and wind power plants. Through its ongoing digital transformation, PPC aspires to become a data-driven organization, thus improving its operation and services. In the context project, PPC aims to exploit the NOUS framework in order to: (i) access the HPC network and uncover ways to exploit it for running computational-intensive tasks, (ii) improve its data management through the standardisation and blockchain services of #data, (iii) improve its data confidentiality and integrity through the cybersecurity service of #compute, and (iv) use the federated learning offered by #edge.
EXISTING WORK IN THE DOMAIN OF THE PROJECT: Two particularly data-intensive applications are relevant to the NOUS project. The first one is *the energy prediction task*, which processes the past energy and weather data and along with the sensor data and weather predictions for the near future to predict the intra-day and the day-ahead power consumption, the intra-day and the day-ahead power generation and the optimal selling prices in the intra-day and the day-ahead electricity markets. This application currently relies on heavy human involvement suffering from excessive cost and limited accuracy, while being time-consuming. The second application is *the energy data lifecycle management*. This involves reporting tasks, which summarize customer, supplier and production data at various levels of granularity to present useful insights to the organization's executives, facilitating them in their decisions. The validity of these reports depends on the lifecycle of these data types. These data types are originally generated in isolated, auxiliary systems, but are moved to central information systems on a regular basis, so that

they can be retrieved by reporting systems. For this application, the main requirement for the reporting tasks is to ensure the well-known CIA triad (Confidentiality, Integrity, Availability) for the customer, supplier and production data in transit, throughout its lifecycle.
DATA, DATA MANAGEMENT PROCESSES & DATA TO BE USED IN NOUS TASKS: The first application leverages production data from PPC's power plants as well as energy data that are publicly available through the following data sources: HEnEx (https://www.enexgroup.gr), IPTO (https://www.admie.gr/en) and DAPEEP (https://www.dapeep.gr). These data sources are combined with past meteorological data and weather forecast data that are specialized for wind and solar farms, e.g., from Meteologica (https://www.meteologica.com). The main data types of the second application pertain to customer, supplier, production, and energy data. Each data type resides in multiple, isolated information systems, e.g., supplier data are stored in the ERP system, the procurement system, etc. A separate department is typically responsible for each information system. In the context of its digital transformation, PPC intents to move all data silos to the cloud, interconnecting them to increase their value, while improving their availability throughout their lifecycle. Their management will be simplified, too, through a centralized, cloud-based data management department, which will also enhance data consistency, through standardisation and by providing a single point of truth.
PARTNERS INVOLVED: AETHON (Leading), PPC (Use Case Provider), ARCTUR, NCSRD, POLITO, UNIMORE, AEGIS, ITML

Table 5: Use Case #3 – Crisis Management and Civil Protection Platform

IDENTITY, ACTIVITIES & OBJECTIVES: CS GROUP - France is a large and very innovative company based in Paris - France, with subsidiaries in Europe, Asia, as well as in North America. It is listed on the Paris Stock Exchange and employs 2100 people worldwide, among which 90% are PhDs or Engineers, for an annual turnover of 237 M€ in 2021, of which ~20% are allocated to RTD activities. In this project, CS GROUP expects to improve the capacities of its cloud-based CRIMSON product (http://crimson.eu), a European reference platform for crisis management, by porting it to the NOUS platform and exploiting its data-life-cycle management capabilities to handle the access and traceability of the mass of data exchanged by all the CRIMSON users, who can be thousands from multiple organisations during major crises.
EXISTING WORK IN THE DOMAIN OF THE PROJECT: CRIMSON enables the sharing of geospatial and situational information on a need-to-know basis between multiple stakeholders in several interconnected command centres and first-responders on the field (using the CRIMSON mobile app). It is used for the surveillance of the environment and critical infrastructures, and the management of crises offering improved situational awareness and operation management capacities. For this, it aggregates massive multiformat 2D/3D/4D information from all available sources (GIS, COPERNICUS, EFFIS, drones, sensors, other crisis management systems) by supporting numerous standards. All CRIMSON instances (Command Centers on stakeholders' premises, Mobile Command Centers on the field, as well as mobile apps) are interconnected using edge devices and centralised (cloud or data servers) data gateways that also enable the interoperation with numerous legacy systems such as Video Management Systems, Sensor networks, other C2s... A backbone for the fine-grained monitoring and administration of data accessibility and traceability, in a fast, robust, secured and transparent way, is currently needed to accompany the continuous increase of organisations that adopt the solution and the number of third-party and legacy systems with which CRIMSON interoperates.
DATA, DATA MANAGEMENT PROCESSES & DATA TO BE USED IN NOUS TASKS: CRIMSON enables the sharing of terabytes of information when protecting sites or managing crisis operations. This includes, first, the digital twin of the environment that can span over large regions. CRIMSON digital twin subsystem can aggregate all the information known about this space. This includes massive geospatial datasets including 3D terrain, aerial imagery, cadastral information, BIM models, without any limitation in terms of surface and resolution, for open space, urban and indoor environments. Second, live sensor data about the ongoing situation such as video streams from surveillance cameras or drones, temperature or hygrometry probes, weather data, earth observation data. Third, tactical information (commands, situation reports, pictures, videos, audio recordings) exchanged between the numerous first responders and command chains of the multiple organisations involved in the management of a crisis. The creation, modification, deletion of any information shall be managed by a robust access right and right to know management system. Any edition shall also be traceable during the whole data life cycle for later after-action analysis and potential judicialization of the incident.
PARTNERS INVOLVED: CS (Leading – Use Case Provider), POLITO, AIR

Table 6: Use case #4 – Scientific Data HPC Storage and AI Analytics

IDENTITY, ACTIVITIES & OBJECTIVES: Founded in July 1961 as a Research Centre for Nuclear Research, Demokritos (NCSRD) has grown to be the largest multidisciplinary Research Centre of Greece. The Centre consists of five independent Institutes focusing on different scientific fields (as per the institute names): informatics and telecommunications, nuclear and particle physics, nanoscience and nanotechnology, biology, nuclear & radiological sciences and technology, energy and safety. The setting is complemented by the in-campus Lefkippos technology park, which hosts more than 40 innovative companies. The variety of scientific fields treated within the walls of NCSRD leads to an explicit, strong need for data management and handling, as well as demanding computation workloads. The NCSRD environment exhibits all the properties of a complex data setting, for the analysis of which HPC constitutes everyday practice. Furthermore, the same ecosystem has formed successful cross-disciplinary research groups working on the application of AI and especially machine learning. The aim of this interaction is the use of AI throughout the whole scientific study and discovery lifecycle: e.g., from related work discovery through natural language processing and reinforcement learning, to predictive models for experiment or simulation speed-up and cost reduction, to improved modelling of processes and multi-scale analysis.

In the frame of the project, the aim is to apply data lifecycle management through NOUS tools and services, while also catering for efficient and effective computation and AI applications to empower scientific and technological excellence. The expected gain from this effort is a set of efficient and effective scientific workflows, allowing the use and combination of different scientific methods and data to significantly improve the time-to-publication on a computationally heavy and complex data setting.
EXISTING WORK IN THE DOMAIN OF THE PROJECT: This use case will rely on the existing HPC knowhow of NCSRD, aiming at: material design and especially Zeolitic and Metal-Organic Frameworks design; molecular multi-scale simulations with an emphasis on coarse-graining; nano-metrology and nanotechnology; advanced material structural modeling. We should stress that this existing work has been reflected in (a) cross-domain publications, (b) common Marie Curie project undertakings connecting AI with natural sciences and technology (ML-MULTIMEM, SmartDeZIgn); (c) the NanoAI community which is leading related multidisciplinary research; (d) the AINST workshop series showcasing the related scientific efforts to a wider scientific audience. The INN part of NCSRD team contributes its world-wide expertise on the natural sciences and technology aspect, while the IIT part contributes its leading role to HPC (EuroCC and EuroCC 2 Project), data management and AI expertise (contribution to the AI on-demand platform of the EU and multiple related efforts), as well as multi-disciplinary related previous projects (e.g. DARE). All these are finely complemented by the ahedd DIH understanding of how this expertise can be transferred to an industrial innovation setting.
DATA, DATA MANAGEMENT PROCESSES & DATA TO BE USED IN NOUS TASKS: The types of data that will be examined, generated and handled within this use case include (non-exhaustively): molecular simulation data, MOF design functional parameters data, nano-metrology data (including synthetic and real data from surface scans), structural material analysis data (such as e.g., damage data in multi-layered advanced materials), but also textual data relating to one or more of the above domains (e.g., related publications, dataset descriptions, etc.). The data management lifecycle in this use case covers data generation/gathering, storage, efficient distribution (as per case requirements), analysis and computation, data combination/integration and enrichment, as well as (FAIR-principled) access and reuse. Currently, an ongoing data management plan is being finalized and piloted within NCSR Demokritos, to support horizontally all science and technology data on campus. NOUS can bring this plan to a new level of applicability and potential.
PARTNERS INVOLVED: NCSRD (Leading – Use Case Provider), HPE, PPC, UNIPI, KUL, AIR

1.2.3 National and International research and innovation activities linked with NOUS

Table 7: Research and Innovation activities linked with NOUS

Project	Relation to NOUS	Partner
DSSC	Interoperable data sharing environment to enable data reuse and secondary use within and across sectors, fully respecting EU values, and contributing to the European economy and society.	KUL
SmartCLIDE	Smart cloud-native development environment to support the creation, testing and deployment of full-stack data-centred services in the cloud. The results are used for deploying highly dynamic cloud-based solutions such as the NOUS project.	AIR ECL NET-INTRA UNP
DIGIS3	Digital one-stop-shop service whose central core of knowledge and training is structured around Trustworthy AI, High Performance Computing and Advanced Digital Skills.	AIR (C) USAL
EXCELLERAT	EXCELLERAT will increase the European competitiveness in HPC-applications, hardware design and HPC service provisioning to exascale level.	ARCTUR
FF4EUROHPC	Advanced High-Performance Computing (HPC) technologies and Cloud-based services for the outreach and support of Europe's SMEs. SMEs are part of NOUS' target groups and will be considered in requirements elicitation as well as invited in workshops of the project.	ARCTUR UNIMORE
EUROCC2	Network of National Competence Centres in the most efficient way while addressing the differences in the maturity of HPC deployment in Europe.	ARCTUR NCSRD INESC TEC
DATES	The main objective of DATES is to explore approaches and options for the deployment of a secure and trusted tourism data space, ensuring transparent control of data access, use and re-use.	ARCTUR
EDIH DIGI-SI	EDIH DIGI-SI's purpose is to actively promote the digital transformation for public administration and industry in 4 core areas: agri-food, healthcare, manufacturing and tourism.	ARCTUR
AI4EU	Open, impartial, and collaborative platform that offers interoperable services, data, and tools from related communities and provides solutions to facilitate research productivity, reproducibility, and collaboration.	F6S, NCSRD
MobiData Lab	Cloud solutions, models and tools that use data sharing, supporting the spread of this culture within the European space. Creation of portals designed to provide information to the players operating in the mobility sector.	UNIMORE KUL AETHON
FENIX Network	First European federated data-sharing architecture to support public and private actors in logistics and mobility infrastructures.	UNIMORE
AEOLIX	Architecture for a collaborative IT infrastructure that supports information systems and the implementation of data access management models in Europe.	UNIMORE
PHOENI2	A Cyber Resilience Framework providing Artificial AI - assisted orchestration, automation &	PPC

X	response capabilities for business continuity and recovery, incident response, and information exchange, tailored to the needs of Operators of Essential Services and of the EU Member State National Authorities entrusted with cybersecurity. Related to the Cybersecurity component of #compute.	AEGIS
DOME	NOUS will utilise the results of DOME regarding market composition and marketplace impact in definition of user groups and personas, and the ontology-based documentation.	NET-INTRA
i3-MARKET	Technologies for trustworthy data-driven collaboration and federation of existing and new future marketplace platforms, with special attention on industrial data and particularly on sensitive commercial data assets.	UNP
AnticipatoryLedgers	Anticipatory design and ethical framework for Distributed Ledger Technologies (blockchain or DAG) and applications (smart contracts, IoTs and supply chain).	USAL (C)
5G-LOGINNOV	Innovative technologies and devices that use 5G technology, IoT, data analytics, next-generation traffic management, Cooperative, Connected and Automated Mobility and 5G logistics corridors, to optimise traffic operations in ports.	UNIMORE POLITO
OpenContinuum	The European Cloud, Edge & IoT Continuum is supported by the effort of two Coordination and Support Actions (CSAs), namely Open Continuum and UNLOCK-CEI, which will cooperate focussing respectively on the supply and demand sides of the CEI Continuum.	ECL
AI4TRUST	AI-based-technologies for trustworthy solutions against disinformation. Innovation technologies to monitor, in nearly real time, multiple online social platforms, filtering out social noise and analyzing multimodal (text, audio, visual) content in multiple languages (up to 70% of coverage in EU) with novel AI algorithms.	NCSRD
NEMO	Optimized to IoT-to-Edge-to-Cloud continuum. Network management for existing/emerging IoT/5G/6G technologies. "By-design" Secure Execution Environment (SEE). Decentralized, cybersecure FML/DRL for IoT node decision or autonomous operations. Cybersecurity and unified/federated Access Control.	AEGIS
Green.Dat.AI	Harness the value of Data Spaces whilst accounting for energy efficiency concerning large scale AI-based systems. Dynamic data sharing in industrial ecosystems through the federation of services and the execution of distributed large scale data pipelines. Energy-Efficient Large-Scale Data Analytics Services including Federated Learning Services (FLS)	AEGIS
CONCORDIA	The project aims to interconnect Europe's cybersecurity capabilities into an expertise network to help build a secure, trusted, resilient and competitive ecosystem. The group of POLITO leads the deployment of Blockchain solutions for the automotive battery supply chain.	POLITO
PrepDSpace4Mobility	PrepDSpace4Mobility lays the foundation for a secure and controlled way of pooling and sharing mobility data across Europe. The 12-month Coordination and Support Action (CSA) contributes to the development of the common European mobility data space by mapping existing data ecosystems, identifying gaps and overlaps within, and proposing common building blocks and governance frameworks found in existing data space architectures.	KUL
ATTRACT DIH	ATTRACT DIH's aims to promote and support experimentation, testing, development and adoption of solutions using AI and HPC, as well as to address training needs, while boosting the innovation ecosystem in these fields.	INESC TEC

1.2.4 Project interdisciplinarity and SSH considerations

Interdisciplinary research has emerged as a key mode of knowledge production and transfer. NOUS takes this into account by bringing together a multidisciplinary consortium and blending different science fields to achieve the investigation of complex and differentiated needs and points of view by multiple stakeholders. The NOUS consortium follows an efficient interdisciplinary approach and has a group of experts in different disciplines that, combining several skills and emphasizing particularly in SSH, will cooperate to develop the architecture for a European Cloud Service and its various components. In addition, the different expertise will be fundamental to test and prove the viability of NOUS and to assure its technical, economic, legal and social feasibility. The combination of different skills will allow to proper communicate with future end-users and stakeholders and a conscious, welcoming, and stimulating environment will be created so that everyone in the group will feel comfortable to share visions, doubts and needs.

The development of the NOUS architecture and its components requires specialised expertise and knowledge, as well as experience, from natural and applied disciplines: **Computer science** (Computer architecture, Software engineering, Algorithms, Artificial Intelligence, Machine Learning, High Performance Computing, Quantum Computing, Cloud Computing, Edge Computing, Federated concepts, Cybersecurity), **Information and Data science** (Data Engineering, Data Management, Data lifecycle, Data Spaces, Data Economy, Blockchain, Standardisation) and **Systems engineering** (Communication and systems engineering, Infrastructure management, Systems architecture). However, to complement the technical side and ensure the feasibility of the proposed technologies, additional aspects also will be covered: **Economics and Business** (Data Economy, Business model), **Social Sciences** (sociology, human in the loop), **Law** (Legal parts of the components, Guidelines and recommendations) and **Ethics** (new technologies, data management). Moreover, the contribution of the

Administrative side (management, governance) for the successful implementation and monitoring of the project tasks is paramount, while the **Media & Communications** part (dissemination of the project results, knowledge exchange and transfer, Open Science, marketing, networking, synergies with other initiatives and projects) is also fundamental. Finally, the use case partners provide insights in 3 additional domains: **Transport Engineering and Mobility**, **Energy** and **Civil Protection**. These disciplines are interwoven within the NOUS activities and will be brought together and integrated in pursuit of the project objectives.

Social Sciences and Humanities The social, ethical, and cultural implications of technological advancements and the needs and perspectives of people should be considered when designing and deploying data-driven solutions. SSH play a crucial role in shaping the human in the loop for agile-data economy, understanding how people interact with technology and data, how they make decisions, and how they are affected by algorithmic decision-making. Moreover, SSH help to identify and mitigate potential harms associated with data-driven solutions, such as algorithmic bias, discrimination, and privacy violations.

1.2.5 Gender analysis in the project of NOUS

Gender issues are often under-considered when dealing with computer science and data technologies. However, technological innovations may have a pivotal role in making these fields more inclusive, especially in terms of job creation and decision-making processes. The active participation, supported by the approach used in NOUS, presupposes that all genders can interact within the activities carried out, without discrimination. Therefore, the gender dimension constitutes a fundamental aspect of the methodology followed by the NOUS project.

The consortium is aligned with the European Policy of equal opportunities regardless sex and genders, as enshrined in the EU Treaties, the EU Charter of Fundamental Rights, the EU Gender Equality Strategy 2020-2025, and aware of the Directives of the EU on this topic. The consortium respects the legal requirements and principles set by EU law and will integrate a gender perspective into all project activities.

The project management strategy addresses Gender issues by maximising gender balance in the project's consortium (the current line-up suggests the active involvement of both males and females) and offering equal opportunities to women and men. The Consortium is acutely aware of the need to bring women into the centre of research and the resultant challenges faced by women in making progress in an overwhelmingly male field. Despite the legacy of gender imbalance in previous EU Frameworks, acknowledged by EC, NOUS has made great progress in ensuring a more gender-balanced consortium. NOUS is fully aware of the Checklist for Gender in Research produced by Yellow Window Management Consultants in association with the European Commission to optimise the gender balance in the consortium. Among others, this has resulted in allocating positions of responsibility to women with females holding key roles in the project. Additionally, while it is not foreseen that any specific gender issues will arise within the scope of the proposed NOUS work, as part of the project management handbook we will collaborate with partners to encourage recruitment of women in this field of research.

The NOUS project will follow the principles for mainstreaming gender into research, innovation and development measures defined by "A gender summit report" together with "A union of equality; gender equality strategy 2020-2025" created by the European Union. NOUS will follow the guidelines that suggest integrating the gender perspective into the design and implementation of strategies and projects. As a general principle, the project will collaborate, ask, and involve all genders through dialogue and discussion: the most important stakeholders' engagement activities will be carried out by challenging gender stereotypes and working for a real economic and societal inclusion.

In particular, NOUS will provide a new standardised and unified approach that combines all tracking information to allow all genders to be end-users of the deployed solutions. All tools developed and used in the project and all the different components will be designed to avoid any kind of gender bias. Addressing in texts and material written in English should be done according to the guidelines for gender-neutral language. The evaluation strategy and the impact assessment will address social and gender considerations. NOUS aims to train researchers: at this scope, the project will promote training activities to specific networks, such as "Women in STEM". In terms of dissemination and clustering strategy, a collaboration with gender neutral initiatives will be established. Finally, women of the project willing to disseminate NOUS findings, will be invited in registering as potential speakers in congresses and events.

1.2.6 Open science practices

The **NOUS** consortium is committed to open science practices, open-source principles and establishing an open ecosystem and will apply the principle of "as open as possible – as closed as necessary". Apart from the mandatory practices, outlined in article 17 of the model grant agreement (e.g., open access research articles), we implement other open science practices as an integral part of our work plan.

In terms of software, **NOUS** will implement an open-source strategy to maximize uptake and achieve sustainability of project results beyond the project lifetime. Project partners are committed to open innovation models and to publish

the **NOUS** Platform and components under an OSI-approved, business-friendly open-source license (e.g., EPL-2.0, Apache v2.0, BSD or MIT). The Eclipse Foundation will establish the framework for open, vendor-neutral collaboration during and after the project and support the creation of an active ecosystem of developers and early adopters. During the project, the Eclipse Foundation will continue educating the partners about the different business models enabled by open collaboration on an open-source platform and the creation of either open or proprietary added value on top of the platform.

The Eclipse business-friendly open-source model is suitable with the open-source strategy for the **NOUS** ecosystem. The Eclipse Foundation will provide (i) guidance in creating and managing open-source projects, (ii) expertise and support in community building initiatives, (iii) Eclipse Cloud ecosystem visibility, and interaction with the wider Eclipse and other open-source ecosystems, (iv) support in on open collaboration best practices and (v) infrastructure and community governance for open collaboration.

Dissemination at scientific journals will aim at publishers with either a gold open-access policy that is putting publications on the publisher's page at free access or at journals with a green open access policy that allows the author to provide a preprint or post-print for download at their private or institutional webpage. A dedicated section of the project website will reference all publication and provide access when publications are available.

1.2.7 Data management and management of other research outputs

NOUS generates primary datasets with different types of data (numeric/quantitative, text/qualitative, videos). To make sure that the data will be in a format to be used by all the consortium partners, project dedicates two tasks to bring data in a shape that follows the FAIR principles. By applying these principles from the early beginning, the project ensure that the data will be openly available after the completion of the project, will be easily accessible and highly utilised by interested actors. **T1.4** is dedicated to the Data Management Plan (DMP) with main purpose to produce a DMP, using the FAIR Data Management Plan Template that will guide the data storage and handling throughout the project and monitor the progress. **T7.2** is responsible with data collection (from use cases), **T5.3** will provide technologies for data storage while each task involving technology creation also executing data collection per their specific needs (e.g., **T3.3**, **T4.3**, **T5.4**). The DMP will include the following:

- **Findability**: Data of the project will be hosted by project technologies developed in **T5.3**. These technologies will be connected to data repositories (predominantly, EOSC and Zenodo). If interfacing is not possible (e.g., virtualisation of API upload) and to ensure that others can find the NOUS data, the data will be uploaded directly to open said repositories and manually assigned a globally unique persistent identifier (i.e., DOI). Same process applies for project deliverables (direct upload). The data will be accompanied by dictionaries describing the data and any other appropriate metadata. Using such a repository and identifier ensures that the data warehouse will continue to be available to both humans and machines in a useable form after the completion of the project.
- **Accessibility**: The project will carefully review the possibilities for data access and assign the proper licenses to the data. This will be determined by the DMP for the data (and the ethics committee) and by **T6.2** for source code. Licenses will be considered to facilitate data reuse. Open-access publications will be implemented in the project.
- **Interoperability**: Data will be stored in non-proprietary open file formats, standardised using approved (by CEN/CENELEC) data standards when possible and described using a standard vocabulary (for example, each data source will be provided with a specific name that is composed by different parts/elements, containing information about the location, data type or format and naming structures).
- **Reusability**: Reusability is a significant concern for the project. It will be ensured given the open workshops that the project will hold inviting the open-source and cloud communities (see ch. **2.2.1** and **T8.4**) promoting the services created during the project and the data associated with them and the project.

Curation and storage/preservation costs will be managed by NET-INTRA (leading **T5.3**). The usage of open repositories and sources will ensure minimum-to-no cost for data storage and preservation.

2 Impact

2.1 *Project's pathways towards impact*

NOUS' ambition is to create a European Cloud service for data processing and connection to Data Spaces, new services and real-life implementations. This architecture will generate a verifiable, transferable, and sustainable result, outcomes, and impacts. The development of the European Cloud service will create innovations that will have great impacts on companies and organisations giving them access to an unprecedented computational power to innovative backend services. However, it must be considered that the overall impact of NOUS touches the most diverse areas of our society: from the environment to the scientific world, from standardization to infrastructure, from the economy to culture, from trust in data sharing to legal-bureaucratic organisation.

2.1.1 *Impacts related to the call requirements and the topic outcomes*

NOUS has defined a specific approach that will allow to produce several outcomes, results and innovations. In the following tables we summarized the NOUS approach, the expected results and the impacts that NOUS aims to reach through the whole project along with related KPIs that will be used as a guide for the NOUS' evaluation.

Table 8: Impacts related to the call requirements and the topic outcomes

Call Requirement - Expected Outcome 1: Ability to process vast volumes data as one of the key enablers for other technological developments, supporting the competitiveness of the EU's industrial ecosystems.
***NOUS approach expected results and impacts*:** NOUS will provide a framework to connect the HPC European infrastructure and a cloud environment. The open-source dimension of NOUS allows its users to deploy the NOUS framework to develop, test, validate, trust and adopt innovative data-driven, edge computing and federated learning services, thus, increasing the competitiveness of European technological companies. The NOUS services developed during the project will support different stakeholders in different applications during the adaptation of the technologies to the specific use-cases (see ch. 1.2.2). The roadmap as well as documentation per task, together form a manual that will guide the further uptake by potential users. ***Connected WPs/tasks*:** T3.1, T3.2, T3.3, T4.2, T4.3, T5.2, T5.4, T6.4. ***KPIs*: KPI1.1** Users engaged and trained in demonstration site areas for using the NOUS framework and the platform for new services creation purposes. {30 people per pilot site}. **KPI1.2:** Individuals satisfied by the testing activities. {At least 90%}. **KPI1.3**: Individuals engaged in demonstrating the NOUS Platform services. {At least 50 people per pilot site}. **KPI1.4**: Users' satisfaction rate from the services in testing areas. **KPI1.5**: Users' satisfaction rate from the services in testing areas. {At least 95% satisfied users}
Call Requirement - Expected Outcome 2: Successful deployment of Data Spaces involving several sectors of economy or society.
***NOUS approach and expected results and impacts*:** The connection of the NOUS to different Data Spaces and the blockchain solution will give the data further added value, thanks to the high interoperability and the possibility of using the information available in diverse sectors. In a brief time, the NOUS approach will facilitate the set-up of workable solutions characterised by reasonable investments and quick time-to-market. The expected benefit will be more competitive and innovative data-driven services and products. ***Connected WPs/tasks*:** T4.3, T4.4, T5.3, T5.4, T6.4, T6.5, T7.1, T7.2, T7.4. ***KPIs*: KPI2.1:** Number of Sectorial Data Spaces connected to the NOUS during the project {4}. **KPI2.2**: Number of data sources and storages connected to NOUS {10}. **KPI2.3:** Number of industrial stakeholders consuming data through NOUS services at the end of the project {6 from at least 4 European countries}.
Call Requirement - Expected Outcome 3: Improve data access (in line with the FAIR principles), data sovereignty, data interoperability and data protection as an essential factor in the development of sustainable value chains respecting all stakeholder interests, particularly SMEs, but also the public sector as data providers and innovation/market ecosystem enablers. The European Strategy for Data calls for actions to support and promote data sharing and the use of data for social and economic benefit.
***NOUS approach and expected results and impacts*:** NOUS will enable data access, fusion and sharing in digital infrastructure, facilitating data processing and exchange. The NOUS will ensure compliance with the security, privacy, and data-space requirements through cross-domain standards and a blockchain decentralised database. The ability to share data and improve access to cloud and HPC services and Data Spaces represents one of the main scopes of the project, which aims at accelerating the development of data-driven and AI-based digital services and their deployment by public and private users. ***Connected WPs/Tasks*:** T2.4, T3.3, T4.1, T4.4, T5.2, T5.3. ***KPIs*: KPI3.1** Development of cybersecurity software to be used in the cloud environment {1}. **KPI3.2** Smart contracts aimed to interoperability with the DLT and guaranteeing fair exchanges with Data Spaces {60}.
Call Requirement: Improved European leadership in the global data economy
NOUS approach and expected results* and impacts*: The European strategy for data implies the creation of common Data Spaces and a European market for data. The innovative data applications and services will benefit the economy and society. NOUS will directly and concretely contribute to this strategy thanks to the connection to several Data Spaces and allowing the development of applications and services based on data. These innovations will create safer transport, reduce energy consumption and improve the reuse of resources. ***Connected WPs/Tasks*:** T2.1, T2.2, T2.4, T5.3, T5.4, T7.2, T8.4. ***KPIs*: KPI4.1** Perceived rate of effectiveness of the solutions for commercialisation {90%}. **KPI4.2** Development of technologies for 4 use-cases solving industrially relevant issues and published open-source. {4 (use-cases)}. **KPI4.3** Workshops with Data Spaces Support Centre. {2}.
Call Requirement: Maximised social and economic benefits from the wider and more effective use of data

NOUS approach and expected results* and impacts*: The Organisation for Economic Cooperation and Development has defined data use and reuse across the economy as a new form of capital. Furthermore, sharing and reusing open data is creating new opportunities for economic growth apart from generating benefits for society for several actors, such as data providers, suppliers, data users, and the overall economy (for example, see Deloitte's study on assessing Transport for London Open Data policies). On that basis, the NOUS will create software for cybersecurity which will allow the handling and anonymising of personal data so that they can be deployed directly for innovative and useful services that have the potential to reduce the impact on the environment and to create a more inclusive and safer society. Furthermore, the decentralised blockchain database will enable data sharing thus facilitating the fusion of cross-sectoral data. ***Connected WPs/Tasks*: T2.4, T3.3, T4.1, T4.2, T4.3, T4.4, T5.2, T5.4, T6.2, T6.4, T7.1, T7.2, T6.5**. ***KPIs:* KPI5.1** Increased road safety in CAM thanks to the adoption of Intelligent Transport Systems HPC edge computing applications {-10% accidents}. **KPI 5.2** Number of innovative applications and services developed thanks to the NOUS platform {4}. **KPI5.3** Reduction of energy consumption thanks to the use of intelligent data-driven digital services {-10%}. **KPI5.4** Reduction of carbon emissions because of the use of low-energy distributed DLTs {-20%}.

Call Requirement: Reinforced Europe's ability to manage urgent societal challenges (e.g., data for crisis management, digital for clean)

NOUS approach and expected results* and impacts*: Leveraging big data and data sharing during crises helps to manage resources and supports decisions for optimal management of resources. NOUS architecture has immense potential in managing big data when disruptive events such as the pandemic, energy crises or natural disasters occur. The blockchain decentralised database allows data sharing while ensuring privacy and compliance with the GDPR. Furthermore, the connection of the NOUS to several Data Spaces enables cross-sectoral data fusion for the maximal exploitation of information. Finally, the possibility to exploit the computational capability offered by the HPC network and the potentialities of federated learning create new opportunities for improved crisis management (i.e., health climate, etc.). NOUS has specifically included a Crisis Management use case led by CS. ***Connected WPs/tasks*: T2.4, T3.1, T5.2, T5.4, T6.5**. ***KPIs*: KPI6.1** Increased level of transparency in data sharing and processing when using applications or services developed with the NOUS for crisis management {+20%}. **KPI6.2** Reduced time to exchange data due to standards supported by the data lake {3 standards}.

Call Requirement: Developing and enabling the uptake of the next generation computing and data technologies and infrastructures (including space infrastructure and data)

NOUS approach and expected results* and impacts*: NOUS will connect to the HPC network to allow unprecedented computational capabilities. Most importantly, NOUS will provide interfaces to Quantum computers to implement Quantum Machine Learning in the context of Federated Learning. **Connected WPs/tasks: T3.1, T3.2, T4.1, T4.3**. ***KPIs*: KPI7.1** Reduced resource footprint thanks to algorithms for decentralised ML {- 50%}. **KPI7.2** Reduced learning time {-70%}. **KPI7.3** Identification of non-real time constraints {100%}. **KPI7.4** Maximum guaranteed delay {95%}. **KPI7.5** Experiments on HPC Quantum Computers developed through NOUS at the end of the project {50}.

Call Requirement: Enabling the European single market for data with the corresponding Data Spaces and a trustworthy artificial intelligence ecosystem

NOUS approach and expected results* and impacts*: It is possible to accomplish a trustworthy AI ecosystem thanks to the collaboration between actors, data sharing, standardisation and the development of an appropriate legal framework. NOUS will contribute to all these aspects by providing software for data standardisation translation, chatbots/voicebots, Quantum ML, Federated Learning and legal frameworks. The Data Economy aspects are studied in **WP7**. ***Connected WPs/tasks*: T4.3, T4.4, T5.2, T5.3, T5.4, T6.5, T7.1, T7.5**. ***KPIs*: KPI8.1** Increased level of trust perceived when using applications or services developed with the NOUS. {90%}. **KPI8.2** Perform cross-platform data exchanges using the NOUS tools for syndication {Exchanges with 3 platforms}.

Call Requirement: Address the entire data life cycle from data generation/collection to the final use and disposal/deletion of data

NOUS approach and expected results* and impacts*: NOUS platform will ensure data life cycle transparency and monitoring from their creation to the final usage thanks to DLT and blockchain. Each use case will tackle different challenges for the data life cycle. **Connected WPs/tasks: T2.1, T2.4, T5.1, T5.2, T6.3, T7.5. KPIs: KPI9.1** Testing PoC implementation of a DLT for monitoring data accessibility, ownership and processing {1}. **KPI9.2** Address legal concerns for cloud and blockchain {100% concerns addressed}. KPI9.3 Compare different life cycle methodologies {6 methodologies compared}.

Call Requirement: Build on existing and emerging standards, models and architectures

NOUS approach and expected results and impacts: NOUS will standardise data in mobility, energy and Green Deal sectors and connect data to existing Data Spaces. Furthermore, NOUS will enable the connection to the HPC network for an increase computational capability. **Connected WPs/tasks: T2.2, T2.4, T5.4. KPIs: KPI10.1** Development of software (Auto-Standardiser) for data translation into a standard {1}. **KPI10.2** Studies in harmonisation between standards {4 standards' comparisons}.

Call Requirement: Enable correct allocation and enforcement of data-related rights, obligations and responsibilities across the life cycle

NOUS approach and expected results* and impacts*: NOUS will map the relevant data-related rights under the relevant legislation and assess their application in the context of cloud and edge technologies. Furthermore, the NOUS consortium will ensure that the project innovations do not infringe patents. **Connected WPs/tasks**: **T2.4, T4.4, T5.1, T6.3, T7.5. KPIs**: **KPI11.1** Identification of data-related rights under relevant laws, Number of relevant patents identified and checked {+10}.

Call Requirement: Process human-generated and human-related data and put data into context.

NOUS approach and expected results* and impacts*: The increased computational capabilities offered by the NOUS platform will allow the storage and processing of human-generated data such as videos or recordings. Most importantly, the auto-standardiser will tackle NLP aspects while considering the cultural specificities. **Connected WPs/tasks: T6.4, T7.5. KPIs:**

KPI12.1 Number of processes which foresee the human intervention {3}. **KPI12.2** Number of multi-modal query interfaces aware of language specificities and cultural background of the users for the interaction with NOUS {2 (data search; cloud service allocation)}.
Call Requirement: The seamless integration of "human in the loop" should be considered and implemented where applicable.
ESC approach and expected results* and impacts*: NOUS will tackle the integration of humans in the loop in the context of federated learning and other supervised ML techniques that require training of the algorithms and for which there is the need for human intervention to ensure the ethical development of automated processes. Human-centred practises will also be explored for a data agile economy **Connected WPs/Tasks: T4.3, T7.5**. **KPIs: KPI13.1** Number of supervised learning algorithms that require the human intervention during the training process {2}. **KPI13.1** Number of human-centred practices established {4}.

2.1.2 NOUS wider impacts

NOUS consortium is aware that its work on the integration of data life cycle, architectures and standards will have much wider impacts compared to those we have summarized in **Table 9**. Since we believe it is important to monitor impacts that are less obviously connected with NOUS - but no less important - we have decided to report the wider impacts and their related KPIs in the table below.

Table 9: NOUS wider impacts structured per different domains

Environment: The knowledge and understanding of environmental problems have gradually increased thanks to the growing data availability, meteorological evidence and awareness about environmental issues. NOUS will improve the ability to use data to find impactful applications and services to decrease CO2 and NOX emissions, monitor the progress of the biosphere, and intervene in specific problems in R&D projects. **KPIs:** number of actors providing data on environmental emissions; 20 kW energy saving. Reduction of 20 kg PM. 20% of fuel consumption reduction; number of datasets provided.
Society: The value of data is not only economical but also sociological: in fact, data improve social well-being both in healthcare [33], in public safety and security [34], in governments [35] and employment [36]. NOUS will maximise the data value thanks to merging different datasets from diverse sectors. Especially in crisis management, this will have a profound impact because of the potential to save lives, better manage existing resources or improve real-time decision-making. **KPIs:** Number of applications and services developed thanks to the NOUS platform
Economy: Greater sharing of data and easier access to the latter allow the use of potentially infinite economic resources, capable of generating positive returns for producers and users of this information. NOUS will contribute by using data in the best conceivable way, reducing the costs related to information management, improving transparency in their use, and creating new business opportunities, thus fostering competitiveness among actors and generating benefits for institutional and economic players. **KPIs:** Reduced costs of information management; number of business opportunities created.
Standardization and Trust: By establishing a common set of APIs, values, and principles, members of the NOUS open-source community will agree on good development practices, security protocols, and design patterns that ensure the maintainability and scalability of NOUS. This approach will result in de facto standards. **KPIs:** Number of open-source project contributors to the NOUS platform
Science and knowledge: Despite living in the Age of Data, there are still many challenges to be solved before we get to talk about a shared culture of data: to reach an awareness of the importance of data and the value associated with it, there is still little knowledge of the necessary organizational aspects [37]. NOUS will increase the ability of research institutions to collaborate, share knowledge and support the production of high-quality patents thanks to enhanced access to publications and research. One of the use cases is focused on more efficient exploitation of science data. **KPIs:** Number of datasets that are exchanged in real-time;
Policy and Law: The European Commission has put in place the legal and regulatory framework to enable data sharing in line with its EU Data Strategy and the EC Communication on Data Spaces. However, several legal questions pertain. The NOUS will seek to clarify some of these questions, thus contributing to raising legal transparency and clarifying several legal aspects, facilitating thus the implementation of the EU Data legislation and Data Spaces. **KPIs:** Clarification of legal obscurities in at least two legal frameworks.

2.1.3 Impacts on NOUS key target groups

Table 10: Impacts on NOUS key target groups

Target Groups	ESC's impact
Research institutions and universities	Universities and research institutions will benefit from applied research concerning HPC, quantum and edge computing, data sharing, data security and the impact on safety and efficiency. The focus on connection to Data Spaces and data-driven innovative services will boost interdisciplinary research in the safety, energy and transport sectors. NOUS will enable the digital transformation of several domains and better coordination of research and innovation. Key Performance Indicators for scientific impact include: **i)** Development of Technical Demonstrator at TRL 4, **ii)** Training of Early Career Researchers: at least 6, **iii)** Training of Educators: at least 20, **iv)** Training Sessions: 4, **v)** End users involved in the Training: at least 100, **vi)** Open access training material: 10 ECTS equivalent.
Policy makers	This impact will consider ethical, data protection, social and economic assessment in a policy framework while considering the public and civil society organisations across the EU asking for a single data market. The policy

	brief will include a comparative analysis of EU laws, ethics, and sector requirements. Expected outcomes: **i)** An innovative, efficient, consistent, and resilient system for validated safety and security, improved robustness, and resilience of data sharing and computing architecture and tools in four scenarios during the project, **ii)** Secure and trustworthy interaction between stakeholders (e.g., SMEs, end users, public authorities, etc.) to achieve better use of NOUS architecture validated in two scenarios throughout the project., i**ii)** Increased perception of European leadership in the HPC and cloud computing systems, ensuring long-term growth and jobs through the engagement of at least ten policymakers in supporting the adoption and dissemination of the findings.
Private companies	Industry partners and private companies will benefit thanks to the possibility of developing new applications and services using the NOUS architecture. NOUS's innovative assets will stimulate and enhance the innovative capabilities of private companies, competitiveness and attractiveness. NOUS will enable private companies to contribute to and benefit from future data services to improve safety, reduce energy consumption and improve recycling practices. Some of the KPIs for impact on private companies include: **i)** Delivery of 3 pilots demonstrating the benefit of NOUS's innovation, **ii)** Implement at least one program of dissemination activities at the EU level, **iii)** Participate in 4 EU programs, promoting updates of technical capabilities on the data and HPC market, **iv)** Deliver at least one workshop to facilitate collaborations between private companies, **v)** Organise at least three seminars for the private sector (60/90 minutes each, with at least three presenters) to facilitate engagement and collaboration.
Public companies	Thanks to NOUS, there will be an improvement in the capacity to detect user needs and the development of active policies. There will be an increased ability to share data with other public bodies and private entities and better cooperation in the management/organization of activities.
Legal and policy experts	Legal researchers as well as lawyers and consultants working on legal and policy matters will benefit from the in-depth legal research conducted in the context of NOUS. The legal consequences and challenges of applying blockchain in the cloud are issues still at their infancy. The same applies for legal issues pertinent to cloud and edge, namely the many open questions deriving from the application of the new EU Data law suite (DGA, DA proposal) and the recent initiatives on Data spaces. The NOUS will seek to tackle those questions, thus contributing to the legal and policy debates, advancing the knowledge in this field and motivating further legal research.

2.1.4 Potential barriers, obstacles, and risks and ways to overcome them

NOUS potential barriers, obstacles and risks to innovation are assessed according to the PESTLE framework as they may be **P**olitical, **Ec**onomic, **S**ocietal, **T**echnological, **L**egislative, **En**vironmental as summarised below.

Description	Type	Mitigation measures in NOUS project
Difficulties for the pilots in using the NOUS platform.	T, L, Ec, En	To use the NOUS platform, the users will be carefully supported by all the research institutes and partners involved in the project. The goal is that all the doubts and concerns can be overcome through collaboration so that the pilots can successfully demonstrate the innovative, sustainable and inclusive infrastructure.
Connect databases to the NOUS platform to be usable and not require too much time for consultation and their use.	T	To overcome this challenge, NOUS will develop software for standardising information, and structuring the connections between the models and the tools created.
Disagreement between partners on IPR / Confidentiality breaches by a consortium partner.	Ec, L	The Consortium Agreement (CA) will define IPR more clearly (also, T8.3 tackles this subject) to identify a solution for all partners. Concerning breaches of confidentiality, they are handled by the Project Board, which convenes on a case-to-case basis to decide whether actions ought to be taken.
Delays during the software development and untimely delivery of the final software.	T	Using an agile software development approach, which is verified and adapted as needed on a regular basis, it is possible to continuously adapt the workplan to the project timeline and obstacles.
Stakeholders are hesitant about using the NOUS standard to exchange data and providing access to their data or systems.	T, L	The benefits and added value of using the NOUS platform and access to their data/systems can be demonstrated to stakeholders by presenting the pilot evaluation results, the business models and business cases, the compliance with ethical as well as legal requirements.
SMEs find it difficult to use the NOUS components for developing new services and sharing data.	S, T	The NOUS will be developed in a way that hides complexity, does not require extensive programming and documentation review to be used and it will be user centric.
Designed use cases and collected requirements do not represent a wide range of stakeholders, thus not attracting a lot of actors.	Ec, L	Task T6.1 and T7.4 will ensure the replicability of the NOUS solutions in various cases, while T8.4 will engage as many stakeholders as possible and encourage them to adopt these solutions.
The NOUS framework faces several cyber-attacks (DDoS, insider threats, etc.)	T	Task 3.3 will (i) perform research to progress beyond the State-of-The-Art in darknets in conjunction with the Zero Trust model and (ii) apply lessons learnt from the research to fortify the NOUS framework both from outside as well as insider threats

A darknet overlay designed on a zero-trust security model hinders the efficiency and productivity of the NOUS framework	T	Other than implementing NOUS's Cybersecurity component, **T3.3** (in collaboration with **T7.2** and **T7.3**) will focus on security evaluations and test runs of the developed module, gradually scaling its deployment, starting from the most sensitive and critical workflows.

2.2 Measures to maximise impact – Dissemination, exploitation and communication

Measures to maximize impact are divided into: i) activities aimed at promoting the action, at awareness raising and communication beyond the project's internal as well as external communities to wider audiences, including the media and the general public; ii) activities aimed at raising interest among stakeholders and the exploitation-oriented dissemination of the benefits provided by the innovative technologies proposed in the project toward potential target end-users/adopters; iii) activities focused on engaging key stakeholders to participate in specific project activities; and iv) activities aimed at the exploitation of the project's results.

2.2.1 Dissemination & Communication

The success of NOUS is strongly dependent on well-coordinated dissemination, communication, and exploitation activities as well as activities to build synergies and cohesion with other projects, initiatives, and the partnership on AI, Data & Robotics. Therefore, the beneficiaries of the NOUS project have decided to include a specific Work Package for this purpose (**WP8**). The development of a detailed exploitation and business plan, as well as exploitation-dependent dissemination and communication activities will ensure widespread knowledge transmission while protecting exploitation interests. The coordinated activities ensure that the different target groups are addressed in an appropriate manner. Focus will be put on disseminating project findings to the European cloud sector, which will be the main beneficiary of the novel technologies, data and knowledge.

The dissemination and communication plan seeks to define the goals (why), identification/classification of the target audience (who/what), the dissemination methods (how), the schedule and the complementarity of the activities (when), the measures to assess the impact of the dissemination and communication activities, and the conditions to ensure the proper spread of the generated knowledge, confidentiality, publication, and use of knowledge.

To ensure a smoother replication and adoption of project results, NOUS has defined a task dedicated to connecting the project with other projects, initiatives and organisations, with the most prominent and important for project success being, the Data Spaces Support Centre (DSSC) and the EuroHPC. NOUS will connect with the DSSC build synergies/ collaborations with the Data Spaces that would best benefit from NOUS outcomes.

Table 11: Dissemination strategies for NOUS

Means / channel	Objective, target and quantifiable indicators
Project technical & publication	A project e-publication will be produced during the second phase of the project to collect relevant results in the form of data and observations from the validation site and from Business simulation activities, thus giving comprehensive evidence of the advantages of the developed enabling technologies and of the concept. ● Project e-publication downloads: <35 = poor; 35-70 = good; >70 = excellent
Scientific/ technical publications and oral/ poster presentations at conferences, symposia, seminars, workshops, etc.	The project's results will be published in the international scientific/technical literature, such as IEEE Transactions on Computers, IEEE Transactions on Cloud Computing, IEEE Transactions on Parallel and Distributed Systems, IEEE Transactions on Software Engineering, IEEE/ACM transactions on Networking, International Journal of Parallel and Distributed Computing, Future Generation Computer Systems, Journal of Network and Computer Applications as well as in relevant scientific/ technical literature at national level mainly in the member states where the partners are established. Results will also be presented at relevant conferences, symposia, seminars, workshops, and other events, such as International ACM Symposium on High-Performance Distributed Computing (HPDC), IEEE International Parallel and Distributed Processing Symposium (IPDPS), IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID), ACM SIGCOMM, IEEE CLOUD; Intern. Conference on Cloud Computing; KubeCon + CloudNative Con Europe; IEEE/ACM Intern. Conference on Utility and Cloud Computing; European Conference on Service-Oriented and Cloud Computing; Data Innovation Summit; Intern. conference cloud-computing and super-computing; Cloud Expo Europe (2023) ● At least 20 papers will be published on Open Research Europe, AIOD platform. ● Collaborative research work with at least 3 different common European Data Spaces (Mobility, Energy, Green Deal) and 3 projects working in the domains (achieved through NOUS partners). ● Organisation of at least 3 special sessions in conferences organised by the partners, at DCAI or BLOCKCHAIN organised by AIR and USAL and ECLIPSECON organised by ECL. ● All publications will be collected in the project's website for open-access/download. Open Access to peer-reviewed scientific publications will be provided. ● Number of papers submitted: <15 =poor; 15-30 =good; >30 =excellent
Video clips, Newsletters,	The technical brochures will provide information about the technical and scientific achievements to the relative stakeholders, while the non-technical brochures will describe the potential project applications and services in a more open and less scientific way. The project videos will cover NOUS general ideas,

Brochures/flyers	demonstrations and presentations and talks that will also include non-technical information about the project, targeting non-expert public and policy makers. Newsletters will be distributed to different mailing lists to foster inter-communication with other relevant research actions, related H2020/HE and other research projects and technical communities. The newsletters, available at the project's website, will provide information regarding the project activities, achievements and results, targeting cross-fertilization. ● Number webinars: <5 = poor; 5-7 = good; >7 = excellent
Exhibitions and Trade Fairs	Events with booths where participants can engage with participating organisations are an excellent way for the project to present its results and future steps. NOUS will consider participating in various events with a booth such as DCAI, BLOCKCHAIN, ECLIPSECON, Cloud Expo Europe, HiPEAC Conference, ESOCC, CloudFest, European Identity and Cloud Conference, Dublin Tech Summit, Big Data Expo, TechEx and Web Summit.
Liaison with EU communities	The consortium will seek liaison with most relevant EU communities on NOUS topics, including the relevant ADRA ppp, AIOD platform (AI4Europe CSA), DAIRO/ BDVA (Big Data Value Association), Gaia-X, FIWARE, EDIHs, European Cloud, Edge & IoT Continuum, HiPEAC, EUCloudEdgeIoT.eu, EDIH network and the DSSC funded under the DEP. ● Number co-organize/participation in events: <3 =poor; 3-5 =good; >5 =excellent
Collaboration with relevant projects & initiatives	The consortium will seek liaison and collaboration with other AI, Data & Robotics tech driven projects as well as business-oriented connections that could complement project activities and provide synergies, also to effectively disseminate project results, especially with DSSC. ● No. of projects engaged with NOUS: <5 =poor; 5-8 =good; >8 =excellent

Communication activities - The communication strategy of NOUS will be adapted to the main target groups and stakeholders listed in the KER Table to ensure that the progress of the project is communicated. Actions will be designed, using different channels of communication (website, events, leaflets, newsletters, social media, video, and webinars) summarized in below table. The overall project objective will guide all communication activities, raising awareness about the affordability and reliability of NOUS products and solutions.

Table 12: Indicative key events, relevant to NOUS (to be reviewed and extended within Activity)

Target groups	Measure for Communication	KPI	Impact
Research community	Publications at international conferences	30	Communicate technical achievements. Setup collaborations for follow up activities
	Publications in international journals	20	
	Organization of joint events (workshops, conferences special sessions, etc.) with sister projects and initiatives	4	
	Open-source code releases (in SLAM, planning, ML for defect detection, navigation policies)	4	
Broad public	Project Website (M3): Number of Visits	6000	Create awareness about the project, its objectives and impact on the EU community
	Public deliverables online – No. of downloads	300	
	Non-scientific publications (articles, press releases)	10	
	Social media followers (aggregated in LinkedIn, Twitter)	500	
	Project video	2	
	Non-technical Flyers/Brochures distributed at conferences etc.	3000	
Industries and users (NOUS users)	NOUS workshops	5	
	White Papers	2	
	Exhibitions and trade fairs	5	Create market needs. Direct contacts at our booth. Discussion on project sustainability, Exploitation
	Request for technology features and its potential use	50	
	Interest of industrial customers on Technology Exploitation via partnerships and/or license agreements	10	
	Participation in EU commission's consultations and policymaking process in the field of interest	1	
Regulation bodies	Co-organize workshops with DSSC	2	Influence on future policy setting
	Creation of guidelines to EuroHPC	1	

2.2.2 Exploitation

2.2.2.1 Key Exploitable Results

Table 13: Key Exploitable Results (KER), Partners (Main Partner: MP, Participants: Part) & Time to Market (TtM)

KERs, MPs, Parts & TtM	NOUS Key Exploitable Results Preliminary Plan
KER1 New and advanced materials investigations	**Main end-user:** Industrial manufacturers, Digital Factories. **Market aimed:** Any industrial manufacturer. **Value proposition:** New knowledge in materials properties, materials reliability,

and predictive maintenance Main partner: NCSRD **MP**: PPC. **TtM:** 1 year after the project.	predictive maintenance, saving raw materials and environmental sustainability. The lack of interoperable, HPC-enabled and data-lifecycle handling unified solutions with support for scientific workflows reduce access to and reusability of scientific data and computation components. **Next steps for commercialization:** Provision of services to material and device manufacturers, digital factories leveraging scientific and IoT data for materials' functionalities and predictive maintenance.
KER2 Plug'n'Interoperate component **MP:** UNP. **Part:** ITML, USAL. **TtM:** 2-3 years after the project.	**Main end-user:** Data marketplace developers/owners, data middleware providers, data integrators. **Market aimed:** Data-driven domains where interoperability is a need e.g., data marketplaces, data technologies (middleware), etc. **Value proposition:** An advanced component for providing data interoperability in complex, large-scale and heterogeneous data sharing environments. **Next steps for commercialization:** Move from TRL6 to TRL9 for having a complete and fully qualified system actually proven in operational environment.
KER3 E^2 analytics **MP:** ITML. **Part:** UNIPI, UNP. **TtM:** 2-3 years after the project.	**Main end-user:** Any owner/ user of complex data infrastructures; data providers. **Market aimed:** Any data-driven domain looking for the full exploitation of data during the whole data lifecycle. **Value proposition:** E^2 analytics will provide a paradigm shift in management of collected data and processed automatically towards the edge, focusing also on distribution run-time and sovereignty aspects through data-aware algorithms for provisioning of resources and services across the distributed computing continuum to serve data object access, pipeline analytics, HPC and federated ML needs. **Next steps for commercialization:** Validate proof of concept in real / realistic environments.
KER4 Standalone Cybersecurity component **MP:** AEGIS. **Part:** AIR, USAL, NCSRD, TIM, ITML, HPE, PPC, ARCTUR, CS, INESC TEC. **TtM:** 2-3 years after the project.	**Main end-user:** Cloud infrastructure providers, Cloud/network environments developers/users. **Market aimed:** Cloud/network environments market, where complete segregation of data/services and traceability/accountability of users' actions are necessary. **Value proposition:** An advanced standalone component that provides data/services compartmentalization in complex, large-scale and heterogeneous networking environments. **Next steps for commercialization:** Validation of proof of concept in real/realistic environments. Provision of services to cloud/network environments developers/users towards securing their infrastructure.
KER5 The Auto-Standardiser **MP:** AETHON. **Part:** NET-INTRA, UNP. **TtM:** ~1 year after the project.	**Main end-user:** Transport Service Providers (users of Mobility Data Space), other providers in other data spaces (secondarily). **Market aimed:** Data-driven domains where interoperability is a need either to exchange or publish data (e.g., for travellers). **Value proposition:** The auto-standardiser can transform Transport Service Provider data to a data standard, enhancing publications (e.g., Google Maps) and collaboration between providers by enabling semantic understanding. **Next steps for commercialization:** The tool will be integrated to TransiTool and exploited as a part of TransiTool's offering.
KER6 Virtual Lab chatbot/voicebot **MP:** USAL. **Part:** AETHON, F6S. **TtM:** 1-2 years after the project.	**Main end-user:** Citizens, professionals, government officials (users of Living/Virtual Labs). **Market aimed:** Living/Virtual Labs, Forums, chat interfaces. Users that want to search and retrieve information faster. **Value proposition:** User of Virtual Lab can now query datasets, posts, questions, faster and easier using chat and voice interaction. **Next steps for commercialization:** USAL will consider knowledge transfer to AETHON and create an agreement for that purpose.
KER7 Connected Vehicles perception module **MP:** POLITO. **Part:** TIM. **TtM:** >3 years after the project.	**Main end-user:** Vehicle Manufacturers, (Mobile) Network Operators, Drivers. **Market aimed:** The vehicles market moving towards autonomous vehicles. Cities or Network Operators that install cameras and other sensors to be used to V2I and I2V. **Value proposition:** The product can provide a better recognition of pedestrians and vulnerable road users using an advanced AI module based on NOUS edge computing implementation. **Next steps for commercialization:** Demonstration and improvement of successful recognition of vulnerable road users leading to trust in the system. Agreement with a city/operator/vehicle manufacturer.
KER8 Energy Prediction and Energy Data Lifecycle Management **MP:** AETHON. **Part:** PPC. **TtM:** ~1 year after the project.	**Main end-user:** Energy producers and distributors that participate in the wholesale energy market. **Market aimed:** The energy domain where data fusion of diverse data sources is required (for reporting and price estimation). **Value proposition:** The tool will facilitate optimal selling price estimation using diverse data sources and reporting of customer, supplier and production data at various levels of granularity to present useful insights. **Next steps for commercialization:** Integration of tool to PPC production.
KER9 Data Space connector and stream handler platform **MP:** NET-INTRA. **Part:** ITML. **TtM:** 2-3 years after the project.	**Main end-user:** Data space providers/owners, data space users and data consumers, data integrators. **Market aimed:** Data-driven domains and spaces. **Value proposition**: The tool provides the connectors and hooks for interconnecting, storing, transforming, and processing data from various sources, facilitating the seamless integration of various data spaces. **Next steps for commercialization**: Test and validate for the domain addressed in the project.
KER10 Data Economy Evaluator **MP:** POLITO. **Part:** AETHON. **TtM:** 1 year after the project.	**Main end-user:** Data space providers/owners, data space users and data consumers, data integrators. **Market aimed:** Data-driven domains and spaces. **Value proposition**: The tool provides the sustainability assessment of Data Spaces, providing a viable tool for assessing Data Spaces projects. **Next steps for commercialization**: Test and validate for the domain addressed in the project.

KER11 Blockchain-based Data Life Cycle **MP:** POLITO. **Part:** AETHON. **TtM:** 1 year after the project.	**Main end-user:** Data space providers/owners, data space users and data consumers, data integrators. **Market aimed:** Data-driven domains and spaces. **Value proposition**: We provide a secured and trustable Blockchain-based system for the information related to the data lifecycle, including storage, modifications and usage. **Next steps for commercialization**: Test and validate for the domain addressed in the project.

Additional KERs may be identified during the project and exploited within the Plan for Exploitation.

2.2.2.2 Protection of results

The NOUS consortium has defined roles to allow proper exploitation of results.

Exploitation team. NOUS will appoint an Exploitation Manager (F6S), to effectively manage the exploitation of project results (see Section 3.2). Exploitation rights will be considered. Furthermore, NOUS will appoint an Innovation Manager (NCSRD) to ensure the project's innovation and long-term sustainability.

Exploitation strategy of all the partners. The competitiveness of organizations is increasingly depending on their ability to collaborate in networks, and for such collaborations (knowledge-based ones), strong IPR management is a prerequisite for efficient trade and transfer of IPR and technology.

The proposers are aware that, once the project is complete, the results will be of great interest and benefit to industry, in Europe and globally. In NOUS, significant information and data will belong to participating SMEs. Consideration will be given to this and the needs of dissemination and communication.

Confidential information will be protected, while non-confidential information will be openly published.

- *Ownership of Results.* Bearing in mind the significance of the Intellectual Property Rights (IPR) in fostering innovations, the work plan for the dissemination and exploitation of the results foresees a continuous review of the important and relevant IP issues. The management of IPR issues will be managed by AIR according to the Project Grant Agreement with the EC, as well as the Collaboration Agreement.
- *Intra-partner licensing of Results.* Access rights to results and background needed for the performance of a partner in SPEAR will be granted on a royalty-free basis. The international market penetration and commercial rights will be defined amongst the participants in the Exploitation Plan during and beyond the project: In case of commercial use by one partner or in case of licensing to third-parties, the other partners shall be entitled to receive financial compensation based on fair and non-discriminatory market conditions.

The NOUS IPR strategy is informed by Articles 16 and Annex 5 of the Annotated Model Grant Agreement. The DESCA Model Consortium Agreement provisions regarding access rights (Section 9) will be negotiated and signed concurrently with the Grant Agreement to ensure clarity around ownership/licensing of background, sideground, and foreground IP ensuring all partners have the necessary access rights to complete the action. In line with the Open Access and Open Science principles, relevant scientific results will be distributed as openly as possible, unless there are specific commercial reasons for patent protection. The exploitation plan is described in **WP8** (within **D8.1**) and i nvolves: 1) the IPR management strategy where the Plan for Exploitation and Dissemination of the Project Results (PEDR) will be articulated; 2) the identification of background, sideground, and foreground IPR where all consortium partners identify their exploitation intentions regarding patents, licensing trade agreements, consultancy services and others; and 3) informing the partners of potential IP strategies and assist in identifying the most suitable IP protection mechanism. Towards the end, a results ownership list will be produced clarifying single or joint ownership, country origin of the IP owner(s), the Key Exploitable Results (KERs), and a determination of whether or not the owner will exploit the KERs. If any given KERs will not be exploited by the owner(s) one year after project completion, they will be added to the Horizon Results Platform (HRP). The Partners will define the rights and obligations in context with the Joint IPR before applying for patents.

2.2.2.3 The NOUS Exploitation Target and Approach

NOUS is an ambitious project that aims to assist Europe define its cloud, technologically and procedurally. We want a strong and sovereign European Digital Market which clearly cannot be achieved without a common vision on the main enabler: cloud services. The approach of NOUS exploitation can be characterised for its **openness**, **reproducibility** and **excellence**, it aims to foster collaborations through open-source technologies and architectures.

NOUS aims to follow the best practices of Gaia-X, FIWARE and EuroHPC and is targeting to create traction for the creation of a Joint Undertaking focusing on European Cloud. While this objective seems ways away, it is a clear target that the consortium partners agree to follow. Such an ambitious target cannot be achieved by only one project, but by a coordinated movement towards a direction, being a secure, sovereign and innovative European Cloud. To that end, partners will sign a Memorandum of Understanding at the end of the project to agree and state the path that NOUS will follow. Project partners have expressed their interest to provide more inputs to NOUS through their other projects and by submitting to new opportunities together.

2.3 Summary

<table>
<tr><th>SPECIFIC NEEDS</th><th>EXPECTED RESULTS</th><th>D & E & C MEASURES</th></tr>
<tr><td rowspan="3">● Process vast volumes data.
● Deployment of data spaces.
● Improve data access (in line with FAIR), data sovereignty, data interoperability and data protection.
● Support and promote data sharing.
● Support the competitiveness of EU's industrial ecosystems.
● Trustworthy data infrastructure and trustworthy artificial intelligence ecosystem.
● Improved European leadership in the global data economy.
● Maximise social and economic benefits from the wider and more effective use of data.
● Reinforce Europe's ability to manage urgent societal challenges.
● Develop the next generation computing and data technologies and infrastructures.
● Address the entire data life cycle from data generation/collection to the final use and disposal/deletion of data
● Enable correct allocation and enforcement of data-related rights, obligations and responsibilities across the life cycle.
● Build on existing and emerging standards, models and architectures.
● Process human-generated and human-related data and integrate "human in the loop".</td><td>● A connectivity framework for HPC and NOUS.
● An HPC-QC interface.
● A cybersecurity software based on darknets and Zero Trust model.
● A Federated Learning and Inference framework.
● An interoperability solution for seamless data sharing and exchange at the edge.
● A DLT module for data life-cycle management.
● Data services for integrating and managing data spaces.
● A tool for translating datasets to specific data standards.
● A high-level architecture for cloud services combining the above results.
● Analyse and tackle legal challenges across all results of NOUS.
● A platform used for data scientists to collaborate on data analytics/fusion/usage.
● Tools for syndicating various NOUS instances and with other platforms.
● A Blockchain Sustainability Index.
● Testing of components through real-life use-cases.
● Guidelines for EuroHPC towards the creation of cloud services.
● Human-centred practices within a data-agile economy.</td><td rowspan="3">**Dissemination:**
● Project technical & publication.
● Scientific publications and poster presentations at conferences, symposia, seminars, workshops, etc.
● Video clips, Newsletters, Brochures/flyers.
● Exhibitions and Trade Fairs.
● Liaison with EU communities.
● Collaboration with relevant projects & initiatives.
Communication:
● Publications in international conferences & journals.
● Organization of joint events (workshops, conferences special sessions, etc.) with sister projects and initiatives.
● Open-source code releases (in SLAM, planning, ML for defect detection, navigation policies).
● Project Website & project video.
● Public deliverables online – No. of downloads.
● Non-scientific publications (articles, press releases).
● Social media followers (aggregated in LinkedIn, Twitter).
● Non-technical Flyers/Brochures distributed at conferences
● NOUS workshops and White Papers.
● Exhibitions and trade fairs.
● Request for technology features and its potential use.
● Interest of industrial customers on Technology Exploitation via partnerships and/or license agreements.
● Participation in EU commission's consultations and policymaking process in the field of interest.
● Co-organize workshops with DSSC.
● Creation of guidelines to EuroHPC.
KERs:
● New and advanced materials investigations and predictive maintenance.
● Plug'n'Interoperate component.
● E^2 analytics.
● Standalone Cybersecurity component.
● The Auto-Standardiser.
● Virtual Lab chatbot/voicebot.
● Connected Vehicles perception module.
● Energy Prediction and Data Lifecycle Management.</td></tr>
<tr><th>OUTCOMES</th></tr>
<tr><td>● Use the HPC and Quantum network to manage and process big amounts of data
● Establish a connection between HPC and cloud services.
● Deployment of data spaces in several sectors.
● Enhance data sharing and interoperability, sovereignty and protection.
● Increase Europe's competitiveness.
● Create a trustworthy cloud service environment.
● Provide open-source cloud architecture and software tools.
● Contribute to standardisation.
● Implement blockchain technologies for addressing data related issues (across the entire data life cycle).
● Establish a Legal Framework for cloud, edge and blockchain aspects.
● Facilitate knowledge transfer regarding data collaboration.
● Create insights about data economy.
● Create and human-centred practises for future technologies.
● Facilitate the exploitation of science data.
● Foster collaborations towards creating the European Cloud.</td></tr>
<tr><th>TARGET GROUPS</th><th>IMPACTS</th></tr>
<tr><td>● Research institutions and universities
● Policy makers</td><td>● Address environmental issues by increasing resources efficiency.
● Improve social well-being by ensuring data sovereignty and protection.
● Economic benefits enhanced data sharing and analytics.</td></tr>
</table>

● Private companies ● Public companies ● Legal and policy experts	● Innovative tools for **standardisation** to support data sharing and interoperability. ● Promote **science** and knowledge transfer through the use of new technologies. ● Provide insights about the **legal** aspects of new technologies.	● Data Space connector and stream handler platform. ● Data Economy Evaluator. ● Blockchain-based Data Life Cycle.

3 Quality and efficiency of the implementation

3.1 Work plan and resources

3.1.1 Overall structure of the work plan

NOUS Pert diagram is located in **Figure 3**. In the figure, the numbering next to the arrows aims to provide a sense of flow of information which is equivalent to the flow shown in **Figure 2** (ch. **1.2.2**).

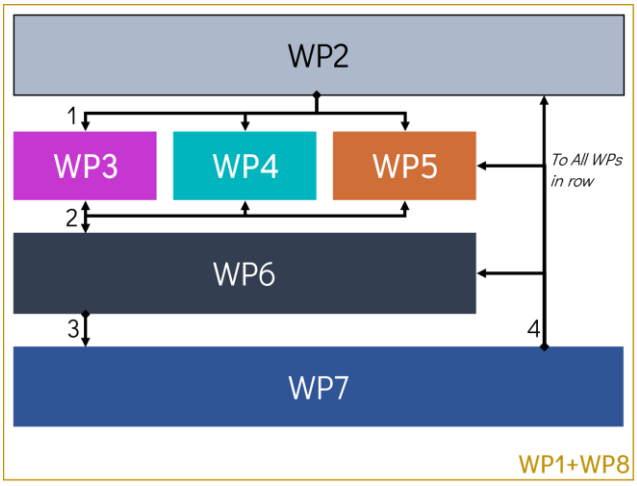

*Figure SEQ Figure * ARABIC3: NOUS Pert Diagram*

The Work Plan, which includes the Work Packages (WPs) and Tasks, was derived from the Methodology as presented in ch. **1.2.2** and more precisely, the specific (lower layer) methodology. The project is initiated with **WP2** which executes requirements elicitation, trends and status quo analysis while creating the evaluation framework of the project. This WP will also deliver a novel data life cycle framework that will govern the creation of data related technologies and the NOUS architecture.

In **WP3**, **WP4** and **WP5** the work regarding the components of #compute, #edge and #data will be executed respectively. Each of these WPs is structured to execute the necessary research for delivering the key achievements of the project, per relevant Objective, i.e., **O1** (**WP3**), **O2** (**WP4**), **O3**+**O4** (**WP5**). **WP6** will lay the foundations for executing WPs 3 to 5 by creating the architecture in two versions, the first in order to facilitate the work of said WPs (top-down, from requirements) and the latter following the comments from the testing. Syndication of NOUS with other EU platforms is also tackled in this WP. **WP6** is working in parallel and independently of WPs 3 to 5 however, always receiving inputs from them that affect the architecture in a bottom-up way (from technological development). Legal aspects for the cloud and the Collaboration Platform (Virtual Lab) are addressed in **WP6**. **WP6** will deliver the key achievements linked with **O5** and **O6**.

Since **WP2**, going up the WPs until **WP6**, the project is in **divergence**, exploring different research paths that are coming whole in the architecture. **WP7** presents the **convergence** part for the project. In **WP7**, a tool for assessing the impact of the data economy will be created. Most importantly, **WP7** includes the testing of the project's technologies with use-case providers. The tools created in previous WPs will be adapted, using the use case providers' data, to address the use-cases' objectives. Following this, **WP7** will provide additional (to **WP2**) requirements leading to improvements of the technologies and other (e.g., the data life cycle framework) outcomes. Finally, **WP7** will perform project assessment and deliver guidelines and a roadmap to promote continuation and adoption of NOUS outcomes.

The project is supported by the administrative **WP1** and the disseminative **WP8**. **WP8** is also responsible for delivering **O7** since, as a dissemination and exploitation WP, it has an outward view and will provide connections with other initiatives and organisations, acting as a liaison for the project and other initiatives.

3.1.2 Gantt chart

#	Code	Description	Leader	Start Month	End Month
1	WP1	Project management and coordination	AIR	1	36
1	T1.1	Administrative and financial management	AIR	1	36
2	T1.2	Technical management	AIR	1	36
3	T1.3	Quality and risk management	AIR	1	36
4	T1.4	Ethics, legal and data management	AETHON	1	36
2	WP2	Gap analysis and requirements	AIR	1	36
1	T2.1	Status quo and gap analysis of existing and emerging standards, models and architectures	NCSRD	1	6
2	T2.2	Requirements elicitation and prioritisation for the NOUS architecture and NOUS components	AETHON	4	10
3	T2.3	Identify, adapt and implement a suitable evaluation framework	USAL	4	10
4	T2.4	A novel data life cycle framework	AIR	4	36
3	WP3	#compute - The computational component of NOUS	NCSRD	6	33
1	T3.1	Leveraging the HPC Network and its resources for the NOUS	ARCTUR	6	12
2	T3.2	Seamless interaction between Quantum Computers and HPC	NCSRD	12	28
3	T3.3	Cybersecurity of the NOUS - A darknet approach and ZeroTrust architecture	AEGIS	18	33
4	WP4	#edge - The edge component of NOUS	ITML	6	36
1	T4.1	Distributed architectures working with a traditional cloud service	ITML	6	12
2	T4.2	Computing on the edge, the far edge and the cloud	UNIPI	13	36
3	T4.3	Federated Learning and Federated (Distributed) Inference framework	ITML	18	36
4	T4.4	Interoperability in the Fog using a delegated network of edge mediators	UNP	22	36
5	WP5	#data - The data component of NOUS	NET-INTRA	1	36
1	T5.1	Legal evaluation of Blockchain technologies	KUL	1	16
2	T5.2	A Blockchain-based framework for cloud and edge storage	POLITO	6	36
3	T5.3	Data spaces, data pools and data warehouses interconnection over an operational NOUS	NET-INTRA	8	36
4	T5.4	The Auto-Standardiser tool for translation to data standards	AETHON	20	36
6	WP6	Towards the creation of the NOUS architecture	AETHON	6	36
1	T6.1	Design and develop a high level architecture that brings #compute, #edge and #data together	NET-INTRA	6	36
2	T6.2	Utilising and delivering open source software and open architecture with and for the NOUS	ECL	6	36
3	T6.3	Cloud and edge legal challenges	KUL	16	36
4	T6.4	A Virtual Lab for data collaboration	AETHON	20	36
5	T6.5	Tools for Syndication between NOUS instances and other platforms	AIR	24	36
7	WP7	The data economy and use-cases	POLITO	18	36
1	T7.1	The Economy of Data - A quantification framework for data economy assessment	POLITO	18	36
2	T7.2	Setting up the use-cases and testing the developed technologies	NCSRD	18	33
3	T7.3	Evaluation and assessment	UNIMORE	27	36
4	T7.4	Roadmap and guidelines enabling the future of European Cloud that fosters data sharing	AIR	30	36
5	T7.5	Fostering human-centered practices for a data-agile economy	INESC TEC	18	36
8	WP8	Communication, dissemination, exploitation and liaison	F6S	1	36
1	T8.1	Communication, dissemination, exploitation, innovation plans & strategies	F6S	1	6
2	T8.2	Communication and dissemination implementation and monitoring	F6S	6	36
3	T8.3	Exploitation and innovation management, implementation and monitoring	F6S	8	36
4	T8.4	Clustering, networking, synergies with other projects & international cooperation	UNIMORE	1	36

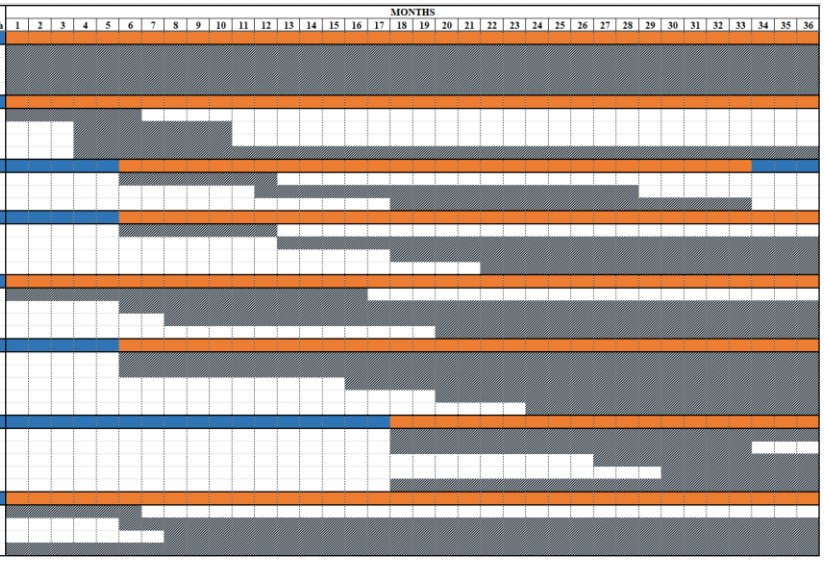

Figure 4: Gantt Chart of the NOUS project

3.1.3 *'Subcontracting costs' items*

Not applicable

3.1.4 *'Purchase costs' items (travel and subsistence, equipment, and other goods, works and services)*

4 – NCSRD		
	Cost (€)	Justification
Travel and subsistence	20.000,00	Attending project meetings and international conferences (accommodation, travel) over the project duration
Equipment	25.000,00	Storage facility and server for fast connection with HPC
Other goods, works and services	26.500,00	Open access costs: 3.000,00€, Organising 2 Workshops for T2.1 & T7.2: 8.000,00€ (4.000,00€ each), Purchase of market apps and cost of measurements on materials: 10.000,00€, Conferences registrations: 2.500,00€, CFS costs: 3.000,00€
Total	71.500,00	

3.1.5 *'Other costs categories' items (e.g., internally invoiced goods and services)*

Not applicable

3.1.6 *'In-kind contributions' provided by third parties*

21 – INESC TEC			
Third party name	**Category**	**Cost (€)**	**Justification**
Universidade do Porto (UP)	Seconded personnel	20.000,00	INESC TEC envisages the use of in-kind contributions provided by third parties free of charge (Article 9.2 of the General Model Grant Agreement). For all contractual purposes INESC TEC assumes full responsibility for the involvement of these researchers in the project, since they are members of its research team. INESC TEC will declare the costs incurred by the third parties for the seconded persons as eligible in accordance with Article 6.1. of the General Model Grant Agreement.
Universidade de Trás-os-Montes e Alto Douro (UTAD)	Seconded personnel	58.000,00	

3.2 *Capacity of participants and consortium as a whole*

The NOUS consortium consists of partners with diverse expertise and of different backgrounds to match the ambitious character of the project and the high standards in technical, scientific and professional terms. All members of the consortium have considerable expertise in successfully managing large R&D projects, acquired through participation in previous and on-going European and national R&D programmes.

The NOUS consortium was created to balance the high-level research that will be performed, the deep industry knowledge, the solutions development, implementation and testing, and the commercial exploitation of project results. The choice of the specific partners was based on criteria such as expertise, matching the roles required in the project, strategic impact, dissemination potential and strong industrial background, previous collaboration, commitment, availability, flexibility and adaptability.

The consortium consists of 21 partners spread across 11 EU countries (Spain, Belgium, Luxembourg, Germany, France, Italy, Portugal, Greece, Cyprus, Slovenia, Ireland), as shown in Figure 5. The partners are from Research, Academia, Industry, IT and Knowledge Transfer sectors. The inclusive consortium reflects the persistent quest to bring together a multidisciplinary partnership – scientific, technological, industrial – to design, develop and conduct activities that correspond to real needs and can generate lasting impact.

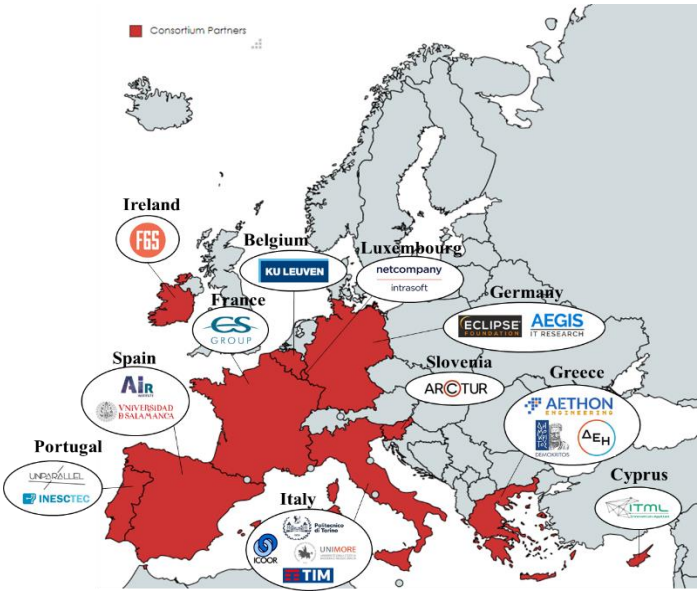

*Figure SEQ Figure * ARABIC5: Geographic repartition of*

In particular, the NOUS **coordinator** (AIR) has great experience from previous EU projects and the capacity to manage the project **[WP1]**. Several partners will work

horizontally, contributing **technically** and ensuring the quality of the project results (AIR, USAL, NCSRD, UNIMORE, AETHON) **[WP2]**. The **research and technology development** will be performed by technical experts in each field (#compute **[WP3]**: NCSRD, ARCTUR, #edge **[WP4]**: ITML, UNIPI, UNP, #data **[WP5]**: NET-INTRA, POLITO, Cloud architecture **[WP6]**: AETHON, AIR, NET-INTRA, HPE). The **Cybersecurity** aspects (AEGIS), the **Legal** ones (KUL), as well as the **SSH and Gender** aspects (INESC TEC) will be tackled across the project by the relevant partners. Certain partners will be focused on the 4 **use cases [WP7]** (Mobility: POLITO, UNIMORE, TIM, Energy: PPC, AETHON, Civil Protection: CS, Science data: NCSRD, HPE, PPC). A **Virtual Lab** for data collaboration will be delivered by AETHON, while the **Open Science** aspects will be addressed by ECL. Finally, **Clustering, Networking and Synergies** and **Communication, Dissemination and Exploitation [WP8]** (F6S, AIR, ECL, ICOOR, UNP). are extremely important for NOUS and will be led by F6S, yet the activities include the participation of all partners.

4 Ethics self-assessment

4.1 Ethical dimension of the objectives, methodology and likely impact

NOUS will design and monitor procedures and protocols for handling ethical issues related to the project, to assure that the project carries out responsible research practice.

There are three main ethical issues to be dealt with within the project:

- Involvement of participants in the co-creation activities
- Collection and processing of personal and sensitive data
- The use of artificial intelligence.

All these aspects will be addressed during the project by:

- Respecting all applicable laws and regulations (e.g., GDPR)
- Respecting ethical principles and values (e.g., FAIR).

There is a dedicated task for setting out the principles for conducting all the project actions with a high level of integrity and ethics (T1.4) led by AETHON. This task will be dealing with security, ethics and data issues in a proper manner. Responsible and ethical guidelines will be provided to the consortium to ensure the appropriate treatment of data protection and privacy issues, providing the Data Management Plan (DMP) and ensuring the FAIR principles are pursued.

In any case, special care will be given to address any ethical concerns and avoid potential misuse of Data and AI. Moreover, the project activities will not cause (physical or mental) harm to participants, or any environmental damage, stigmatisation of particular social groups, political or financial adverse consequences or misuse.

Moreover, during stakeholders engagement, emphasis will be given to including vulnerable population groups (i.e., women, disabled, elderly) to make sure that the results we provide do not exclude these sensitive groups.

4.2 Compliance with ethical principles and relevant legislations

The European Code of Conduct for Research Integrity (by Allea) will be used as framework for the research settings, together with Data Protection legislation and regulation. NOUS will ensure compliance with the EU GDPR and national data protection legislation throughout its activities.

The Consortium will ensure that all the procedures concerning personal data will be carried out with a high protection of the rights of persons who are taking part in these actions, avoiding potential problems in advance, and ensuring the compliance with the EU/ National legislation and principles. All consortium partners will comply, among others, with the requirements of the General Data Protection Regulation (GDPR).

The FAIR principles will be implemented for said data to provide indability, Accessibility, Interoperability and Reusability, as described in ch. 1.2.7 of Part B.

There is a dedicated task for setting out all these principles for conducting all the project actions with a high level of integrity and ethics (**T1.4**) led by AETHON and a specific deliverable that establishes the procedures to be followed by the NOUS partners (**D1.3**).

The partners' Ethics Committees will also ensure the compliance with ethical principles, while the partners' legal departments will ensure the compliance with relevant legislations.

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ANNEX 2

ESTIMATED BUDGET FOR THE ACTION

	Estimated eligible[1] costs (per budget category)										Estimated EU contribution[2]			
	Direct costs								Indirect costs	Total costs	EU contribution to eligible costs			Maximum grant amount[6]
	A. Personnel costs			B. Subcontracting costs	C. Purchase costs			D. Other cost categories	E. Indirect costs[3]		Funding rate %[4]	Maximum EU contribution[5]	Requested EU contribution	
	A.1 Employees (or equivalent) A.2 Natural persons under direct contract A.3 Seconded persons		A.4 SME owners and natural person beneficiaries	B. Subcontracting	C.1 Travel and subsistence	C.2 Equipment	C.3 Other goods, works and services	D.2 Internally invoiced goods and services	E. Indirect costs					
Forms of funding	Actual costs	Unit costs (usual accounting practices)	Unit costs[7]	Actual costs	Actual costs	Actual costs	Actual costs	Unit costs (usual accounting practices)	Flat-rate costs[8]					
	a1	a2	a3	b	c1	c2	c3	d2	e = 0,25 * (a1 + a2 + a3 + c1 + c2 + c3)	f = a + b + c + d + e	U	g = f * U%	h	m
1 - AIR	429 400.00	0.00	0.00	0.00	20 000.00	0.00	28 000.00	0.00	119 350.00	596 750.00	100	596 750.00	596 750.00	596 750.00
2 - AETHON	405 900.00	0.00	0.00	0.00	20 000.00	4 000.00	13 500.00	0.00	110 850.00	554 250.00	100	554 250.00	554 250.00	554 250.00
3 - USAL	155 800.00	0.00	0.00	0.00	15 000.00	0.00	4 000.00	0.00	43 700.00	218 500.00	100	218 500.00	218 500.00	218 500.00
4 - NCSR "D"	450 000.00	0.00	0.00	0.00	20 000.00	25 000.00	26 500.00	0.00	130 375.00	651 875.00	100	651 875.00	651 875.00	651 875.00
5 - ICOOR	30 000.00	0.00	0.00	0.00	4 000.00	0.00	0.00	0.00	8 500.00	42 500.00	100	42 500.00	42 500.00	42 500.00
5.1 - POLITO	306 000.00	0.00	0.00	0.00	8 000.00	0.00	6 000.00	0.00	80 000.00	400 000.00	100	400 000.00	400 000.00	400 000.00
5.2 - UNIMORE	222 000.00	0.00	0.00	0.00	8 000.00	0.00	9 000.00	0.00	59 750.00	298 750.00	100	298 750.00	298 750.00	298 750.00
6 - TIM	266 500.00	0.00	0.00	0.00	15 000.00	0.00	0.00	0.00	70 375.00	351 875.00	100	351 875.00	351 875.00	351 875.00
7 - NET-INTRA	426 000.00	0.00	0.00	0.00	20 000.00	0.00	23 000.00	0.00	117 250.00	586 250.00	100	586 250.00	586 250.00	586 250.00
8 - ITML CY	379 500.00	0.00	0.00	0.00	20 000.00	0.00	4 000.00	0.00	100 875.00	504 375.00	100	504 375.00	504 375.00	504 375.00
9 - UNIPI	236 500.00	0.00	0.00	0.00	20 000.00	0.00	4 000.00	0.00	65 125.00	325 625.00	100	325 625.00	325 625.00	325 625.00
10 - UNPARALLEL	281 250.00	0.00	0.00	0.00	15 000.00	0.00	4 000.00	0.00	75 062.50	375 312.50	100	375 312.50	375 312.50	375 312.50
11 - KU Leuven	360 000.00	0.00	0.00	0.00	20 000.00	0.00	10 000.00	0.00	97 500.00	487 500.00	100	487 500.00	487 500.00	487 500.00
12 - HPE Hellas	412 500.00	0.00	0.00	0.00	20 000.00	0.00	4 000.00	0.00	109 125.00	545 625.00	100	545 625.00	545 625.00	545 625.00
13 - ECL	252 300.00	0.00	0.00	0.00	20 000.00	0.00	15 000.00	0.00	71 825.00	359 125.00	100	359 125.00	359 125.00	359 125.00
14 - F6S IE	235 000.00	0.00	0.00	0.00	15 000.00	0.00	20 000.00	0.00	67 500.00	337 500.00	100	337 500.00	337 500.00	337 500.00
15 - PPC	251 750.00	0.00	0.00	0.00	15 000.00	0.00	0.00	0.00	66 687.50	333 437.50	100	333 437.50	333 437.50	333 437.50
16 - ARCTUR d.o.o.	243 800.00	0.00	0.00	0.00	15 000.00	0.00	0.00	0.00	64 700.00	323 500.00	100	323 500.00	323 500.00	323 500.00
17 - AEGIS	303 400.00	0.00	0.00	0.00	15 000.00	0.00	0.00	0.00	79 600.00	398 000.00	100	398 000.00	398 000.00	398 000.00
18 - CS GROUP-FRANCE	263 500.00	0.00	0.00	0.00	15 000.00	0.00	0.00	0.00	69 625.00	348 125.00	100	348 125.00	348 125.00	348 125.00
19 - INESC TEC	275 000.00	0.00	0.00	0.00	20 000.00	8 000.00	9 000.00	0.00	78 000.00	390 000.00	100	390 000.00	390 000.00	390 000.00
Σ consortium	6 186 100.00	0.00	0.00	0.00	340 000.00	37 000.00	180 000.00	0.00	1 685 775.00	8 428 875.00		8 428 875.00	8 428 875.00	8 428 875.00

[1] See Article 6 for the eligibility conditions. All amounts must be expressed in EUR (see Article 21 for the conversion rules).
[2] The consortium remains free to decide on a different internal distribution of the EU funding (via the consortium agreement; see Article 7).
[3] Indirect costs already covered by an operating grant (received under any EU funding programme) are ineligible (see Article 6.3). Therefore, a beneficiary/affiliated entity that receives an operating grant during the action duration cannot declare indirect costs for the year(s)/reporting period(s) covered by the operating grant, unless they can demonstrate that the operating grant does not cover any costs of the action. This requires specific accounting tools. Please immediately contact us via the EU Funding & Tenders Portal for details.
[4] See Data Sheet for the funding rate(s).
[5] This is the theoretical amount of the EU contribution to costs, if the reimbursement rate is applied to all the budgeted costs. This theoretical amount is then capped by the 'maximum grant amount'.
[6] The 'maximum grant amount' is the maximum grant amount decided by the EU. It normally corresponds to the requested grant, but may be lower.
[7] See Annex 2a 'Additional information on the estimated budget' for the details (units, cost per unit).
[8] See Data Sheet for the flat-rate.

ANNEX 2a

ADDITIONAL INFORMATION ON UNIT COSTS AND CONTRIBUTIONS

SME owners/natural person beneficiaries without salary (Decision C(2020) 7115[1])

Type: unit costs

Units: days spent working on the action (rounded up or down to the nearest half-day)

Amount per unit (daily rate): calculated according to the following formula:

{EUR 5 080 / 18 days = **282,22**}
multiplied by
{country-specific correction coefficient of the country where the beneficiary is established}

The country-specific correction coefficients used are those set out in the Horizon Europe Work Programme (section Marie Skłodowska-Curie actions) in force at the time of the call (see Portal Reference Documents).

HE and Euratom Research Infrastructure actions[2]

Type: unit costs

Units[3]: see (for each access provider and installation) the unit cost table in Annex 2b

Amount per unit*: see (for each access provider and installation) the unit cost table in Annex 2b

* Amount calculated as follows:

For trans-national access:

$$\frac{\text{average annual total trans-national access costs to the installation (over past two years[4])}}{\text{average annual total quantity of trans-national access to the installation (over past two years[5])}}$$

For virtual access:

$$\frac{\text{total virtual access costs to the installation (over the last year[6])}}{\text{total quantity of virtual access to the installation (over the last year[7])}}$$

Euratom staff mobility costs[8]

Monthly living allowance

Type: unit costs

---

1 Commission Decision of 20 October 2020 authorising the use of unit costs for the personnel costs of the owners of small and medium-sized enterprises and beneficiaries that are natural persons not receiving a salary for the work carried out by themselves under an action or work programme (C(2020)7715).

2 Decision of 19 April 2021 authorising the use of unit costs for the costs of providing trans-national and virtual access in Research Infrastructure actions under the Horizon Europe Programme (2021-2027) and the Research and Training Programme of the European Atomic Energy Community (2021-2025).

3 Unit of access (e.g. beam hours, weeks of access, sample analysis) fixed by the access provider in proposal.

4 In exceptional and duly justified cases, the granting authority may agree to a different reference period.

5 In exceptional and duly justified cases, the granting authority may agree to a different reference period.

6 In exceptional and duly justified cases, the granting authority may agree to a different reference period.

7 In exceptional and duly justified cases, the granting authority may agree to a different reference period.

8 Decision of 15 March 2021 authorising the use of unit costs for mobility in co-fund actions under the Research and Training Programme of the European Atomic Energy Community (2021-2025).

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit*: see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

* Amount calculated as follows from 1 January 2021:
{**EUR 4 300** multiplied by
country-specific correction coefficient** of the country where the staff member is seconded}[9]

**Country-specific correction coefficients as from 1 January 2021[10]

EU-Member States[11]

Country / Place	Coefficient (%)
Bulgaria	59,1
Czech Rep.	85,2
Denmark	131,3
Germany Bonn Karlsruhe Munich	101,9 95,8 98 113,9
Estonia	82,3
Ireland	129
Greece	81,4
Spain	94,2
France	120,5
Croatia	75,8
Italy Varese	95 90,7
Cyprus	78,2
Latvia	77,5
Lithuania	76,6
Hungary	71,9
Malta	94,7
Netherlands	113,9
Austria	107,9
Poland	70,9
Portugal	91,1
Romania	66,6
Slovenia	86,1

[9] Unit costs for living allowances are calculated by using a method of calculation similar to that applied for the secondment to the European Commission of seconded national experts (SNEs).

[10] For the financial statements, the amount must be adjusted according to the actual place of secondment.
The revised coefficients were adopted in the Decision authorising the use of unit costs for the Fusion Programme co-fund action under the Research and training Programme of the European Atomic Energy Community 2021-2025. They are based on the 2020 Annual update of the remuneration and pensions of the officials and other servants of the European Union and the correction coefficients applied thereto (OJ C 428, 11.12.2020) to ensure purchasing power parity. The revised coefficient are applied as from 1 January 2021 through an amendment to the grant agreement.

[11] No correction coefficient shall be applicable in Belgium and Luxembourg.

Slovakia	80,6
Finland	118,4
Sweden	124,3

Third countries

Country/place	Coefficient (%)
China	82,2
India	72,3
Japan	111,8
Russia	92,7
South Korea	92,3
Switzerland	129.2
Ukraine	82.3
United Kingdom	97.6
United States	101,4 (New-York) 90,5 (Washington)

Mobility allowance

Type: Unit costs

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit: **EUR 600** per person-month; see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

Family allowance

Type: unit costs

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit: **EUR 660** per person-month; see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

Education allowance

Type: Unit costs

Units: months spent by the seconded staff member(s) on research and training in fission and fusion activities (person-month)

Amount per unit*: see (for each beneficiary/affiliated entity and secondment) the unit cost table in Annex 2b

*Amount calculated as follows from 1 January 2021:
{EUR 283.82 x number of dependent children[12]**}**

[12] For the estimated budget (Annex 2): an average should be used. (⚠ For the financial statements, the number of children (and months) must be adjusted according to the actual family status at the moment the secondment starts.)

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

AETHON ENGINEERING SINGLE MEMBER PC (AETHON), PIC 918590292, established in EMMANOUIL BENAKI 25, ATHENS 106 78, Greece,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS (‘the Agreement’)

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** (‘EU’), represented by the European Commission (‘European Commission’ or ‘granting authority’),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

UNIVERSIDAD DE SALAMANCA (USAL), PIC 999846610, established in CALLE PATIO DE ESCUELAS 1, SALAMANCA 37008, Spain,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS ('the Agreement')

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** ('EU'), represented by the European Commission ('European Commission' or 'granting authority'),

and mandates

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SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

NATIONAL CENTER FOR SCIENTIFIC RESEARCH "DEMOKRITOS" (NCSR "D"), PIC 999978239, established in END OF PATRIARCHOU GRIGORIOU E AND 27 NEAPOLEOS STREET, AGIA PARASKEVI 15341, Greece,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS ('the Agreement')

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** ('EU'), represented by the European Commission ('European Commission' or 'granting authority'),

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SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

CONSORZIO INTERUNIVERSITARIO PER L'OTTIMIZZAZIONE E LA RICERCA OPERATIVA (ICOOR), PIC 996385068, established in VIALE RISORGIMENTO 2, BOLOGNA 40136, Italy,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS ('the Agreement')

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** ('EU'), represented by the European Commission ('European Commission' or 'granting authority'),

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SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

TELECOM ITALIA SPA (TIM), PIC 999908496, established in VIA GAETANO NEGRI 1, MILANO 20123, Italy,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS (‘the Agreement’)

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** (‘EU’), represented by the European Commission (‘European Commission’ or ‘granting authority’),

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SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

NETCOMPANY-INTRASOFT SA (NET-INTRA), PIC 999702371, established in RUE NICOLAS BOVE 2B, LUXEMBOURG 1253, Luxembourg,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS ('the Agreement')

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** ('EU'), represented by the European Commission ('European Commission' or 'granting authority'),

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SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

IOTAM INTERNET OF THINGS APPLICATIONS AND MULTI LAYER DEVELOPMENT LTD (ITML CY), PIC 912549326, established in 56 GRIVA DIGENI STR ANNA TOWER FLAT/ OFFICE 3, LIMASSOL 3101, Cyprus,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS (‘the Agreement’)

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** (‘EU’), represented by the European Commission (‘European Commission’ or ‘granting authority’),

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SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

UNIVERSITA DI PISA (UNIPI), PIC 999862712, established in LUNGARNO PACINOTTI 43/44, PISA 56126, Italy,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS (‘the Agreement’)

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** (‘EU’), represented by the European Commission (‘European Commission’ or ‘granting authority’),

and mandates

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By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

UNPARALLEL INNOVATION LDA (UNPARALLEL), PIC 953318426, established in RUA DAS LENDAS ALGARVIAS LOTE 123, PORTIMAO 8500-794, Portugal,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS ('the Agreement')

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** ('EU'), represented by the European Commission ('European Commission' or 'granting authority'),

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By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

KATHOLIEKE UNIVERSITEIT LEUVEN (KU Leuven), PIC 999991334, established in OUDE MARKT 13, LEUVEN 3000, Belgium,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS ('the Agreement')

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** ('EU'), represented by the European Commission ('European Commission' or 'granting authority'),

and mandates

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By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

HEWLETT - PACKARD HELLAS ETAIREIA PERIORISMENIS EFTHYNIS (HPE Hellas), PIC 913711871, established in TZAVELLA 1-3, ATHINA 152 31, Greece,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS (‘the Agreement’)

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** (‘EU’), represented by the European Commission (‘European Commission’ or ‘granting authority’),

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SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

ECLIPSE FOUNDATION EUROPE GMBH (ECL), PIC 944758564, established in BERLINER ALLEE 47, DARMSTADT 64295, Germany,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS (‘the Agreement’)

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** (‘EU’), represented by the European Commission (‘European Commission’ or ‘granting authority’),

and mandates

the coordinator to submit and sign in its name and on its behalf any **amendments** to the Agreement, in accordance with Article 39.

By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

F6S NETWORK IRELAND LIMITED (F6S IE), PIC 900885658, established in 39 FITZWILLIAM PLACE, DUBLIN D02 ND61, Ireland,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS (‘the Agreement’)

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** (‘EU’), represented by the European Commission (‘European Commission’ or ‘granting authority’),

and mandates

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SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

DIMOSIA EPICHEIRISI ILEKTRISMOU ANONYMI ETAIREIA (PPC), PIC 999938954, established in CHALKOKONDYLI STREET 30, ATHINA 104 32, Greece,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS (‘the Agreement’)

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** (‘EU’), represented by the European Commission (‘European Commission’ or ‘granting authority’),

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By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

ARCTUR RACUNALNISKI INZENIRING DOO (ARCTUR d.o.o.), PIC 991568048, established in INDUSTRIJSKA CESTA 1A, NOVA GORICA 5000, Slovenia,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS (‘the Agreement’)

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** (‘EU’), represented by the European Commission (‘European Commission’ or ‘granting authority’),

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SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

AEGIS IT RESEARCH GMBH (AEGIS), PIC 911867804, established in HUMBOLDTSTR. 25, BRAUNSCHWEIG 38106, Germany,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS ('the Agreement')

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** ('EU'), represented by the European Commission ('European Commission' or 'granting authority'),

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SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

CS GROUP-FRANCE (CS GROUP-FRANCE), PIC 999938275, established in AVENUE GALILEE 22, LE PLESSIS ROBINSON 92350, France,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS ('the Agreement')

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** ('EU'), represented by the European Commission ('European Commission' or 'granting authority'),

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By signing this accession form, the beneficiary accepts the grant and agrees to implement it in accordance with the Agreement, with all the obligations and terms and conditions it sets out.

SIGNATURE

For the beneficiary

ANNEX 3

ACCESSION FORM FOR BENEFICIARIES

INESC TEC - INSTITUTO DE ENGENHARIADE SISTEMAS E COMPUTADORES, TECNOLOGIA E CIENCIA (INESC TEC), PIC 999513706, established in RUA DR ROBERTO FRIAS CAMPUS DA FEUP, PORTO 4200 465, Portugal,

hereby agrees

to become beneficiary

in Agreement No 101135927 — NOUS ('the Agreement')

between FUNDACION INSTITUTO INTERNACIONAL DE INVESTIGACION EN INTELIGENCIA ARTIFICIAL Y CIENCIAS DE LA COMPUTACION (AIR) **and** the **European Union** ('EU'), represented by the European Commission ('European Commission' or 'granting authority'),

and mandates

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SIGNATURE

For the beneficiary

FINANCIAL STATEMENT FOR [PARTICIPANT NAME] FOR REPORTING PERIOD [NUMBER]

	Eligible[1] costs (per budget category)																	EU contribution[2]				Revenues
	Direct costs															Indirect costs	Total costs	EU contribution to eligible costs			Total requested EU contribution	Income generated by the action
	A. Personnel costs			B. Subcontracting costs	C. Purchase costs			D. Other cost categories								E. Indirect costs[2]		Funding rate %[3]	Maximum EU contribution[4]	Requested EU contribution		
	A.1 Employees (or equivalent) A.2 Natural persons under direct contract A.3 Seconded persons		A.4 SME owners and natural person beneficiaries	B. Subcontracting	C.1 Travel and subsistence	C.2 Equipment	C.3 Other goods, works and services	[D.1 Financial support to third parties]	D.2 Internally invoiced goods and services	[D.3 Transnational access to research infrastructure unit costs]	[D.4 Virtual access to research infrastructure unit costs]	*[OPTION for HE PCP/PPI:* D.5 PCP/PPI procurement costs*]*	*[OPTION for Euratom Programme Cofund Actions:* D.6 Euratom Cofund staff mobility costs*]*	*[OPTION for HE ERC Grants:* D.7 ERC additional funding*]*	*[OPTION for HE ERC Grants:* D.8 ERC additional funding (subcontracting, FSTP and internally invoiced goods and services)*]*	E. Indirect costs						
Forms of funding	Actual costs	Unit costs (usual accounting practices)	Unit costs[5]	Actual costs	Actual costs	Actual costs	Actual costs	[Actual costs]	Unit costs (usual accounting practices)	[Unit costs[5]]	[Unit costs[5]]	[Actual costs]	[Unit costs[5]]	[Actual costs]	[Actual costs]	Flat-rate costs[6]						
	a1	a2	a3	b	c1	c2	c3	[d1a]	d2	[d3]	[d4]	[d5]	[d6]	[d7]	[d8]	e = 0,25 * (a1 + a2 + a3 ~~+ b~~ + c1 +c2 + c3 ~~+ d1a + d2 + d3 + d4 + d5[+ d6]~~ [+d7] ~~[+ d8]~~)	f = a+b+c+d+e	U	g = f*U%	h	m	n
XX – [short name beneficiary/affiliated entity]																						

The beneficiary/affiliated entity hereby confirms that:

The information provided is complete, reliable and true.

The costs and contributions declared are eligible (see Article 6).

The costs and contributions can be substantiated by adequate records and supporting documentation that will be produced upon request or in the context of checks, reviews, audits and investigations (see Articles 19, 20 and 25).

For the last reporting period: that all the revenues have been declared (see Article 22).

ⓘ Please declare all eligible costs and contributions, even if they exceed the amounts indicated in the estimated budget (see Annex 2). Only amounts that were declared in your individual financial statements can be taken into account lateron, in order to replace costs/contributions that are found to be ineligible.

[1] See Article 6 for the eligibility conditions. All amounts must be expressed in EUR (see Article 21 for the conversion rules).

[2] If you have also received an EU operating grant during this reporting period, you cannot claim indirect costs - unless you can demonstrate that the operating grant does not cover any costs of the action. This requires specific accounting tools. Please contact us immediately via the Funding & Tenders Portal for details.

[3] See Data Sheet for the reimbursement rate(s).

[4] This is the *theoretical* amount of EU contribution to costs that the system calculates automatically (by multiplying the reimbursement rates by the costs declared). The amount you request (in the column 'requested EU contribution') may be less.

[5] See Annex 2a 'Additional information on the estimated budget' for the details (units, cost per unit).

[6] See Data Sheet for the flat-rate.

ANNEX 5

SPECIFIC RULES

CONFIDENTIALITY AND SECURITY (— ARTICLE 13)

Sensitive information with security recommendation

Sensitive information with a security recommendation must comply with the additional requirements imposed by the granting authority.

Before starting the action tasks concerned, the beneficiaries must have obtained all approvals or other mandatory documents needed for implementing the task. The documents must be kept on file and be submitted upon request by the coordinator to the granting authority. If they are not in English, they must be submitted together with an English summary.

For requirements restricting disclosure or dissemination, the information must be handled in accordance with the recommendation and may be disclosed or disseminated only after written approval from the granting authority.

EU classified information

If EU classified information is used or generated by the action, it must be treated in accordance with the security classification guide (SCG) and security aspect letter (SAL) set out in Annex 1 and Decision 2015/444[1] and its implementing rules — until it is declassified.

Deliverables which contain EU classified information must be submitted according to special procedures agreed with the granting authority.

Action tasks involving EU classified information may be subcontracted only with prior explicit written approval from the granting authority and only to entities established in an EU Member State or in a non-EU country with a security of information agreement with the EU (or an administrative arrangement with the Commission).

EU classified information may not be disclosed to any third party (including participants involved in the action implementation) without prior explicit written approval from the granting authority.

ETHICS (— ARTICLE 14)

Ethics and research integrity

The beneficiaries must carry out the action in compliance with:

- ethical principles (including the highest standards of research integrity)

[1] Commission Decision 2015/444/EC, Euratom of 13 March 2015 on the security rules for protecting EU classified information (OJ L 72, 17.3.2015, p. 53).

and

- applicable EU, international and national law, including the EU Charter of Fundamental Rights and the European Convention for the Protection of Human Rights and Fundamental Freedoms and its Supplementary Protocols.

No funding can be granted, within or outside the EU, for activities that are prohibited in all Member States. No funding can be granted in a Member State for an activity which is forbidden in that Member State.

The beneficiaries must pay particular attention to the principle of proportionality, the right to privacy, the right to the protection of personal data, the right to the physical and mental integrity of persons, the right to non-discrimination, the need to ensure protection of the environment and high levels of human health protection.

The beneficiaries must ensure that the activities under the action have an exclusive focus on civil applications.

The beneficiaries must ensure that the activities under the action do not:

- aim at human cloning for reproductive purposes
- intend to modify the genetic heritage of human beings which could make such modifications heritable (with the exception of research relating to cancer treatment of the gonads, which may be financed)
- intend to create human embryos solely for the purpose of research or for the purpose of stem cell procurement, including by means of somatic cell nuclear transfer, or
- lead to the destruction of human embryos (for example, for obtaining stem cells).

Activities involving research on human embryos or human embryonic stem cells may be carried out only if:

- they are set out in Annex 1 or
- the coordinator has obtained explicit approval (in writing) from the granting authority.

In addition, the beneficiaries must respect the fundamental principle of research integrity — as set out in the European Code of Conduct for Research Integrity[2].

This implies compliance with the following principles:

- reliability in ensuring the quality of research reflected in the design, the methodology, the analysis and the use of resources
- honesty in developing, undertaking, reviewing, reporting and communicating research in a transparent, fair and unbiased way

[2] European Code of Conduct for Research Integrity of ALLEA (All European Academies).

- respect for colleagues, research participants, society, ecosystems, cultural heritage and the environment
- accountability for the research from idea to publication, for its management and organisation, for training, supervision and mentoring, and for its wider impacts

and means that beneficiaries must ensure that persons carrying out research tasks follow the good research practices including ensuring, where possible, openness, reproducibility and traceability and refrain from the research integrity violations described in the Code.

Activities raising ethical issues must comply with the additional requirements formulated by the ethics panels (including after checks, reviews or audits; see Article 25).

Before starting an action task raising ethical issues, the beneficiaries must have obtained all approvals or other mandatory documents needed for implementing the task, notably from any (national or local) ethics committee or other bodies such as data protection authorities.

The documents must be kept on file and be submitted upon request by the coordinator to the granting authority. If they are not in English, they must be submitted together with an English summary, which shows that the documents cover the action tasks in question and includes the conclusions of the committee or authority concerned (if any).

VALUES (— ARTICLE 14)

Gender mainstreaming

The beneficiaries must take all measures to promote equal opportunities between men and women in the implementation of the action and, where applicable, in line with the gender equality plan. They must aim, to the extent possible, for a gender balance at all levels of personnel assigned to the action, including at supervisory and managerial level.

INTELLECTUAL PROPERTY RIGHTS (IPR) — BACKGROUND AND RESULTS — ACCESS RIGHTS AND RIGHTS OF USE (— ARTICLE 16)

Definitions

Access rights — Rights to use results or background.

Dissemination — The public disclosure of the results by appropriate means, other than resulting from protecting or exploiting the results, including by scientific publications in any medium.

Exploit(ation) — The use of results in further research and innovation activities other than those covered by the action concerned, including among other things, commercial exploitation such as developing, creating, manufacturing and marketing a product or process, creating and providing a service, or in standardisation activities.

Fair and reasonable conditions — Appropriate conditions, including possible financial terms or royalty-free conditions, taking into account the specific circumstances of the request for access, for example the actual or potential value of the results or background to which access is requested and/or the scope, duration or other characteristics of the exploitation envisaged.

FAIR principles — ‘findability’, ‘accessibility’, ‘interoperability’ and ‘reusability’.

Open access — Online access to research outputs provided free of charge to the end-user.

Open science — An approach to the scientific process based on open cooperative work, tools and diffusing knowledge.

Research data management — The process within the research lifecycle that includes the organisation, storage, preservation, security, quality assurance, allocation of persistent identifiers (PIDs) and rules and procedures for sharing of data including licensing.

Research outputs — Results to which access can be given in the form of scientific publications, data or other engineered results and processes such as software, algorithms, protocols, models, workflows and electronic notebooks.

Scope of the obligations

For this section, references to ‘beneficiary’ or ‘beneficiaries’ do not include affiliated entities (if any).

Agreement on background

The beneficiaries must identify in a written agreement the background as needed for implementing the action or for exploiting its results.

Where the call conditions restrict control due to strategic interests reasons, background that is subject to control or other restrictions by a country (or entity from a country) which is not one of the eligible countries or target countries set out in the call conditions and that impact the exploitation of the results (i.e. would make the exploitation of the results subject to control or restrictions) must not be used and must be explicitly excluded from it in the agreement on background — unless otherwise agreed with the granting authority.

Ownership of results

Results are owned by the beneficiaries that generate them.

However, two or more beneficiaries own results jointly if:

- they have jointly generated them and
- it is not possible to:
    - establish the respective contribution of each beneficiary, or
    - separate them for the purpose of applying for, obtaining or maintaining their protection.

The joint owners must agree — in writing — on the allocation and terms of exercise of their joint ownership (‘**joint ownership agreement**’), to ensure compliance with their obligations under this Agreement.

Unless otherwise agreed in the joint ownership agreement or consortium agreement, each joint owner may grant non-exclusive licences to third parties to exploit the jointly-owned results (without any right to sub-license), if the other joint owners are given:

- at least 45 days advance notice and
- fair and reasonable compensation.

The joint owners may agree — in writing — to apply another regime than joint ownership.

If third parties (including employees and other personnel) may claim rights to the results, the beneficiary concerned must ensure that those rights can be exercised in a manner compatible with its obligations under the Agreement.

The beneficiaries must indicate the owner(s) of the results (results ownership list) in the final periodic report.

Protection of results

Beneficiaries which have received funding under the grant must adequately protect their results — for an appropriate period and with appropriate territorial coverage — if protection is possible and justified, taking into account all relevant considerations, including the prospects for commercial exploitation, the legitimate interests of the other beneficiaries and any other legitimate interests.

Exploitation of results

Beneficiaries which have received funding under the grant must — up to four years after the end of the action (see Data Sheet, Point 1) — use their best efforts to exploit their results directly or to have them exploited indirectly by another entity, in particular through transfer or licensing.

If, despite a beneficiary's best efforts, the results are not exploited within one year after the end of the action, the beneficiaries must (unless otherwise agreed in writing with the granting authority) use the Horizon Results Platform to find interested parties to exploit the results.

If results are incorporated in a standard, the beneficiaries must (unless otherwise agreed with the granting authority or unless it is impossible) ask the standardisation body to include the funding statement (see Article 17) in (information related to) the standard.

Additional exploitation obligations

Where the call conditions impose additional exploitation obligations (including obligations linked to the restriction of participation or control due to strategic assets, interests, autonomy or security reasons), the beneficiaries must comply with them — up to four years after the end of the action (see Data Sheet, Point 1).

Where the call conditions impose additional exploitation obligations in case of a public emergency, the beneficiaries must (if requested by the granting authority) grant for a limited period of time specified in the request, non-exclusive licences — under fair and reasonable conditions — to their results to legal entities that need the results to address the public emergency and commit to rapidly and broadly exploit the resulting products and services at fair and reasonable conditions. This provision applies up to four years after the end of the action (see Data Sheet, Point 1).

Additional information obligation relating to standards

Where the call conditions impose additional information obligations relating to possible standardisation, the beneficiaries must — up to four years after the end of the action (see Data Sheet, Point 1) — inform the granting authority, if the results could reasonably be expected to contribute to European or international standards.

Transfer and licensing of results

Transfer of ownership

The beneficiaries may transfer ownership of their results, provided this does not affect compliance with their obligations under the Agreement.

The beneficiaries must ensure that their obligations under the Agreement regarding their results are passed on to the new owner and that this new owner has the obligation to pass them on in any subsequent transfer.

Moreover, they must inform the other beneficiaries with access rights of the transfer at least 45 days in advance (or less if agreed in writing), unless agreed otherwise in writing for specifically identified third parties including affiliated entities or unless impossible under the applicable law. This notification must include sufficient information on the new owner to enable the beneficiaries concerned to assess the effects on their access rights. The beneficiaries may object within 30 days of receiving notification (or less if agreed in writing), if they can show that the transfer would adversely affect their access rights. In this case, the transfer may not take place until agreement has been reached between the beneficiaries concerned.

Granting licences

The beneficiaries may grant licences to their results (or otherwise give the right to exploit them), including on an exclusive basis, provided this does not affect compliance with their obligations.

Exclusive licences for results may be granted only if all the other beneficiaries concerned have waived their access rights.

Granting authority right to object to transfers or licensing — Horizon Europe actions

Where the call conditions in Horizon Europe actions provide for the right to object to transfers or licensing, the granting authority may — up to four years after the end of the action (see Data Sheet, Point 1) — object to a transfer of ownership or the exclusive licensing of results, if:

- the beneficiaries which generated the results have received funding under the grant
- it is to a legal entity established in a non-EU country not associated with Horizon Europe, and
- the granting authority considers that the transfer or licence is not in line with EU interests.

Beneficiaries that intend to transfer ownership or grant an exclusive licence must formally notify the granting authority before the intended transfer or licensing takes place and:

- identify the specific results concerned
- describe in detail the new owner or licensee and the planned or potential exploitation of the results, and
- include a reasoned assessment of the likely impact of the transfer or licence on EU interests, in particular regarding competitiveness as well as consistency with ethical principles and security considerations.

The granting authority may request additional information.

If the granting authority decides to object to a transfer or exclusive licence, it must formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information it has requested).

No transfer or licensing may take place in the following cases:

- pending the granting authority decision, within the period set out above
- if the granting authority objects
- until the conditions are complied with, if the granting authority objection comes with conditions.

A beneficiary may formally notify a request to waive the right to object regarding intended transfers or grants to a specifically identified third party, if measures safeguarding EU interests are in place. If the granting authority agrees, it will formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information requested).

Granting authority right to object to transfers or licensing — Euratom actions

Where the call conditions in Euratom actions provide for the right to object to transfers or licensing, the granting authority may — up to four years after the end of the action (see Data Sheet, Point 1) — object to a transfer of ownership or the exclusive or non-exclusive licensing of results, if:

- the beneficiaries which generated the results have received funding under the grant
- it is to a legal entity established in a non-EU country not associated to the Euratom Research and Training Programme 2021-2025 and
- the granting authority considers that the transfer or licence is not in line with the EU interests.

Beneficiaries that intend to transfer ownership or grant a licence must formally notify the granting authority before the intended transfer or licensing takes place and:

- identify the specific results concerned
- describe in detail the results, the new owner or licensee and the planned or potential exploitation of the results, and
- include a reasoned assessment of the likely impact of the transfer or licence on EU interests, in particular regarding competitiveness as well as consistency with

ethical principles and security considerations (including the defence interests of the EU Member States under Article 24 of the Euratom Treaty).

The granting authority may request additional information.

If the granting authority decides to object to a transfer or licence, it will formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information requested).

No transfer or licensing may take place in the following cases:

- pending the granting authority decision, within the period set out above
- if the granting authority objects
- until the conditions are complied with, if the granting authority objection comes with conditions.

A beneficiary may formally notify a request to waive the right to object regarding intended transfers or grants to a specifically identified third party, if measures safeguarding EU interests are in place. If the granting authority agrees, it will formally notify the beneficiary concerned within 60 days of receiving notification (or any additional information requested).

Limitations to transfers and licensing due to strategic assets, interests, autonomy or security reasons of the EU and its Member States

Where the call conditions restrict participation or control due to strategic assets, interests, autonomy or security reasons, the beneficiaries may not transfer ownership of their results or grant licences to third parties which are established in countries which are not eligible countries or target countries set out in the call conditions (or, if applicable, are controlled by such countries or entities from such countries) — unless they have requested and received prior approval by the granting authority.

The request must:

- identify the specific results concerned
- describe in detail the new owner and the planned or potential exploitation of the results, and
- include a reasoned assessment of the likely impact of the transfer or license on the strategic assets, interests, autonomy or security of the EU and its Member States.

The granting authority may request additional information.

Access rights to results and background

Exercise of access rights — Waiving of access rights — No sub-licensing

Requests to exercise access rights and the waiver of access rights must be in writing.

Unless agreed otherwise in writing with the beneficiary granting access, access rights do not include the right to sub-license.

If a beneficiary is no longer involved in the action, this does not affect its obligations to grant access.

If a beneficiary defaults on its obligations, the beneficiaries may agree that that beneficiary no longer has access rights.

Access rights for implementing the action

The beneficiaries must grant each other access — on a royalty-free basis — to background needed to implement their own tasks under the action, unless the beneficiary that holds the background has — before acceding to the Agreement —:

- informed the other beneficiaries that access to its background is subject to restrictions, or
- agreed with the other beneficiaries that access would not be on a royalty-free basis.

The beneficiaries must grant each other access — on a royalty-free basis — to results needed for implementing their own tasks under the action.

Access rights for exploiting the results

The beneficiaries must grant each other access — under fair and reasonable conditions — to results needed for exploiting their results.

The beneficiaries must grant each other access — under fair and reasonable conditions — to background needed for exploiting their results, unless the beneficiary that holds the background has — before acceding to the Agreement — informed the other beneficiaries that access to its background is subject to restrictions.

Requests for access must be made — unless agreed otherwise in writing — up to one year after the end of the action (see Data Sheet, Point 1).

Access rights for entities under the same control

Unless agreed otherwise in writing by the beneficiaries, access to results and, subject to the restrictions referred to above (if any), background must also be granted — under fair and reasonable conditions — to entities that:

- are established in an EU Member State or Horizon Europe associated country
- are under the direct or indirect control of another beneficiary, or under the same direct or indirect control as that beneficiary, or directly or indirectly controlling that beneficiary and
- need the access to exploit the results of that beneficiary.

Unless agreed otherwise in writing, such requests for access must be made by the entity directly to the beneficiary concerned.

Requests for access must be made — unless agreed otherwise in writing — up to one year after the end of the action (see Data Sheet, Point 1).

Access rights for the granting authority, EU institutions, bodies, offices or agencies and national authorities to results for policy purposes — Horizon Europe actions

In Horizon Europe actions, the beneficiaries which have received funding under the grant must grant access to their results — on a royalty-free basis — to the granting authority, EU institutions, bodies, offices or agencies for developing, implementing and monitoring EU policies or programmes. Such access rights do not extend to beneficiaries' background.

Such access rights are limited to non-commercial and non-competitive use.

For actions under the cluster 'Civil Security for Society', such access rights also extend to national authorities of EU Member States for developing, implementing and monitoring their policies or programmes in this area. In this case, access is subject to a bilateral agreement to define specific conditions ensuring that:

- the access rights will be used only for the intended purpose and
- appropriate confidentiality obligations are in place.

Moreover, the requesting national authority or EU institution, body, office or agency (including the granting authority) must inform all other national authorities of such a request.

Access rights for the granting authority, Euratom institutions, funding bodies or the Joint Undertaking Fusion for Energy — Euratom actions

In Euratom actions, the beneficiaries which have received funding under the grant must grant access to their results — on a royalty-free basis — to the granting authority, Euratom institutions, funding bodies or the Joint Undertaking Fusion for Energy for developing, implementing and monitoring Euratom policies and programmes or for compliance with obligations assumed through international cooperation with non-EU countries and international organisations.

Such access rights include the right to authorise third parties to use the results in public procurement and the right to sub-license and are limited to non-commercial and non-competitive use.

Additional access rights

Where the call conditions impose additional access rights, the beneficiaries must comply with them.

COMMUNICATION, DISSEMINATION, OPEN SCIENCE AND VISIBILITY (— ARTICLE 17)

Dissemination

Dissemination of results

The beneficiaries must disseminate their results as soon as feasible, in a publicly available format, subject to any restrictions due to the protection of intellectual property, security rules or legitimate interests.

A beneficiary that intends to disseminate its results must give at least 15 days advance notice to the other beneficiaries (unless agreed otherwise), together with sufficient information on the results it will disseminate.

Any other beneficiary may object within (unless agreed otherwise) 15 days of receiving notification, if it can show that its legitimate interests in relation to the results or background would be significantly harmed. In such cases, the results may not be disseminated unless appropriate steps are taken to safeguard those interests.

Additional dissemination obligations

Where the call conditions impose additional dissemination obligations, the beneficiaries must also comply with those.

Open Science

Open science: open access to scientific publications

The beneficiaries must ensure open access to peer-reviewed scientific publications relating to their results. In particular, they must ensure that:

- at the latest at the time of publication, a machine-readable electronic copy of the published version or the final peer-reviewed manuscript accepted for publication, is deposited in a trusted repository for scientific publications

- immediate open access is provided to the deposited publication via the repository, under the latest available version of the Creative Commons Attribution International Public Licence (CC BY) or a licence with equivalent rights; for monographs and other long-text formats, the licence may exclude commercial uses and derivative works (e.g. CC BY-NC, CC BY-ND) and

- information is given via the repository about any research output or any other tools and instruments needed to validate the conclusions of the scientific publication.

Beneficiaries (or authors) must retain sufficient intellectual property rights to comply with the open access requirements.

Metadata of deposited publications must be open under a Creative Common Public Domain Dedication (CC 0) or equivalent, in line with the FAIR principles (in particular machine-actionable) and provide information at least about the following: publication (author(s), title, date of publication, publication venue); Horizon Europe or Euratom funding; grant project name, acronym and number; licensing terms; persistent identifiers for the publication, the authors involved in the action and, if possible, for their organisations and the grant. Where applicable, the metadata must include persistent identifiers for any research output or any other tools and instruments needed to validate the conclusions of the publication.

Only publication fees in full open access venues for peer-reviewed scientific publications are eligible for reimbursement.

Open science: research data management

The beneficiaries must manage the digital research data generated in the action (‘data’) responsibly, in line with the FAIR principles and by taking all of the following actions:

- establish a data management plan (‘DMP’) (and regularly update it)

- as soon as possible and within the deadlines set out in the DMP, deposit the data in a trusted repository; if required in the call conditions, this repository must be federated in the EOSC in compliance with EOSC requirements

- as soon as possible and within the deadlines set out in the DMP, ensure open access — via the repository — to the deposited data, under the latest available version of the Creative Commons Attribution International Public License (CC BY) or Creative Commons Public Domain Dedication (CC 0) or a licence with equivalent rights, following the principle 'as open as possible as closed as necessary', unless providing open access would in particular:

    - be against the beneficiary's legitimate interests, including regarding commercial exploitation, or

    - be contrary to any other constraints, in particular the EU competitive interests or the beneficiary's obligations under this Agreement; if open access is not provided (to some or all data), this must be justified in the DMP

- provide information via the repository about any research output or any other tools and instruments needed to re-use or validate the data.

Metadata of deposited data must be open under a Creative Common Public Domain Dedication (CC 0) or equivalent (to the extent legitimate interests or constraints are safeguarded), in line with the FAIR principles (in particular machine-actionable) and provide information at least about the following: datasets (description, date of deposit, author(s), venue and embargo); Horizon Europe or Euratom funding; grant project name, acronym and number; licensing terms; persistent identifiers for the dataset, the authors involved in the action, and, if possible, for their organisations and the grant. Where applicable, the metadata must include persistent identifiers for related publications and other research outputs.

Open science: additional practices

Where the call conditions impose additional obligations regarding open science practices, the beneficiaries must also comply with those.

Where the call conditions impose additional obligations regarding the validation of scientific publications, the beneficiaries must provide (digital or physical) access to data or other results needed for validation of the conclusions of scientific publications, to the extent that their legitimate interests or constraints are safeguarded (and unless they already provided the (open) access at publication).

Where the call conditions impose additional open science obligations in case of a public emergency, the beneficiaries must (if requested by the granting authority) immediately deposit any research output in a repository and provide open access to it under a CC BY licence, a Public Domain Dedication (CC 0) or equivalent. As an exception, if the access would be against the beneficiaries' legitimate interests, the beneficiaries must grant non-exclusive licenses — under fair and reasonable conditions — to legal entities that need the research output to address the public emergency and commit to rapidly and broadly exploit the resulting products and services at fair and reasonable conditions. This provision applies up to four years after the end of the action (see Data Sheet, Point 1).

Plan for the exploitation and dissemination of results including communication activities

Unless excluded by the call conditions, the beneficiaries must provide and regularly update a plan for the exploitation and dissemination of results including communication activities.

SPECIFIC RULES FOR CARRYING OUT THE ACTION (— ARTICLE 18)

Implementation in case of restrictions due to strategic assets, interests, autonomy or security of the EU and its Member States

Where the call conditions restrict participation or control due to strategic assets, interests, autonomy or security, the beneficiaries must ensure that none of the entities that participate as affiliated entities, associated partners, subcontractors or recipients of financial support to third parties are established in countries which are not eligible countries or target countries set out in the call conditions (or, if applicable, are controlled by such countries or entities from such countries) — unless otherwise agreed with the granting authority.

The beneficiaries must moreover ensure that any cooperation with entities established in countries which are not eligible countries or target countries set out in the call conditions (or, if applicable, are controlled by such countries or entities from such countries) does not affect the strategic assets, interests, autonomy or security of the EU and its Member States.

Recruitment and working conditions for researchers

The beneficiaries must take all measures to implement the principles set out in the Commission Recommendation on the European Charter for Researchers and the Code of Conduct for the Recruitment of Researchers[3], in particular regarding:

- working conditions
- transparent recruitment processes based on merit, and
- career development.

The beneficiaries must ensure that researchers and all participants involved in the action are aware of them.

Specific rules for access to research infrastructure activities

Definitions

Research Infrastructures — Facilities that provide resources and services for the research communities to conduct research and foster innovation in their fields. This definition includes the associated human resources, and it covers major equipment or sets of instruments; knowledge-related facilities such as collections, archives or scientific data infrastructures; computing systems, communication networks, and any other infrastructure, of a unique nature and open to external users, essential to achieve excellence in research and innovation. Where relevant, they may be used beyond research, for example

[3] Commission Recommendation 2005/251/EC of 11 March 2005 on the European Charter for Researchers and on a Code of Conduct for the Recruitment of Researchers (OJ L 75, 22.3.2005, p. 67).

for education or public services, and they may be ‘single-sited’, ‘virtual’ or ‘distributed’[4]:

When implementing access to research infrastructure activities, the beneficiaries must respect the following conditions:

- for transnational access:

  - access which must be provided:

    The access must be free of charge, transnational access to research infrastructure or installations for selected user-groups.

    The access must include the logistical, technological and scientific support and the specific training that is usually provided to external researchers using the infrastructure. Transnational access can be either in person (hands-on), provided to selected users that visit the installation to make use of it, or remote, through the provision to selected user-groups of remote scientific services (e.g. provision of reference materials or samples, remote access to a high-performance computing facility).

  - categories of users that may have access:

    Transnational access must be provided to selected user-groups, i.e. teams of one or more researchers (users).

    The majority of the users must work in a country other than the country(ies) where the installation is located (unless access is provided by an international organisation, the Joint Research Centre (JRC), an ERIC or similar legal entity).

    Only user groups that are allowed to disseminate the results they have generated under the action may benefit from the access (unless the users are working for SMEs).

    Access for user groups with a majority of users not working in a EU Member State or Horizon Europe associated country is limited to 20% of the total amount of units of access provided under the grant (unless a higher percentage is foreseen in Annex 1).

  - procedure and criteria for selecting user groups:

    The user groups must request access by submitting (in writing) a description of the work that they wish to carry out and the names, nationalities and home institutions of the users.

    The user groups must be selected by (one or more) selection panels set up by the consortium.

[4] See Article 2(1) of the Horizon Europe Framework Programme Regulation 2021/695.

The selection panels must be composed of international experts in the field, at least half of them independent from the consortium (unless otherwise specified in Annex 1).

The selection panels must assess all proposals received and recommend a short-list of the user groups that should benefit from access.

The selection panels must base their selection on scientific merit, taking into account that priority should be given to user groups composed of users who:

- have not previously used the installation and
- are working in countries where no equivalent research infrastructure exist.

It will apply the principles of transparency, fairness and impartiality.

Where the call conditions impose additional rules for the selection of user groups, the beneficiaries must also comply with those.

- other conditions:

The beneficiaries must request written approval from the granting authority for the selection of user groups requiring visits to the installations exceeding 3 months (unless such visits are foreseen in Annex 1).

In addition, the beneficiaries must:

- advertise widely, including on a their websites, the access offered under the Agreement
- promote equal opportunities in advertising the access and take into account the gender dimension when defining the support provided to users
- ensure that users comply with the terms and conditions of the Agreement
- ensure that its obligations under Articles 12, 13, 17 and 33 also apply to the users
- keep records of the names, nationalities, and home institutions of users, as well as the nature and quantity of access provided to them

- for virtual access:

- access which must be provided:

The access must be free of charge, virtual access to research infrastructure or installations.

‘Virtual access’ means open and free access through communication networks to digital resources and services needed for research, without selecting the users to whom access is provided.

The access must include the support that is usually provided to external users.

Where allowed by the call conditions, beneficiaries may in justified cases define objective eligibility criteria (e.g. affiliation to a research or academic institution) for specific users.

- other conditions:

  The beneficiaries must have the virtual access services assessed periodically by a board composed of international experts in the field, at least half of whom must be independent from the consortium (unless otherwise specified in Annex 1). For this purpose, information and statistics on the users and the nature and quantity of the access provided, must be made available to the board.

  The beneficiaries must advertise widely, including on a dedicated website, the access offered under the grant and the eligibility criteria, if any.

  Where the call conditions impose additional traceability[5] obligations, information on the traceability of the users and the nature and quantity of access must be provided by the beneficiaries.

These obligations apply regardless of the form of funding or budget categories used to declare the costs (unit costs or actual costs or a combination of the two).

[5] According to the definition given in ISO 9000, i.e.: "Traceability is the ability to trace the history, application, use and location of an item or its characteristics through recorded identification data." The users can be traced, for example, by authentication and/or by authorization or by other means that allows for analysis of the type of users and the nature and quantity of access provided.